

GRADENEXT · SMART LAB

Simulation Experiment Book

Grade 7 · Unit C · The Chemistry of Being Alive

California Integrated Science, Grade 7 · Domain: Biology / Biochemistry · 5 topics · 25 experiments

NGSS performance expectations anchored: MS-LS1-6, MS-LS1-7

Specification standard: v1.0 · Assembled 04 September 2026

Build specifications for interactive simulation experiments · one per syllabus subtopic · for developers, artists and curriculum designers

How to read this book. Every subtopic in the GradeNext California Science syllabus has exactly one simulation experiment specification. Each specification is a *build order*: it tells a developer what to code, an artist what to model, and a curriculum designer what the student is meant to discover. Nothing here is a worksheet; every experiment is a running, manipulable scene with a control panel, more than one scenario, and something measurable coming out.

Anatomy of a specification

Section	What it is for	Who reads it first
Header fields	Experiment name, render mode, simulation engine, interaction level, session length, NGSS anchor	Product, curriculum
Theme & scene	What the student sees on load: setting, camera, palette, lighting, foreground/background, where the panel sits	Art, UX
Objects & components	Every entity the engine must instantiate, including invisible ones (fields, colliders, probes). Build notes give geometry, appearance, counts, behaviour. The <i>Editable</i> column says which objects the student can drag, place, swap, resize or connect directly	3D/2D art, engine
How it works — the model	The rules the engine runs: equations, rate laws, state machines, datasets; what updates per tick; what is deliberately simplified and why	Engine, science review
Control panel	Every control with widget type, range or options, default, unit and visible effect. At least one control is <i>structural</i> (changes which objects exist or how they are arranged)	Engine, UI
Scenarios	Named presets that reconfigure the same simulation to ask a different question. Presets reference control names from the panel	Curriculum, QA
Student activities	Numbered actions the student performs and what they record	Curriculum
Outputs & measurement	Readouts, graphs, tables, badges the sim produces and when they update	Engine, UI, assessment
What the student should realise	The single idea, the misconception it displaces, and the sentence the student should be able to say	Curriculum, assessment

Glossary of tags

Render mode · *2D Canvas* — vector/sprite scene on a flat canvas, the mode of the existing collision and projectile sims · *3D Scene* — camera, lights, meshes and materials · *2.5D Layered* — parallax layers or cutaways with depth ordering but no free camera · *Hybrid 2D+3D* — a 3D viewport alongside a 2D graph, diagram or map panel · *Data Dashboard* — datasets, filters, plot canvases and claim-building tools as the primary scene.

Simulation engine · *Rigid-body* (masses, forces, collisions) · *Particle system* (many small agents with simple rules) · *Field/vector* (scalar or vector fields sampled in space) · *Molecular* (atoms, bonds, valence, CPK colouring) · *Agent-based* (organisms or entities with behaviours) · *Fluid/thermal* (heat flow, buoyancy, convection) · *Ray/wave* (light and sound propagation) · *Data-driven model* (real or realistic datasets driving the scene) · *State machine* (staged processes with transitions) · *Procedural geology* (layers, uplift, erosion, plate motion).

Interaction level · *Explore* (look, zoom, toggle overlays) · *Manipulate* (drag, place, swap objects) · *Investigate* (control variables, measure, record) · *Design* (build, test, score, iterate) · *Argue-from-data* (select evidence, build and score claims).

Object class · *Actor* (moves or acts) · *Structure* (static or slowly changing body) · *Particle* (instanced small entity) · *Field* (invisible spatial quantity) · *Instrument* (measures something) · *Overlay* (visual layer on top of the scene) · *Environment* (backdrop, terrain, container) · *UI-Probe* (cursor-following tool).

Control widgets · Slider · Stepper · Dropdown · Toggle · Radio · Colour · Dial · Drag-handle · Multi-select · Numeric field · Timeline scrubber.

Engineering conventions

- **Experiment ID** is G<grade>-<subtopic> (for example G6-B2.1). It is stable and unique across the whole programme.
- **Scenario presets** use control=value pairs separated by semicolons. Control names in presets are shortened versions of the panel labels; the JSON companion resolves them.
- **Every control carries a unit.** Where a control is unitless the unit column reads "—".
- **The JSON companion** for this unit (json/G<grade>_Unit<letter>.json) contains every field of every specification in structured form, with numeric ranges parsed into min/max and option lists into options[], ready for ingestion by the Smart Lab content pipeline.
- **Ages 11–14.** All simulated hazards are clearly virtual. No procedure here is a real-lab instruction.

Unit map

ID	Subtopic	Experiment	Render	Engine	Level	Session
C1.1	Naming the inputs	Supply Lines: What a Leaf Takes In	Hybrid 2D+3D	Molecular + Particle system	Investigate	12–18 min
C1.2	Naming the outputs	The Harvest: Collecting What a Leaf Makes	Hybrid 2D+3D	Molecular + State machine	Investigate	18–25 min
C1.3	Where in the cell this happens	Zoom In: Hunting the Green Machine	3D Scene	Molecular + Ray/wave	Investigate	18–25 min
C1.4	Photosynthesis as atoms rearranged	Nothing New: The Atom Ledger	Hybrid 2D+3D	Molecular	Investigate	18–25 min
C1.5	Conditions that affect the rate	Bubble Count: Finding the Limiting Factor	Hybrid 2D+3D	Particle system + Data-driven model	Investigate	18–25 min
C2.1	The intuitive but wrong answer	The Soil Story: Run Your Own Model	Hybrid 2D+3D	Data-driven model + Particle system	Investigate	12–18 min
C2.2	Van Helmont's willow tree experiment	Van Helmont's Willow: Five Years on a Balance	3D Scene	State machine + Data-driven model	Investigate	18–25 min
C2.3	What the soil-mass evidence actually shows	Weigh the Ash: What the Tree Is Made Of	3D Scene	State machine + Molecular	Investigate	18–25 min
C2.4	Building the case for air and water	Labelled Atoms: Tracing the New Mass	Hybrid 2D+3D	Molecular + Data-driven model	Investigate	18–25 min
C2.5	Evaluating, not asserting, the conclusion	The Warrant Test: Does Your Evidence Hold?	Data Dashboard	Data-driven model + Agent-based	Argue-from-data	18–25 min
C3.1	Naming the inputs	Supply Line: What a Mitochondrion Orders	3D Scene	Molecular + Particle system	Investigate	18–25 min
C3.2	Naming the outputs	The Exhaust Test: Proving What Comes Out	Hybrid 2D+3D	Fluid/thermal + State machine (reagent chemistry)	Investigate	18–25 min
C3.3	Where in the cell this happens	All the Way In: Hummingbird to Crista	3D Scene	Molecular + State machine (pathway tracker)	Investigate	18–25 min
C3.4	Comparing photosynthesis and respiration side by side	Mirror Image: One Set of Atoms, Two Directions	Hybrid 2D+3D	Molecular + Data-driven model	Investigate	18–25 min
C3.5	Respiration in plants and animals alike	The Plant in the Dark	Data Dashboard	Data-driven model + Fluid/thermal	Argue-from-data	18–25 min
C4.1	Modeling a food molecule broken down	Taking Glucose Apart	3D Scene	Molecular	Manipulate	18–25 min
C4.2	Modeling the atoms rearranged into products	Building the Exhaust	3D Scene	Molecular	Design	18–25 min
C4.3	Energy released, not created	The Bond Bank	Hybrid 2D+3D	Molecular + Data-driven model	Investigate	18–25 min
C4.4	Comparing respiration to a familiar reaction	Two Ways to Burn the Same Sugar	Hybrid 2D+3D	Fluid/thermal + Molecular + Data-driven model	Investigate	18–25 min
C4.5	Applying the model to an unfamiliar food molecule	Mystery Fuel: Balancing a Fat	Hybrid 2D+3D	Molecular + Data-driven model	Argue-from-data	18–25 min
C5.1	A carbon atom enters a plant	Through the Stoma: One Carbon Comes In	3D Scene	Molecular + State machine	Explore	18–25 min
C5.2	A carbon atom moves into an animal	Swallowed: One Carbon Changes Owner	2.5D Layered	Agent-based + State machine	Manipulate	18–25 min
C5.3	A carbon atom returns to the air	Exhaled: One Carbon Leaves the Deer	Hybrid 2D+3D	Molecular + Fluid/thermal	Investigate	12–18 min

ID	Subtopic	Experiment	Render	Engine	Level	Session
C5.4	Photosynthesis and respiration as linked opposite processes	Round Trip: Atoms Cycle, Energy Doesn't	2D Canvas	Molecular + Data-driven model	Argue-from-data	12–18 min
C5.5	Why plants run both processes	Night Shift: A Sealed Seedling, 24 Hours	Hybrid 2D+3D	Data-driven model + Particle system	Investigate	18–25 min

C1 · Photosynthesis: inputs and outputs · MS-LS1-6

C1.1 · Naming the inputs

Experiment name: Supply Lines: What a Leaf Takes In

Render mode: Hybrid 2D+3D

Simulation engine: Molecular + Particle system

Interaction level: Investigate

Session length: 12–18 min

NGSS anchor: MS-LS1-6

THEME & SCENE

A single coast live oak leaf lies across the frame, sliced like a wedge of cake so the student looks into its thickness. The cut face is the star: a glassy cuticle on top, a rank of colourless brick epidermal cells, then tall palisade columns packed with jade lozenges, then spongy cells with open air caves. Sunlight falls from the upper left as visible golden darts. Beneath the leaf the air is a thin haze of red-and-charcoal CO₂ molecules queuing at two open stomatal pores. Rising from the lower right, a glass-clear xylem vessel carries a column of water molecules up into the vein. Palette is jade, straw and pale sky. The control panel is docked right; three brass gate valves stand on the left margin, one per supply line.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Leaf cutaway block	Environment	900×420 px wedge cut at 45°, tissue depth 180 µm to scale, bevelled cut face, printed 50 µm scale bar; rotates ±20° on drag	Yes: drag
2	Waxy cuticle	Structure	4 µm glossy transparent film across the upper surface with one specular streak; gas-tight, so no molecule ever crosses it	No
3	Upper epidermis	Structure	Single rank of 14 flat brick cells, 1 µm colourless walls, deliberately drawn with no chloroplasts	No
4	Palisade mesophyll cell	Actor	8 columns 60×20 µm, thin cellulose wall, one large central vacuole, 38 jade chloroplast lozenges pressed against the outer wall; cytoplasm streams clockwise at 2 µm/s	Yes: swap
5	Spongy mesophyll and air space	Structure	20 rounded cells at 45 percent void; the void is one connected air network tinted pale grey so CO ₂ paths stay visible	No
6	Guard cell pair	Actor	Two kidney-shaped cells 35 µm long, thick inner wall, thin outer wall, 6 chloroplasts each; swell and bow apart to open the pore	Yes: drag
7	Stomatal pore	Structure	Lens-shaped gap between the guard cells, aperture 0–12 µm; the only route CO ₂ has into the leaf	No
8	Xylem vessel	Structure	Hollow tube, 25 µm bore, annular lignin thickening rings, glass-clear; continuous water column; a clamp collar can pinch it shut	Yes: connect
9	Phloem sieve tube	Structure	Parallel tube with perforated sieve plates, amber sucrose dots moving downward; carries product out, takes no input	No
10	Chloroplast	Actor	5×2 µm jade lozenge, double envelope, 12 dark grana stacks resolvable at this zoom; a small hopper gauge on its flank fills with arriving molecules	No
11	CO ₂ molecule	Particle	Linear O=C=O, charcoal C sphere r 0.77 u between two red O spheres r 0.66 u, double-bond rod pairs 116 pm; 400 instanced, Brownian drift, serial-tagged	No
12	Water molecule	Particle	Bent triad at 104.5°, red O r 0.66 u with two white H r 0.31 u, O–H rods 96 pm; 1200 in the xylem column, rising at 4 mm/s scaled	No
13	Photon packet	Particle	Golden 6 px dart carrying a wavelength tag and zero mass; annihilates on a chlorophyll hit; 200 per second at default intensity	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
14	Mineral ion set	Particle	Violet K^+ , blue NO_3^- and green Mg^{2+} spheres riding the xylem stream at 1 ion per 400 water molecules; the deliberate distractor	Yes: swap
15	Supply valve rig	Instrument	Three brass gate valves on the left margin labelled LIGHT, AIR, WATER; closing one greys its whole route and stops its counter	Yes: drag
16	Input tally board	Overlay	Docked strip counting molecules delivered to the chloroplast per minute per species, plus atoms of C, H and O arriving per minute	No

HOW IT WORKS — THE MODEL

Three independent supply lines feed one consumer. Light arrives as photon packets at a flux set by the intensity slider. CO_2 enters only through open stomata, so its delivery rate is proportional to aperture area multiplied by the outside-to-inside concentration difference. Water arrives by mass flow up the xylem at a rate set by soil availability and by transpiration through those same pores, which is why closing the stomata throttles two lines at once. The chloroplast hopper consumes strictly 6 CO_2 to 6 H_2O to 48 photons per glucose; each tick it completes whatever combinations it can and queues the rest visibly. Rate is therefore set by the scarcest line, never by the total. Two honesty rules the build must keep: photons must never enter the atom columns of the tally, because light supplies energy and not matter, and mineral ions must be visible yet must always tally under 0.1 percent of arriving atoms.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Light intensity	Slider	0–2000	600	$\mu\text{mol m}^{-2} \text{s}^{-1}$	Photon packet flux onto the leaf surface
Stomatal aperture	Slider	0–12	7	μm	Guard cell shape and the CO_2 entry rate
Air CO_2 concentration	Slider	0–1200	420	ppm	Molecule density in the air below the leaf
Soil water availability	Slider	0–100	80	percent	Height and speed of the xylem water column
Xylem clamp	Toggle	Open / Clamped	Open	—	Cuts the water line at the petiole without touching the other two
Mineral supply	Dropdown	None / Trace / Full	Full	—	Ion density in the xylem stream; tests the soil-food idea
Cutaway plane	Drag-handle	0–100	40	percent of blade width	Slides the slice across the leaf to expose vein or lamina
Molecule labels	Toggle	On / Off	On	—	Chemical formulae float beside each particle species
Playback speed	Dial	0.25×–4×	1×	—	Time scaling of the whole scene

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Open for business	light=600; aperture=7; CO_2 =420; water=80; clamp=Open	Three things arrive at the chloroplast. Which two are made of atoms, and which one is not?
S2	Sealed pores	aperture=0; light=600; water=80	Closing one 12 μm gap stops the leaf almost completely. Which input line did that gap belong to?
S3	Rich soil, no water	minerals=Full; water=0; clamp=Clamped; light=600	The soil is packed with minerals. Why does the chloroplast hopper still stop filling?
S4	Central Valley midday	light=1900; CO_2 =420; aperture=2; water=25	On a hot dry afternoon the leaf shuts its pores to save water. What does it lose by doing so?

STUDENT ACTIVITIES

- Set all three valves open at default and run 60 s. Record the tally board figures for CO_2 , H_2O and photons delivered.
- Close the LIGHT valve only. Re-run 60 s and record which counters fall to zero and which keep running.

3. Drag the stomatal aperture from 7 μm to 0 and back in four steps, recording CO_2 delivery at each step; state which other counter moved with it.
4. Set minerals to Full and water to 0. Record the atom tally and write one sentence on whether minerals can be the source of the leaf's mass.
5. Predict, before touching it, what raising CO_2 to 1200 ppm does when light is set to 100. Then test and explain the result.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Molecules delivered	Live numeric	count/min	Separate counters for CO_2 , H_2O and mineral ions reaching the chloroplast, refreshed every 2 s
Photon arrival rate	Live numeric	packets/s	Energy input, displayed in its own panel away from the atom columns
Atom inventory in	Data table	atoms/min	C, H and O arriving, split by which molecule carried them
Glucose assembly progress	Live numeric	percent	Fill level of the chloroplast hopper toward the next complete glucose
Limiting input	Pass-fail badge	—	Names the scarcest of the three lines and turns amber when it changes
Run log	Data table	mixed	One row per run with all control values and all counters; exportable CSV

WHAT THE STUDENT SHOULD REALISE

Students believe a plant takes food from the soil and that sunlight is another ingredient. The atom tally separates the claims: the only atoms crossing into the leaf come from carbon dioxide and water, minerals arrive but stay below a thousandth of it, and photons register in the energy column with no atoms. The student should be able to say: *"A leaf takes in carbon dioxide and water as materials, and light as energy — those are different jobs."*

C1.2 · Naming the outputs

Experiment name: The Harvest: Collecting What a Leaf Makes

Render mode: Hybrid 2D+3D

Simulation engine: Molecular + State machine

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS1-6

THEME & SCENE

A school prep-room bench under cool daylight, everything laid out in the order the procedure runs. At the centre stands a potted variegated geranium, cream-margined leaves over deep green centres, with one leaf clipped between two halves of a matte-black foil stencil cut into a star. On the left a borosilicate bell jar waits on a greased plate, an oxygen probe through its collar and a glowing splint clamped beside it. To the right runs the testing line: a hotplate carrying a beaker of boiling water, a smaller ethanol beaker standing inside it, a white glazed tile, and an amber dropper of iodine. Floating above the bench is a circular molecular window into one chloroplast, where glucose rings are being threaded onto a growing starch grain. Control panel docked right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Variegated geranium	Actor	24 cm potted plant, 9 leaves, cream margin over green centre, turgid stems; any leaf can be detached with forceps	Yes: drag
2	Foil stencil mask	Structure	Two 0.1 mm matte-black aluminium plates clipped either side of one blade, cut-out shapes star / letter G / circle / half-leaf	Yes: place
3	Bell jar and base plate	Instrument	30 cm borosilicate dome, ground rim on a greased glass plate, side port with a rubber septum, sealed volume 9.4 L	No
4	Oxygen probe	Instrument	8 mm optical O ₂ sensor through the collar, range 0–30 percent v/v, samples every 1 s, digital head	No
5	Soda-lime sachet	Structure	20 g of white pellets in a mesh bag inside the jar; drops jar CO ₂ below 5 ppm within 10 min when fitted	Yes: place
6	Glowing splint	Actor	10 cm wooden splint with an orange ember; flares back to flame with a visible white burst above 24 percent O ₂	Yes: swap
7	Hotplate and water bath	Instrument	18 cm ceramic top, dial 0–200 °C, 400 mL beaker of water held at 100 °C to kill the leaf and stop it changing	No
8	Ethanol beaker	Instrument	100 mL beaker standing inside the water bath, never over a flame; ethanol runs clear to deep bottle-green as chlorophyll leaves the leaf	No
9	White tile and forceps	Instrument	12 cm glazed tile plus stainless forceps; the decolourised leaf lands brittle and pale straw on the tile	Yes: drag
10	Iodine dropper	Instrument	Amber bottle of iodine in potassium iodide, orange-brown drops that turn blue-black on starch within 3 s	Yes: drag
11	Chloroplast inset window	Overlay	240 px circular view: double envelope, 14 grana stacks, stroma, one growing starch grain, faint scanline rim	No
12	Glucose molecule	Particle	Six-membered pyranose ring, 6 charcoal C, 6 red O, 12 white H, ball-and-stick, all 24 atoms serial-tagged	No
13	Starch grain	Structure	Cream-white ovoid in the stroma growing from 0.2 to 4 µm as glucose rings link into chains, concentric growth rings visible	No
14	Oxygen molecule	Particle	Two red spheres r 0.66 u joined by one 121 pm double-bond rod; released at a granum, drifts out through the stoma into the jar	No
15	Sucrose export stream	Particle	Amber paired rings entering the phloem and moving down the stem, showing sugar leaving rather than piling up	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Product tally board	Overlay	Running counts of O ₂ released, glucose assembled, starch mass gained, and the C, H and O atoms now locked in products	No

HOW IT WORKS — THE MODEL

The leaf runs a three-condition gate at every 2 mm patch: light reaching that patch, chlorophyll present in that patch, and CO₂ available inside the jar. Product rate at a patch is the product of the three, so any one of them at zero gives zero, which is what makes the stencil and the cream margin readable. Fixed carbon accumulates first as glucose, then polymerises into a starch grain at a rate that saturates, and drains during dark pre-treatment at a slower fixed rate — this is why 48 h of darkness empties the leaf. Oxygen is released stoichiometrically at one O₂ per CO₂ fixed and mixes into the jar volume, so the probe trace and the iodine map come from the same event, not two separate stories. The iodine test is destructive and irreversible: once a leaf is boiled the sim must lock it out of further exposure runs.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Dark pre-treatment	Slider	0–72	48	h	How completely existing starch is drained before the run starts
Light exposure	Slider	0–12	6	h	How long the masked leaf sits under the lamp before testing
Light intensity	Slider	0–1200	500	μmol m ⁻² s ⁻¹	Photon flux on the canopy, and the depth of colour in the iodine map
Mask shape	Dropdown	None / Star / Letter G / Circle / Half-leaf	Star	—	Which cut-out appears in the stain; swaps the stencil object
Leaf chosen	Dropdown	All-green / Variegated / Chlorotic yellow	Variegated	—	Structural swap of the leaf that enters the testing line
Jar atmosphere	Dropdown	Normal air / CO ₂ -free with soda lime / CO ₂ -enriched 1000 ppm	Normal air	—	Whether CO ₂ reaches the leaf; fits or removes the sachet
Ethanol bath time	Slider	0–10	5	min	How completely chlorophyll is removed; under 3 min leaves green that hides the stain
Iodine drops	Stepper	0–10	4	drops	Stain coverage across the tile
Inset zoom	Dropdown	Leaf / Cell / Chloroplast / Molecule	Chloroplast	—	Which scale the molecular window shows

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	The stencil test	leaf=All-green; mask=Star; dark=48; exposure=6; atmosphere=Normal air	Where does blue-black appear, and what was missing everywhere it stayed straw-coloured?
S2	The cream margin	leaf=Variegated; mask=None; exposure=6; intensity=500	Every part of this leaf got the same light and the same air. Why does only part of it stain?
S3	Air taken away	leaf=All-green; mask=None; atmosphere=CO ₂ -free with soda lime; exposure=6	Light and chlorophyll are both present. Why is there no starch and almost no oxygen?
S4	Overnight control	leaf=All-green; mask=None; dark=48; exposure=0	Nothing stains at all. Does the glowing splint agree, and what does that tell you about the two outputs?

STUDENT ACTIVITIES

1. Set the dark pre-treatment to 48 h and run the full testing line on an unexposed leaf. Record the iodine result as your zero.
2. Place the star stencil on an all-green leaf, expose for 6 h, then run the test. Sketch the stain pattern and label which region got light.

3. Measure the jar oxygen at the start and end of the same 6 h run, and test with the glowing splint. Record both results side by side.
4. Swap to CO₂-free air and repeat. Record starch result and oxygen change, then state which single input you removed.
5. Zoom the inset to Molecule and count the atoms in one glucose ring. Compare that count with the product atom inventory after a full run.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Iodine result map	Heat map	—	False colour over the leaf outline from straw (no starch) to blue-black (dense starch), sampled on a 2 mm grid
Oxygen in the jar	Line graph	percent v/v	Probe trace across the whole exposure, 1 s samples, with the start value marked
Glowing splint test	Pass-fail badge	—	Relight or no relight, stamped with the O ₂ percentage at the moment of testing
Oxygen released	Counter	count and cm ³	Cumulative O ₂ molecules leaving the leaf, also shown as gas volume at room conditions
Starch grain mass	Live numeric	pg per chloroplast	Rises during exposure, falls during the dark pre-treatment, updated every 5 simulated minutes
Product atom inventory	Data table	atoms	C, H and O locked into glucose and released as O ₂ per run; exportable CSV

WHAT THE STUDENT SHOULD REALISE

Students describe a plant's outputs as "oxygen for us" and imagine sugar being eaten as it is made. Both tests here fire from one event: starch appears only where light, chlorophyll and CO₂ all reached, and the jar oxygen rises over exactly the same hours. Oxygen is a leftover of the reaction, not a service; sugar is a store the plant builds. The student should be able to say: *"The leaf makes sugar and lets the oxygen go."*

C1.3 · Where in the cell this happens

Experiment name: Zoom In: Hunting the Green Machine

Render mode: 3D Scene

Simulation engine: Molecular + Ray/wave

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS1-6

THEME & SCENE

The scene opens on a sunlit coast live oak leaf in a soft-jawed clamp, underside to the camera, a 2 cm scale bar in the corner. Turning the zoom dial drives a single continuous flight with no cuts: the surface resolves into interlocking jigsaw epidermal cells, a stoma parts between two kidney-shaped guard cells, the camera slips through the gap into a moist air cave, brushes past spongy cells, enters the flank of a palisade column, and finally crosses the double envelope of one chloroplast to hover between two grana. The palette cools as it descends, oak-green outside to deep teal within, with wet highlights on every membrane. A micro-beam spotlight and a needle-thin oxygen microsensor hang in frame throughout. Control panel docked right; the scale bar updates every frame.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Leaf blade and cutaway wedge	Environment	6 cm oak leaf, 210 µm thick, held in a clamp; a slicing plane opens a wedge so the camera can enter the interior	Yes: drag
2	Cuticle and epidermis	Structure	4 µm glassy wax film over one rank of jigsaw-edged cells 30 µm across, colourless walls, no chloroplasts	No
3	Guard cell pair and stoma	Actor	Two kidney cells 35 µm long, thickened inner walls, 6 small chloroplasts each; bow apart to open a 0–12 µm pore	Yes: drag
4	Mesophyll cells	Actor	8 palisade columns 60×20 µm plus 20 rounded spongy cells at 45 percent air space; palisade holds 38 chloroplasts, spongy holds 12	Yes: swap
5	Central vacuole and nucleus	Structure	Vacuole filling 80 percent of cell volume, faint straw tint; nucleus 8 µm with visible pores and a dense nucleolus; both pigment-free	No
6	Mitochondrion	Actor	2×0.6 µm capsule, russet, folded cristae inside a double membrane; drifts on cytoplasmic streaming; the contrast organelle	No
7	Chloroplast outer envelope	Structure	Smooth continuous membrane, 5×2 µm lens, pale jade, freely permeable, drawn as a distinct skin	No
8	Chloroplast inner envelope	Structure	Second membrane 20 nm inside the first with a visible intermembrane gap; controls what enters the stroma	No
9	Thylakoid disc	Structure	Flattened sac 500 nm across, 20 nm lumen, teal membrane with a wet sheen; the only surface carrying pigment	No
10	Granum	Structure	Stack of 12–18 thylakoid discs like a roll of coins, 14 grana per chloroplast, spaced 10 nm apart	Yes: resize
11	Stroma lamella	Structure	Unstacked thylakoid tube 40 nm wide bridging neighbouring grana at a shallow angle, 2–3 per granum pair	No
12	Stroma	Structure	Viscous olive-green fluid filling the envelope, with 70S ribosomes as 20 nm dots and one circular DNA loop	No
13	Chlorophyll molecule	Particle	Flat porphyrin head 1.5 nm with a central Mg atom plus a phytol tail anchored in the lipid bilayer; 2400 instanced per disc face	Yes: swap
14	Starch grain	Structure	Cream ovoid 2–4 µm in the stroma, concentric rings, grows during a lit run	No
15	Micro-beam and O ₂ microsensor	Instrument	1–20 µm spotlight of chosen wavelength plus a 3 µm sensor tip 5 µm downstream reading oxygen evolution	Yes: place

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Zoom dial and scale bar	UI-Probe	Continuous camera track with a live field-width readout that steps from cm to μm to nm without ever cutting	Yes: drag

HOW IT WORKS — THE MODEL

This is a localisation assay in the style of Engelmann's micro-beam. A spotlight of chosen diameter and wavelength is aimed at one named structure, and the microsensor reads oxygen evolved from that spot. The rule the engine runs is $\text{signal} = \text{beam power} \times \text{chlorophyll surface density at the hit point} \times \text{action-spectrum weight for that wavelength}$. Chlorophyll surface density is non-zero only on thylakoid membranes, so wall, vacuole, nucleus, mitochondrion and stroma fluid all return a flat zero, grana return maximum, stroma lamellae return roughly 40 percent because they carry less membrane area per unit volume, and guard cells return a small positive. Zoom is one continuous camera path with a continuously updating scale bar and no jump cuts, because the misconception that a chloroplast is a solid green blob comes from cutting straight to a cartoon icon at high magnification.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Zoom	Dial	1×–500 000×, detents at leaf, surface, cell, chloroplast, membrane	1×	—	Camera altitude along the flight path and the scale bar reading
Micro-beam target	Dropdown	Cell wall / Vacuole / Nucleus / Mitochondrion / Stroma / Granum / Stroma lamella / Guard cell	Granum	—	Which structure the spot lands on
Beam wavelength	Slider	400–740	662	nm	Spot colour and the action-spectrum weight applied to the signal
Beam power	Slider	0–100	40	μW	Photon rate into the spot
Spot diameter	Slider	1–20	5	μm	How selectively one structure can be hit without catching its neighbours
Cutaway plane	Drag-handle	0–100	45	percent of leaf thickness	Slices the tissue open so interior structures are reachable
Chlorophyll density	Slider	0–100	100	percent of normal	Strips pigment from the thylakoids while leaving the membranes intact
Leaf sector	Dropdown	Green / Cream margin / Chlorotic yellow	Green	—	Structural swap of which tissue the camera and beam are working in
Stomatal aperture	Slider	0–12	7	μm	Width of the entry route the camera flies through

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Find the source	zoom=chloroplast; target=Granum; wavelength=662; power=40; sector=Green	Fire the beam at six structures in turn. Which ones give an oxygen signal at all?
S2	Membranes without pigment	chlorophyll=0; target=Granum; power=40	The thylakoid stacks are still there and still stacked. Why has the signal vanished?
S3	The cream margin	sector=Cream margin; target=Granum; zoom=cell	The beam reports no such structure here. What are these cells missing that green cells two millimetres away have?
S4	Guard cells count too	sector=Green; target=Guard cell; zoom=surface; spot=5	Guard cells are green and they do give a signal. Why is it so much smaller than a palisade cell's?

STUDENT ACTIVITIES

1. Drag the zoom dial from 1× to 500 000× in one continuous move and record the scale bar value at each of the five detents.
2. Fire the micro-beam at the wall, vacuole, nucleus, mitochondrion, stroma and granum in turn. Record the sensor reading for each in the hit log.
3. Set chlorophyll density to 0 and repeat the granum shot. Record the new reading and state what changed and what did not.
4. Count the chloroplasts in one palisade cell and the grana in one chloroplast, then read off the thylakoid membrane area. Record all three.
5. Sweep beam wavelength from 400 to 740 nm at the granum in 20 nm steps and mark the two wavelengths that give the strongest signal.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Oxygen microsensor	Live numeric	pmol O ₂ cm ⁻² s ⁻¹	Signal at the sensor tip, refreshed every 0.2 s while the beam is on
Structure hit log	Data table	mixed	One row per shot: structure, wavelength, power, spot size, signal; exportable
Signal by structure	Bar chart	pmol O ₂ cm ⁻² s ⁻¹	One bar per structure, built up as the student samples; zeros stay visible as empty bars
Scale readout	Live numeric	m, switchable to μm and nm	Current field width, updating continuously through the zoom flight
Chloroplast census	Counter	count	Chloroplasts in the cell on screen and grana in the chloroplast on screen
Thylakoid membrane area	Live numeric	μm ² per chloroplast	Total pigment-bearing surface in the current chloroplast; rises as grana are resized

WHAT THE STUDENT SHOULD REALISE

Students place photosynthesis vaguely "in the leaf", or wherever the plant looks green. The beam makes location testable: wall, vacuole, nucleus, mitochondrion and stroma fluid all give nothing, the granum gives a strong signal, and stripping pigment from intact thylakoids kills that signal outright. The site is a membrane, not a colour and not a whole organ. The student should be able to say: *"It happens on the thylakoid membranes inside chloroplasts, because that is where the chlorophyll sits."*

C1.4 · Photosynthesis as atoms rearranged

Experiment name: Nothing New: The Atom Ledger

Render mode: Hybrid 2D+3D

Simulation engine: Molecular

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS1-6

THEME & SCENE

A dark, quiet workbench lit only from inside a glass reaction chamber standing at the centre on a steel plinth. To its left, two loading racks: one holding linear red-charcoal-red carbon dioxide molecules, one holding bent white-red-white water molecules, each sitting in a numbered slot. Above the chamber a small lamp drops golden photon packets onto the glass. To the right, two catch trays sit empty at the start, waiting for glucose and for oxygen. Along the bottom edge runs the atom ledger: three columns headed C, H and O, each with a live count and a stack of coloured tokens that physically move as bonds change. Every atom in the scene carries a faint serial tag that stays with it through every rearrangement.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Reaction chamber	Instrument	220 mm borosilicate cylinder on a steel plinth, lit from within, slow internal swirl; a lid ring switches it between sealed and vented	Yes: swap
2	Carbon atom	Particle	Charcoal sphere drawn at r 0.77 u, matte finish, serial tags C-01 to C-12	No
3	Oxygen atom	Particle	Red sphere r 0.66 u, satin finish, serial tags O-01 to O-36	No
4	Hydrogen atom	Particle	White sphere r 0.31 u, glossy, serial tags H-01 to H-24	No
5	Double bond cylinder	Structure	Two parallel rods 0.08 u thick, 116 pm for C=O and 121 pm for O=O; flashes white for 0.3 s as it breaks or forms	No
6	Single bond cylinder	Structure	One rod 0.10 u thick at the correct length: O–H 96 pm, C–H 109 pm, C–C 154 pm, C–O 143 pm	No
7	Carbon dioxide molecule	Actor	Linear at 180°, one C flanked by two O with double bonds; 12 available in the left rack	Yes: drag
8	Water molecule	Actor	Bent at 104.5°, one O with two single-bonded H; 12 available in the left rack	Yes: drag
9	Photon packet	Particle	Golden dart with a wavelength tag and zero mass; enters the energy panel and is barred from the atom columns by construction	No
10	Glucose molecule	Actor	Six-carbon pyranose ring with one CH ₂ OH arm, C ₆ H ₁₂ O ₆ , 24 atoms; assembles atom by atom over 6 s in the chamber	No
11	Oxygen molecule	Actor	Two red spheres joined by one 121 pm double bond; drifts to the right catch tray on release	No
12	Loading racks	Instrument	Two numbered 12-slot racks; a molecule is dragged from a slot into the chamber mouth and cannot be un-made once inside	Yes: place
13	Atom ledger panel	Overlay	Three live columns, C, H and O, each split into In chamber, In products and Unused; plus a total mass row in u	No
14	Serial-tag overlay	Overlay	Small floating ID beside every atom; one selected atom is haloed and its route is drawn as a ribbon	Yes: swap
15	Leftover bin	Structure	Glass dish beneath the chamber where unmatched molecules drop and are counted by species	No
16	Balance plate	Instrument	Weighing plate under the whole chamber reading total contents in u before, during and after the run	No

HOW IT WORKS — THE MODEL

The chamber runs strict conservation. Every atom is a persistent object with an immutable serial number; bonds are separate objects that can be destroyed and created, but atoms can only be re-parented. The reaction proceeds as $6 \text{CO}_2 + 6 \text{H}_2\text{O} + \text{light energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6 \text{O}_2$, and the engine will only complete an assembly when all required atoms are physically present in the chamber. Photon packets deplete an energy budget of 8 packets per CO_2 fixed and are logged in a separate energy panel that has no atom columns at all. Anything unmatched is pushed to the leftover bin, still carrying its serials. The failure the build must avoid is the classic cheat of spawning product atoms and despawning reactant atoms: products must be assembled from the very spheres the student loaded, visibly, or the ledger teaches nothing.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
CO_2 molecules loaded	Stepper	0–12	6	count	Carbon and oxygen atoms entering the chamber
Water molecules loaded	Stepper	0–12	6	count	Hydrogen and oxygen atoms entering the chamber
Photon packets	Slider	0–96	48	count	Energy budget; 48 packets are needed to complete one glucose
Chamber seal	Toggle	Sealed / Vented	Sealed	—	Vented lets O_2 leave, so the after-tally loses atoms and the badge fails
Build mode	Radio	Guided / Free	Guided	—	Structural: Guided auto-pairs molecules, Free waits for the student to drag every one
Traced atom	Dropdown	None / any serial from C-01 to O-36	O-07	—	Haloes one atom and draws its path from reactant to product
Display style	Dropdown	Ball-and-stick / Space-fill / Skeletal with labels	Ball-and-stick	—	How atoms and bonds are drawn in the chamber
Assembly speed	Dial	0.25×–4×	1×	—	Time scaling of bond breaking and forming
Reverse run	Toggle	Off / On	Off	—	Runs the same chamber backwards as respiration; the ledger must still balance

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	The exact recipe	$\text{CO}_2=6$; water=6; photons=48; seal=Sealed; mode=Guided	Count every element before and after. Which of the three counts changed?
S2	Short of water	$\text{CO}_2=6$; water=2; photons=48; mode=Guided	No complete glucose forms. What is sitting in the leftover bin, and what does it tell you about the recipe?
S3	Too much air	$\text{CO}_2=10$; water=6; photons=48; mode=Free	Four CO_2 are left over. Does the glucose come out bigger, or does something else happen instead?
S4	Follow one oxygen	$\text{CO}_2=6$; water=6; photons=48; traced=O-07	Trace an oxygen that started in a water molecule. Where does it end up, and where do the CO_2 oxygens go?

STUDENT ACTIVITIES

1. Load 6 CO_2 and 6 H_2O , run, and record the C, H and O totals before and after from the ledger.
2. Set water to 2 and re-run. Record what appears in the leftover bin and how many O_2 were released.
3. Switch to Free mode and build one glucose by dragging molecules yourself. Record how many of each you needed.
4. Select a traced atom in a water molecule, run, and write down which product molecule it finished in.
5. Set the chamber to Vented, re-run, and explain in one sentence why the balance badge turns red even though nothing was destroyed.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Atom ledger	Data table	atoms	C, H and O counted before, during and after; the three after-totals must equal the three before-totals
Element tally strip	Bar chart	atoms	Paired before and after bars for each element, updating every frame as bonds change
Balance badge	Pass-fail badge	u	Total chamber mass before against after; green only when identical to the unit
Molecules formed	Counter	count	Glucose and O ₂ produced this run, incremented as each completes
Leftover contents	Live numeric	count	Unmatched molecules and loose atoms in the bin, listed by species
Traced atom path	Vector overlay	—	Ribbon showing the selected atom's journey and every bond it belonged to

WHAT THE STUDENT SHOULD REALISE

Students think a plant makes matter out of sunlight, or that atoms are used up and new ones appear. The ledger makes both impossible: every atom in the glucose and the oxygen was already in the chamber, serial number and all, the element totals never move, and light is logged in an energy panel with no atoms attached. The student should be able to say: *"Nothing new was made. The same atoms were taken apart and put back together differently."*

C1.5 · Conditions that affect the rate

Experiment name: Bubble Count: Finding the Limiting Factor

Render mode: Hybrid 2D+3D

Simulation engine: Particle system + Data-driven model

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS1-6

THEME & SCENE

A school lab bench seen straight on, deliberately plain so the numbers dominate. A boiling tube of pondwater holds three sprigs of Cabomba, cut stems upward, weighted at the base with a paperclip. Silver oxygen bubbles rise from the cut ends in a steady string, funnelled by an inverted glass cone into a graduated collecting tube. A lamp rides on a metre ruler track laid across the bench, its distance printed in centimetres. Between lamp and tube stands an optional glass tank of water, the heat shield. A wheel of coloured filter discs clicks in front of the lamp; a thermostatted water jacket surrounds the boiling tube; a syringe of sodium hydrogencarbonate solution stands ready. A photo-gate straddles the bubble path. Graph panel along the bottom, control panel docked right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Boiling tube and rack	Instrument	25×150 mm borosilicate tube in a white plastic rack, filled to 130 mm with pondwater, faint green tint	No
2	Cabomba sprigs	Actor	1–5 stems 80 mm long, whorls of feathery bright-green leaves every 8 mm, gently swaying; trimmed at the top	Yes: place
3	Cut stem end	Structure	Freshly cut 2 mm bore at the top of each sprig, pale cream core; the single emission point for bubbles	Yes: resize
4	Oxygen bubble	Particle	Silvered sphere 1.6 mm across, volume 2.1 mm ³ , rises at 22 mm/s with a slight wobble, coalesces at the funnel neck	No
5	Inverted funnel and collecting tube	Instrument	60 mm glass cone over the sprigs feeding a 5 cm ³ graduated tube marked every 0.1 cm ³	No
6	Lamp on ruler track	Instrument	25 W LED lamp in a reflector on a sliding carriage; metre ruler graduated in cm runs the bench width	Yes: drag
7	Colour filter wheel	Structure	Six-position disc of 50 mm gel filters, clear / blue 450 / green 550 / yellow 590 / red 660 / far-red 730	Yes: swap
8	Heat shield tank	Structure	120×80×60 mm glass tank of clear water between lamp and tube; absorbs infrared without dimming much	Yes: place
9	Thermostatted water jacket	Instrument	Clear sleeve around the boiling tube fed by a bath, setpoint 5–45 °C, with a digital thermometer probe in the water	No
10	Hydrogencarbonate syringe	Instrument	10 mL syringe of sodium hydrogencarbonate solution; each dose visibly disperses as a shimmer through the tube	Yes: drag
11	Dissolved CO ₂ field	Field	Invisible scalar field through the tube volume, in mmol/L, depleted locally near the leaves and replenished by stirring	No
12	Photo-gate bubble counter	Instrument	Infrared beam and detector clamped across the bubble path, 8 mm above the stems; increments on each break	No
13	Dissolved oxygen probe	Instrument	6 mm optical probe through a bung, range 0–20 mg/L, 1 s sampling, digital head on the rack	Yes: place
14	Light meter	Instrument	Flat sensor taped to the tube face reading incident photon flux, so intensity is measured and not assumed	No
15	Rate graph panel	Overlay	Dark-slate panel along the bottom plotting the chosen factor on x and bubble rate on y, points added per run	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Results logbook	Overlay	Scrolling table of every counting window with all control settings attached; exportable	No

HOW IT WORKS — THE MODEL

Two independent supply terms compete. The light-driven supply is $L = k \cdot I \cdot w(\lambda)$, where I is the measured flux at the tube face, falling with the inverse square of lamp distance, and $w(\lambda)$ is an action-spectrum weight: 1.00 at 440 and 662 nm, 0.15 at 550 nm, 0.05 beyond 720 nm. The carbon supply is $C = m \cdot [\text{CO}_2]$ from the dissolved hydrogencarbonate. Gross rate is the smaller of L and C , multiplied by a temperature factor with Q_{10} of 2.0 from 5 to 30 °C, flat to 35 °C, then falling to zero by 48 °C as enzymes denature. Net oxygen output is gross minus respiration, which also rises with temperature, so below the light compensation point no bubbles appear at all. Bubble count is net volume divided by 2.1 mm³. The plateaus are never drawn in: they emerge because one term overtakes the other.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Lamp distance	Drag-handle	5–100	20	cm	Carriage position on the ruler; incident flux follows the inverse square law
Lamp power	Slider	5–60	25	W	Brightness at the source, and heat output if the shield is removed
Filter	Dropdown	Clear / Blue 450 / Green 550 / Yellow 590 / Red 660 / Far-red 730	Clear	nm	Wavelength reaching the sprigs and its action-spectrum weight
Hydrogencarbonate	Slider	0.0–2.0	0.2	percent w/v	Dissolved CO ₂ available to the leaves
Water temperature	Slider	5–45	20	°C	Jacket setpoint, so enzyme rate and respiration rate
Number of sprigs	Stepper	1–5	3	count	Structural: total leaf area, so the height of every plateau
Heat shield	Toggle	Fitted / Removed	Fitted	—	Structural: blocks lamp heat, separating the light variable from temperature
Counting window	Slider	30–300	120	s	How long the photo-gate counts before reporting a rate
Settling time	Slider	0–10	5	min	Acclimatisation allowed after any change before counting starts
Plot mode	Dropdown	Rate vs light / Rate vs CO ₂ / Rate vs temperature / Rate vs wavelength	Rate vs light	—	Which factor the graph panel builds along its x-axis

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Build the light curve	distance stepped 100 → 50 → 30 → 20 → 10 → 5; hydrogencarbonate=0.2; temperature=20; shield=Fitted	The curve rises then flattens. What is running out at the flat part, given that light is still increasing?
S2	Lift the ceiling	distance=10; hydrogencarbonate stepped 0.2 → 1.0 → 2.0; temperature=20	Light was already bright and did not change. Why does the rate climb again?
S3	Bright but useless	filter=Green 550; distance=10; hydrogencarbonate=1.0; temperature=20	The tube looks brightly lit to your eye. Why do the bubbles almost stop?
S4	Delta canal in August	temperature=38; distance=10; hydrogencarbonate=1.0; shield=Fitted	Warm water and strong light, yet the rate is below the 30 °C result. What has gone wrong inside the leaf?

STUDENT ACTIVITIES

1. Set the lamp at 100 cm, allow 5 min settling, and count bubbles for 120 s. Record the rate and the light meter reading together.
2. Repeat at 50, 30, 20, 10 and 5 cm. Plot rate against measured flux and mark the point where the curve stops rising.
3. At 10 cm, raise hydrogencarbonate from 0.2 to 1.0 to 2.0 percent, counting after each change. Record what happens to the plateau height.
4. Remove the heat shield, re-run the 5 cm reading, and record both bubble rate and water temperature. Explain why this run cannot be used as a light result.
5. Step the temperature from 5 to 45 °C in 5 °C steps at fixed light and CO₂. Record the rate at every step and mark the temperature where it peaks.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Bubble rate	Counter	bubbles/min	Photo-gate count divided by the counting window, reported at the end of each window
Oxygen collected	Live numeric	cm ³	Cumulative gas in the graduated tube, readable to 0.1 cm ³
Dissolved oxygen	Line graph	mg/L	Probe trace against time for the whole session, 1 s samples
Rate against chosen factor	Line graph	bubbles/min vs factor unit	Built point by point as the student steps a control; plateaus appear without being drawn in
Limiting factor	Pass-fail badge	—	Names whichever of light, CO ₂ or temperature currently caps the rate, and flags when it changes
Results table	Data table	mixed	Every counting window with all ten control values attached; exportable CSV

WHAT THE STUDENT SHOULD REALISE

Students believe more light always means more photosynthesis, and treat the factors as adding up. The curve refuses to keep rising, and only raising CO₂ lifts the ceiling, so addition cannot be the rule. Green light being bright yet useless separates energy delivered from energy absorbed. The student should be able to say: *"The rate is set by whichever condition is in shortest supply, and adding more of anything else does nothing until you fix that one."*

C2 · Evidence for where a plant's mass comes from · MS-LS1-6

C2.1 · The intuitive but wrong answer

Experiment name: The Soil Story: Run Your Own Model

Render mode: Hybrid 2D+3D

Simulation engine: Data-driven model + Particle system

Interaction level: Investigate

Session length: 12–18 min

NGSS anchor: MS-LS1-6

THEME & SCENE

A Central Valley greenhouse bench in flat morning light, split down the middle by a brass rail. On the right stands the real plant: a sunflower seedling in a terracotta pot on a large digital balance, a fine steel pin gauge resting on the soil surface against a millimetre scale, and a cutaway window in the pot wall showing the root ball. On the left is the model board, four glass source tanks labelled SOIL, WATER, AIR and SUNLIGHT, each with a brass tap, a mass gauge and its own colour: umber, blue, grey, gold. When the run starts, coloured mass parcels stream from whichever taps the student opened into the growing plant, and every tank's gauge falls by what it gave. Control panel right, timeline scrubber along the foot.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Terracotta pot and soil column	Environment	22 cm rim pot, 3.10 kg of dry loam filled to a scored line, crumb texture, darkens when watered	Yes: resize
2	Soil-level pin gauge	Instrument	Steel needle resting on the surface against a 0–50 mm vernier scale; reads to 0.5 mm and never moves on its own	Yes: drag
3	Sunflower plant	Actor	Six growth stages from 4 cm seedling to 1.4 m plant with 14 leaves and one head; stem thickens, leaves unfurl on schedule	No
4	Root ball	Structure	Branching white root system visible through a glass cutaway in the pot wall; extends but does not consume the soil crumbs	No
5	Digital balance (whole pot)	Instrument	300×300 mm platform, capacity 20 kg, resolution 0.1 g, backlit display, tare button	No
6	Soil-only balance stand	Instrument	Second pan beside the bench used at harvest, after the plant is lifted out and the soil is sieved and oven-dried	Yes: place
7	Soil source tank and tap	Instrument	Umbre glass cylinder holding 3100 g, brass tap, gauge falling as mass parcels leave	Yes: drag
8	Water source tank and tap	Instrument	Blue cylinder with a graduated watering can beside it, gauge in grams and in millilitres	Yes: drag
9	Air source tank and tap	Instrument	Pale grey cylinder of drifting CO ₂ molecules, gauge in grams; refills from the greenhouse air	Yes: drag
10	Sunlight source tank and tap	Instrument	Gold cylinder of photon darts with a locked tap; the lock tooltip states that energy carries no mass	No
11	Mass parcel	Particle	1 g cube tinted to its source tank, streams along a visible route into the stem and disappears into the plant	No
12	Allocation dial set	UI-Probe	Four linked sliders that always renormalise to 100 percent, with the running total shown above them	Yes: drag
13	Sieve and drying oven	Instrument	2 mm mesh sieve and a small 105 °C oven for recovering and re-drying the soil at harvest	Yes: place
14	Prediction versus measurement chart	Overlay	Paired bars per source: what your model spent against what was actually measured, updated each simulated week	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Mismatch meter	Instrument	Single dial in grams showing the gap between your model and the measurement, turning red beyond 50 g	No
16	Timeline scrubber	UI-Probe	0–16 week track with weekly tick marks; scrubbing replays growth and rewinds every gauge	Yes: drag

HOW IT WORKS — THE MODEL

Two engines run side by side on one clock. The belief engine takes the four allocation percentages and moves literal 1 g mass parcels out of those tanks into the plant, decrementing each gauge; if soil is set to 100 percent, the pot must visibly surrender 1.4 kg and the pin gauge must sink accordingly. The measurement engine runs the same plant on logged greenhouse data: fresh mass follows a logistic curve to 1.60 kg over 16 weeks, dry mass is 12 percent of that, applied water is tallied, and the soil, sieved and re-dried at harvest, has lost 2.9 g through mineral uptake alone. Each week the two engines are compared source by source and the gap is reported in grams. The sunlight tap stays locked by construction. The build must never quietly rescale the belief engine to fit the plant; it has to fail visibly, because the failure is the lesson.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Mass from soil	Slider	0–100	60	percent	Share of new plant mass drawn from the soil tank; the other three renormalise
Mass from water	Slider	0–100	25	percent	Share drawn from the water tank
Mass from air	Slider	0–100	10	percent	Share drawn from the air tank
Mass from sunlight	Slider	0–100	5	percent	Share requested from the locked tank; the sim shows the request being refused
Extra soil added	Slider	0–2000	0	g	Structural: raises the soil column and moves the pin gauge up
Fertiliser	Dropdown	None / Low NPK / Full NPK	Low NPK	—	Mineral supply, which caps growth when absent but supplies almost no mass
Watering	Slider	0–500	250	mL/day	Water applied, logged in millilitres and grams
Run length	Timeline scrubber	0–16	16	weeks	How far the growth run plays out before harvest
View mode	Radio	Your model / Measured plant / Side by side	Side by side	—	Which pane the bench shows

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	The soil story	soil=100; water=0; air=0; sunlight=0; run=16 weeks	Your model spends 1.4 kg of soil. How much soil did the pot actually lose, and by what factor were you out?
S2	The sunlight story	sunlight=100; soil=0; water=0; air=0; run=16 weeks	The gold tap will not open. What can sunlight supply to the plant, and what can it not?
S3	Feed it more soil	extra soil=2000; fertiliser=Low NPK; watering=250; run=16 weeks	You have doubled the soil in the pot. Does the plant finish heavier, and by how much?
S4	Starve the minerals	fertiliser=None; extra soil=0; watering=250; run=16 weeks	The plant grows badly without fertiliser. Does that prove its mass came from the soil?

STUDENT ACTIVITIES

1. Set the allocation to soil=100 and run all 16 weeks. Record the pin gauge reading at week 0 and week 16 in your model pane.
2. Switch to Measured plant and record the same two pin gauge readings, plus the harvested soil mass from the oven.

3. Set your allocation to what you personally believe, run, and record the mismatch meter value for each of the four sources.
4. Add 2000 g of extra soil and re-run. Record the final plant fresh mass and compare it with the 0 g run.
5. Adjust your allocation until the mismatch meter falls below 50 g and write down the four percentages that got you there.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Plant fresh mass	Live numeric	g	Whole-plant mass on the balance, updated weekly and at harvest
Plant dry mass	Live numeric	g	Mass after oven drying at harvest, with water removed
Soil mass change	Live numeric	g	Dry soil at harvest minus dry soil at planting, to 0.1 g
Soil level	Live numeric	mm	Pin gauge reading, shown for the model pane and the measured pane side by side
Predicted versus measured	Bar chart	g	Paired bars per source across the whole run
Mismatch per source	Data table	g	Model spend minus measured change for soil, water, air and sunlight; exportable

WHAT THE STUDENT SHOULD REALISE

Almost every student starts with the idea that a plant eats soil, and this sim makes them commit to it numerically before testing it. Running the soil model drains 1.4 kg from a pot that really lost under 3 g, and the pin gauge does not move. Minerals matter without supplying mass. The student should be able to say: *"The pot barely changed while the plant gained more than a kilogram, so the plant is not made of soil."*

C2.2 · Van Helmont's willow tree experiment

Experiment name: Van Helmont's Willow: Five Years on a Balance

Render mode: 3D Scene

Simulation engine: State machine + Data-driven model

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS1-6

THEME & SCENE

A seventeenth-century Brussels garden room seen from a low angle: lime-washed walls, an oak bench, tall leaded windows throwing shafts of daylight across a stone floor. At the centre, on a low plinth, stands a wide glazed earthenware pot, a perforated tin lid collaring the stem where it leaves the soil. The willow begins as a bare sapling barely taller than the pot rim. Behind it waits a brass beam balance with two pans and a rack of graded weights; beside it a rainwater butt with a copper dipper and a measuring jug, and a leather-bound ledger open on a lectern. A seasons wheel turns on the wall, driving the five-year clock, and leaf litter gathers on the flagstones each autumn. Palette: ochre, brass, wet green. Control panel docked right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Earthenware pot and plinth	Environment	60 cm rim, cream salt glaze, two lug handles, on a 30 cm stone plinth; holds the soil charge to a scored fill line	No
2	Oven-dried soil charge	Structure	90.72 kg of pale dried loam, visible crumb texture, surface scored with a fill line so any change in level is readable	Yes: resize
3	Drying oven	Instrument	Brick kiln with a sliding iron door and a pyrometer; dries soil at 105 °C and plots mass against time to constant mass	Yes: place
4	Willow tree	Actor	Growth from a 2.27 kg bare sapling to a 76.74 kg tree over five years: trunk girth 3 → 28 cm, 4 → 112 branches, full canopy each summer	No
5	Perforated tin lid	Structure	Beaten tin disc with a stem collar and 60 pinholes; admits water and air, keeps windblown dust off the soil	Yes: swap
6	Rainwater butt and jug	Instrument	Oak butt with a copper dipper and a 2 L graduated glass jug; every pour is logged to the ledger	Yes: drag
7	Beam balance	Instrument	Brass beam, two 50 cm pans, agate knife edge, pointer against an ivory scale; readable to 1 oz as built	Yes: place
8	Graded weight set	Structure	Brass weights from 2 oz to 56 lb in a fitted mahogany case; stacked on the pan to bring the beam level	Yes: place
9	Ledger book	Overlay	Leather-bound volume on a lectern; the student writes each weighing into a dated row, and nothing is auto-filled	Yes: drag
10	Seasons wheel	UI-Probe	Wall-mounted five-year dial with quarter markers; drives leaf-out, growth, autumn shed and dormancy	Yes: drag
11	Leaf litter pile	Structure	1.1–1.6 kg of shed leaves each autumn, gathering on the flagstones; either swept to the balance or blown away	Yes: place
12	Sieve and root-washing tray	Instrument	2 mm brass sieve over a zinc tray; separates roots from soil at harvest so each can be weighed alone	Yes: place
13	Soil scoop and moisture probe	Instrument	Iron scoop and a needle probe reading soil water as percent by mass, so wet soil is never weighed as dry	Yes: drag
14	Atmosphere field	Field	Invisible CO ₂ and water-vapour field through the room, tagged NOT WEIGHED in the corner of the frame	No
15	Mass ledger board	Overlay	Running columns: soil in, soil out, water in, tree mass, litter mass, and an unaccounted line the student must fill	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Uncertainty band overlay	Overlay	Grey band on every logged value equal to the balance resolution, so a 57 g change can be seen against a 28 g step	No

HOW IT WORKS — THE MODEL

Straight bookkeeping on a five-year clock compressed into about five minutes, scrubbable in both directions. The historical run is reproduced exactly: 90.72 kg of oven-dried soil, a 2.27 kg willow shoot, rainwater only, and after five years a tree of 76.74 kg, a gain of 74.47 kg, with the re-dried soil down by 57 g. Water added accumulates at the watering rate, so 8 L a week for 260 weeks is roughly 2080 kg poured in, almost all of which transpires back into the room. Leaf litter sheds each autumn and enters the ledger only if the student chooses to collect it. Every weighing is quantised to the chosen balance resolution and drawn with an uncertainty band. The engine must never total the unaccounted line itself: that arithmetic, and the discomfort it produces, belongs to the student.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Soil charge	Slider	50–120	90.7	kg	Mass of oven-dried soil placed in the pot at the start
Soil drying time	Slider	0–48	24	h at 105 °C	How completely soil is dried before and after the run; short drying leaves water in the reading
Sapling starting mass	Slider	1.0–5.0	2.27	kg	Mass of the willow shoot planted on day one
Water source	Dropdown	Rainwater / Distilled / Tap / None	Rainwater	—	What is poured, and whether it carries dissolved minerals
Watering rate	Slider	0–20	8	L/week	Volume added and logged each week
Pot lid	Toggle	Perforated tin / None	Perforated tin	—	Structural: keeps windblown dust and debris out of the soil
Leaf litter	Radio	Collect and weigh / Let it blow away	Let it blow away	—	Structural: whether shed mass ever enters the ledger
Balance resolution	Dropdown	1 oz (28 g) / 1 g / 0.01 g	1 oz (28 g)	—	Smallest change any weighing can detect, and the width of every uncertainty band
Elapsed time	Timeline scrubber	0–5	5	years	Position in the run; a weighing can be taken at any point on the track

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	The 1648 run	soil=90.7; sapling=2.27; water=Rainwater; rate=8; lid=Perforated tin; litter=Let it blow away; balance=1 oz	The tree gained 74.47 kg and the soil lost 57 g. Which explanation does that rule out?
S2	Take the water away	water=None; rate=0; elapsed=1 year	The willow dies in its first summer. What does that prove, and what does it fail to prove?
S3	A modern re-run	balance=0.01 g; litter=Collect and weigh; lid=None; water=Distilled; rate=8	With the litter weighed and a far better balance, how much mass is still unaccounted for?
S4	Mineral-rich water	water=Tap; rate=8; balance=0.01 g; litter=Collect and weigh	Does tap water grow a heavier tree than distilled, and is the difference anywhere near 74 kg?

STUDENT ACTIVITIES

1. Weigh the dried soil and the sapling before planting and write both into the ledger with their uncertainty bands.
2. Scrub the seasons wheel to years 1, 2, 3, 4 and 5, weighing the tree at each stop, and record the five masses.
3. At year 5, wash the roots, re-dry the soil for 24 h, weigh it, and record the change from the starting figure.
4. Add up the water poured over the whole run from the ledger and compare that total with the tree's mass gain.

5. Complete the unaccounted line yourself, then switch the balance to 0.01 g and re-run to see whether that line changes.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Mass ledger	Data table	kg and g	Soil before and after, water added, tree mass, litter mass, one dated row per weighing; exportable
Tree mass against time	Line graph	kg vs years	Five-year growth trace with a marker at every weighing the student took
Soil mass change	Live numeric	g	Dry soil at harvest minus dry soil at planting, shown with its uncertainty band
Cumulative water added	Live numeric	L and kg	Running total of everything poured into the pot
Unaccounted mass	Live numeric	kg	Tree gain minus everything the student has actually weighed; stays blank until they complete the ledger
Resolution warning	Pass-fail badge	—	Flags amber whenever a recorded change is smaller than the balance can honestly resolve

WHAT THE STUDENT SHOULD REALISE

Van Helmont's numbers kill the soil explanation: 74 kg of new tree against 57 g of missing soil is not a shortfall but a different order of magnitude. But the sim withholds his conclusion: 2080 kg of water went in and was never tracked out, and the air was never weighed. The student should be able to say: *"The soil answer is dead, but this experiment cannot tell me whether it was the water or the air."*

C2.3 · What the soil-mass evidence actually shows

Experiment name: Weigh the Ash: What the Tree Is Made Of

Render mode: 3D Scene

Simulation engine: State machine + Molecular

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS1-6

THEME & SCENE

A modern analytical bench set up beside the harvested willow, two centuries in one frame: the earthenware pot and beam balance sit out of focus behind. In the foreground, discs cut from trunk, branch, root and leaf lie in labelled porcelain crucibles on a stainless tray. A glass-doored drying oven holds one sample on a hanging pan whose mass trace is drawn live. Behind toughened safety glass a muffle furnace glows dull orange, tagged VIRTUAL FURNACE in the corner so nobody mistakes this for a bench procedure. A combustion tube runs from the furnace through two absorption tubes, white soda-lime granules in one, magnesium perchlorate in the other, each standing on its own four-decimal balance. A small vial labelled 57 g sits at the front.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Willow sample set	Actor	Four discs on a tray: trunk wood with growth rings, branch, washed root, and a fresh leaf; each 40 mm across and separately selectable	Yes: swap
2	Porcelain crucible and tongs	Instrument	30 mL glazed crucible with a lid and a pair of nickel tongs; carries the sample between balance, oven and furnace	Yes: drag
3	Analytical balance	Instrument	200 g capacity, 0.0001 g resolution, inside a glass draught shield with sliding doors and a levelling bubble	No
4	Drying oven	Instrument	40 L oven at 40–120 °C with a glass door and a hanging pan on a load cell; plots sample mass against time to constant mass	Yes: place
5	Muffle furnace	Instrument	550 °C chamber behind toughened glass, dull orange glow, digital setpoint, clearly tagged as simulated	Yes: place
6	Combustion tube and oxygen stream	Instrument	25 mm silica tube through the furnace with a metered oxygen flow, carrying combustion gases forward to the absorbers	Yes: connect
7	Soda-lime absorption tube	Instrument	U-tube of white granules that trap CO ₂ and gain mass; sits on its own balance so the gain is read directly	Yes: connect
8	Magnesium perchlorate tube	Instrument	Second U-tube trapping water vapour and gaining mass; also on its own balance	Yes: connect
9	Ash residue	Structure	Pale grey-white powder left in the crucible, 1–5 percent of the dry mass, visibly real and not a trace	No
10	Flame-test rack	Instrument	Nichrome loop and a burner with reference colour cards for potassium lilac, calcium orange-red and sodium yellow	Yes: drag
11	Soil-loss vial	Structure	Stoppered 50 mL vial holding the 57 g the pot lost in C2.2, labelled and placed for direct comparison with the ash	Yes: drag
12	Element pie board	Overlay	Circular chart splitting dry mass into C, H, O, N and mineral ash, redrawn after each analysis stage	No
13	Mass-flow Sankey overlay	Overlay	Flow diagram from fresh mass, splitting to water and dry mass, then dry mass splitting to CO ₂ , H ₂ O and ash, widths to scale	No
14	Element ledger	Overlay	Grams of each element found, alongside the same figures as percentages of dry mass	No
15	Whole-tree extrapolator	Instrument	Scales any sample result up to the 76.74 kg tree, showing the arithmetic and the assumption it rests on	Yes: drag

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Conclusion scope cards	UI-Probe	Four draggable cards stating candidate conclusions; each turns green, amber or red against the evidence actually gathered	Yes: place

HOW IT WORKS — THE MODEL

Three gravimetric stages run on one sample. Drying at 105 °C to constant mass splits fresh mass into water and dry mass, using tissue-specific water fractions: leaf 0.78, young stem 0.62, trunk wood 0.45, root 0.68 of fresh mass. Ashing at 550 °C splits dry mass into volatile matter and ash, ash being 1.2 percent of trunk dry mass and 4.8 percent of leaf. With the absorption tubes fitted, the volatile matter is caught as CO₂ on soda lime and as H₂O on magnesium perchlorate, and each tube's mass gain converts back to grams of carbon, hydrogen and oxygen. Every mass is quantised to the chosen balance resolution. Above 700 °C some potassium volatilises and the ash reads low, which is a real and instructive error. The build must keep the ash visibly non-zero, because "it all burns away to nothing" is the mirror misconception to "it is all soil".

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Sample tissue	Dropdown	Leaf / Young stem / Trunk wood / Root / Whole-tree composite	Trunk wood	—	Structural: which disc goes into the crucible, and its water and ash fractions
Sample fresh mass	Slider	1–50	10.000	g	How much tissue is analysed, and therefore how large every later reading is
Oven temperature	Slider	40–120	105	°C	Drying setpoint; below 100 °C the sample never reaches constant mass
Drying time	Slider	0–48	24	h	How close the sample gets to constant mass before it is weighed
Ashing temperature	Slider	300–800	550	°C	Furnace setpoint; above 700 °C minerals volatilise and the ash reads low
Absorption tubes	Dropdown	None / CO ₂ only / CO ₂ and H ₂ O	CO ₂ and H ₂ O	—	Structural: which combustion products are captured and weighed
Balance resolution	Dropdown	0.1 g / 0.001 g / 0.0001 g	0.0001 g	—	Smallest mass change the bench can detect
Mineral assay	Dropdown	Off / Flame test / Full elemental	Flame test	—	Which elements the ash is identified as containing
Compare with soil vial	Toggle	On / Off	On	—	Overlays the 57 g soil loss against the whole-tree ash figure

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Take out the water	tissue=Leaf; fresh=10.000; oven=105; drying=24; tubes=None	What fraction of a fresh leaf is water, and does that water belong to the tree's dry mass at all?
S2	Burn what is left	tissue=Trunk wood; fresh=10.000; ashing=550; tubes=None	Only a small grey residue remains. What percentage of the dry wood was ash, and what is that ash?
S3	Catch what escaped	tissue=Trunk wood; fresh=10.000; tubes=CO ₂ and H ₂ O; ashing=550; balance=0.0001 g	Both tubes gain mass. Which elements left the wood, and in what form did they leave?
S4	Match it to the soil	tissue=Whole-tree composite; compare=On; balance=0.0001 g	The whole tree's ash and the pot's 57 g loss are the same order of size. What exactly does that match explain?

STUDENT ACTIVITIES

1. Weigh a 10.000 g fresh leaf disc, dry it for 24 h at 105 °C, and record the dry mass and the water percentage.
2. Ash the dried sample at 550 °C and weigh the crucible again. Record the ash mass and its percentage of dry mass.

3. Fit both absorption tubes, repeat the combustion, and record the mass gained by each tube separately.
4. Convert those two gains into grams of carbon, hydrogen and oxygen using the element ledger, and record all three.
5. Extrapolate to the whole 76.74 kg tree, place the soil vial alongside the ash figure, and sort the four scope cards into supported and unsupported.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Fresh, dry and ash mass	Data table	g	One row per analysis with all three masses to the chosen resolution; exportable
Water and ash percentages	Live numeric	percent	Water as a share of fresh mass and ash as a share of dry mass, updated at each stage
Absorption tube gains	Live numeric	g	Mass gained by the soda-lime tube and the magnesium perchlorate tube, read separately
Element split	Bar chart	percent of dry mass	C, H, O, N and mineral ash side by side, rebuilt after every run
Mass-flow diagram	Heat map	g	Sankey flow from fresh mass through water and dry mass to CO ₂ , H ₂ O and ash, widths drawn to scale
Whole-tree extrapolation	Live numeric	kg	Sample result scaled to 76.74 kg, with the ash total set against the 57 g soil loss
Scope card verdict	Pass-fail badge	—	Green, amber or red for each candidate conclusion, with the reason stated

WHAT THE STUDENT SHOULD REALISE

It is tempting to jump from "the soil barely changed" to "so it came from the air". This bench measures instead: almost half the fresh mass is water, the ash is a percent or two and matches the soil's 57 g loss closely, and the rest leaves as carbon dioxide and water vapour when burned. The student should be able to say: *"The tree is mostly carbon, hydrogen and oxygen, and the soil supplied the minerals, not the mass."*

C2.4 · Building the case for air and water

Experiment name: Labelled Atoms: Tracing the New Mass

Render mode: Hybrid 2D+3D

Simulation engine: Molecular + Data-driven model

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS1-6

THEME & SCENE

A sealed growth chamber the size of a fridge stands in a clean plant-science lab: acrylic on all four faces, a gasketed door, a pressure gauge and a rubber septum on the flank. Inside, a young sunflower stands under a white LED panel, its leaves stirring faintly in the circulation fan's draught. On the left, two gas cylinders sit in a rack, one banded charcoal for ordinary carbon dioxide, one cyan for the labelled kind, feeding brass valves and a glass flow meter. Beneath them, two water reservoirs, one clear and one banded magenta. On the right stands a bench mass spectrometer with a live spectrum on its screen. Along the foot runs the atom map, where labelled atoms glow cyan and magenta against ordinary grey, red and white.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Sealed growth chamber	Environment	900×600×600 mm acrylic cabinet, silicone door gasket, pressure gauge on the flank, septum sampling port, internal circulation fan turning at 400 rpm	Yes: swap
2	Sunflower plant	Actor	35 cm plant with 6 leaves growing to 9 over the run; leaf area and stem thickness both increase on the logged growth curve	No
3	LED grow panel	Instrument	300×300 mm white panel above the plant, 0–800 $\mu\text{mol m}^{-2} \text{s}^{-1}$, on a programmable day-night timer	Yes: place
4	Ordinary CO ₂ cylinder	Instrument	Charcoal-banded steel cylinder with a two-stage regulator and a pressure dial; supplies ¹² C ¹⁶ O ₂	Yes: connect
5	Labelled CO ₂ cylinder	Instrument	Cyan-banded cylinder, identical fittings; supplies either ¹³ CO ₂ or C ¹⁸ O ₂ depending on the label chosen	Yes: connect
6	Ordinary water reservoir	Instrument	2 L clear vessel with a peristaltic feed line to the pot and a graduated sight tube	Yes: connect
7	Labelled water reservoir	Instrument	2 L magenta-banded vessel of H ₂ ¹⁸ O, identical feed line, so the plant cannot tell them apart	Yes: connect
8	Valves, flow meter and septum	Instrument	Brass needle valves, a glass rotameter with a floating ball, and a self-sealing septum for drawing gas samples	Yes: drag
9	CO ₂ analyser	Instrument	Infrared analyser on the chamber return line, 0–2000 ppm, 1 s logging, digital display on the cabinet face	No
10	Oxygen sample loop	Instrument	Stainless loop drawing chamber gas through a drier and into the spectrometer inlet at 20 mL/min	Yes: connect
11	Mass spectrometer	Instrument	Bench unit with a 60 cm flight tube and a screen plotting counts against m/z, markers preset at 32, 34, 36, 44, 45 and 46	No
12	Harvest station	Instrument	8 mm leaf punch, a small freeze-drier, and numbered sample vials on a rack; the punch leaves a visible hole in the leaf	Yes: drag
13	Atom map	Overlay	Molecules from the harvested tissue drawn atom by atom, glucose, starch chain and cellulose; labelled atoms glow cyan or magenta	Yes: drag
14	Label-origin counter	Overlay	Per element, the share of atoms in the sample tagged as coming from chamber gas, from irrigation water, or from the original seedling	No
15	Dark control chamber	Environment	Identical second cabinet with the LED panel off and the same gas and water supply; grows nothing and takes up no label	Yes: swap

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Evidence card printer	Instrument	Small thermal printer that issues a card carrying this run's settings and results, to be used in C2.5	Yes: place

HOW IT WORKS — THE MODEL

Isotopes are the only difference: ^{13}C behaves chemically like ^{12}C and ^{18}O like ^{16}O , so nothing about growth changes when the label is switched on. Every carbon fixed into new tissue inherits the isotope ratio of the chamber gas at the moment it was fixed, and every hydrogen inherits the ratio of the water in the xylem. The engine keeps a per-atom origin tag, so any harvested molecule can be interrogated atom by atom. Two results carry the whole argument and the build must get both right. First, carbon in new dry mass traces to the chamber gas, so labelling the air labels the plant. Second, the oxygen released as O_2 traces to water and not to carbon dioxide, reproducing Ruben and Kamen's 1941 result: label the water and the released O_2 shifts from m/z 32 to 34, but label the oxygen in the CO_2 instead and the released O_2 stays at 32.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
CO_2 label	Dropdown	None ($^{12}\text{C}^{16}\text{O}_2$) / Heavy carbon ($^{13}\text{CO}_2$) / Heavy oxygen (C^{18}O_2)	Heavy carbon ($^{13}\text{CO}_2$)	—	Which atom in the chamber gas carries a traceable tag
Water label	Dropdown	None (H_2^{16}O) / Heavy oxygen (H_2^{18}O)	None (H_2^{16}O)	—	Whether the irrigation water carries a traceable tag
Label strength	Slider	0–100	95	percent enrichment	How strongly the labelled supply is enriched; low values make the peak hard to separate
Chamber CO_2	Slider	100–1500	420	ppm	Gas concentration held inside the cabinet by the regulator
Light period	Slider	0–24	16	h/day	Daily photoperiod under the LED panel
Run length	Timeline scrubber	1–21	14	days	How long the plant grows in the labelled atmosphere before harvest
Chamber seal	Toggle	Sealed / Vented	Sealed	—	Structural: vented admits outside air and dilutes the label over hours
Dark control	Toggle	On / Off	On	—	Structural: runs the identical second cabinet with the lights off
Harvest tissue	Dropdown	New leaf / Old leaf / Stem / Root / Starch grain	New leaf	—	Which tissue is punched, freeze-dried and analysed
Analyse	Dropdown	Chamber O_2 / Chamber CO_2 / Plant tissue	Plant tissue	—	What the sample loop feeds into the spectrometer

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Label the air	CO_2 label=Heavy carbon; water label=None; strength=95; run=14 days; seal=Sealed; tissue=New leaf; analyse=Plant tissue	What share of the carbon in brand-new leaf tissue carries the tag, and where must that carbon have come from?
S2	Where the oxygen comes from	water label=Heavy oxygen; CO_2 label=None; run=7 days; analyse=Chamber O_2	The oxygen leaving the plant is heavy. Which of the two inputs supplied it?
S3	The other way round	CO_2 label=Heavy oxygen; water label=None; run=7 days; analyse=Chamber O_2	Now the carbon dioxide carries the heavy oxygen. Why does the released O_2 stay light?
S4	Broken seal and dark control	seal=Vented; dark control=On; CO_2 label=Heavy carbon; run=14 days; tissue=New leaf	Compare lit against dark, and vented against sealed. What has happened to the strength of your evidence?

STUDENT ACTIVITIES

1. Set the CO₂ label to heavy carbon, seal the chamber, run 14 days, and record the labelled fraction of carbon in a new leaf.
2. Harvest an old leaf and a root from the same plant and record their labelled fractions. Explain the difference in one sentence.
3. Switch to heavy-oxygen water, analyse the chamber oxygen, and record the m/z peaks you see and their relative heights.
4. Repeat with heavy oxygen in the CO₂ instead of the water. Record the peaks again and state which input the released oxygen came from.
5. Print the evidence card for your two strongest runs and note the sample size and the seal state recorded on each.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Mass spectrum	Line graph	counts vs m/z	Live trace with markers at 32, 34 and 36 for oxygen and 44, 45 and 46 for carbon dioxide
Labelled atom fraction	Live numeric	percent	Share of C, H and O atoms in the sampled tissue carrying a label, reported per element
Atom-origin split	Bar chart	percent of atoms	Where each element in new dry mass came from: chamber gas, irrigation water, or the original seedling
Chamber CO ₂ trace	Line graph	ppm vs h	Falls through each light period as carbon is fixed, rises in darkness as the plant respire
Dry mass gain	Live numeric	g	New dry mass over the run, lit chamber and dark control shown side by side
Evidence card	Data table	mixed	Printed record of settings, results, sample size and seal state; carried forward into C2.5

WHAT THE STUDENT SHOULD REALISE

Students who give up on soil land on "the plant breathes carbon dioxide in and the same oxygen out". The labels break it: the carbon in new tissue was in the air, the hydrogen came from water, and the released oxygen carries the water's label, not the carbon dioxide's. The student should be able to say: *"The tree's mass is carbon from the air and hydrogen from water, and the oxygen we breathe was split out of water."*

C2.5 · Evaluating, not asserting, the conclusion

Experiment name: The Warrant Test: Does Your Evidence Hold?

Render mode: Data Dashboard

Simulation engine: Data-driven model + Agent-based

Interaction level: Argue-from-data

Session length: 18–25 min

NGSS anchor: MS-LS1-6

THEME & SCENE

A dark slate review table seen from directly above, lit like a boardroom at night. Dead centre is an empty claim slot, a raised brass frame waiting for a sentence. Down the left runs a card rack holding every run made across this unit: the willow ledger, the ash bench, the isotope chamber, the bubble counter, the dark control, the soil-model board, the starch test. Each card carries a miniature chart, its sample size, whether a control was run, and its instrument resolution. Ribbons drag from any card to the claim slot and glow green, amber or red as they land. Down the right stand three dials under glass: Relevance, Sufficiency and Control. Along the bottom edge a challenger panel deals objection cards face up, one at a time.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Claim slot and builder	UI-Probe	Brass sentence frame with three word wells: strength, verb and source; assembled words snap in and lock the claim's scope	Yes: drag
2	Evidence card	Structure	90×140 px card carrying a thumbnail chart, run ID, n, control state, instrument resolution and the variable measured	Yes: drag
3	Card rack and filters	Instrument	Left-hand rail holding 7 base cards plus any requisitioned ones, with filter chips for topic, controlled and replicated	Yes: place
4	Reasoning ribbon	Overlay	Drawn link from a card to the claim slot, 4 px, green when the card measures a variable inside the claim's scope, amber when adjacent, red when irrelevant	Yes: connect
5	Warrant dials	Instrument	Three brass dials under glass, 0–100 each, needles moving live as ribbons are attached and objections are answered	No
6	Reviewer B	Actor	Challenger agent shown as a small silhouette at the table's foot; deals one objection card whenever a rule trips, and never repeats a closed one	No
7	Objection card set	Structure	Twelve printed objections: no control, sample of one, confound uncontrolled, change below resolution, scope overreach, correlation not cause, and six more	Yes: drag
8	Confound flag pin	Instrument	Red pin the student sticks into any card to declare a known confound; costs Control points until it is answered	Yes: place
9	Plot canvas	Overlay	420×300 px panel that re-plots the raw data behind any selected card as scatter, bar or line	Yes: swap
10	Annotation pin	Instrument	Numbered pin dropped anywhere on the plot canvas with a short typed note attached, saved to the certificate	Yes: place
11	Uncertainty band renderer	Overlay	Draws each card's stated instrument resolution as a grey band on its chart, so unresolvable changes are visible as noise	No
12	Alternative-explanation tray	Structure	Shelf of rival explanation tiles; each stays lit until a card is linked that excludes it	Yes: place
13	Scope-gap ruler	Instrument	Horizontal bar comparing the claim's scope with the combined scope of the linked cards; the unfilled part is the gap	No
14	Missing-evidence detector	Overlay	Lists the specific measurements the claim needs and the rack does not contain, naming the lab that could supply each	No
15	Requisition slip	Instrument	Tear-off slip that sends a request to the rate lab, willow rig, ash bench or isotope chamber and returns a new card	Yes: drag

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Conclusion certificate	Overlay	Printed claim, evidence, reasoning and verdict, exportable as a CER document with every dial reading attached	Yes: drag

HOW IT WORKS — THE MODEL

A rubric engine, not an answer key. Each card carries tags: the variable it measured, whether a control was run, sample size, instrument resolution and provenance. The claim carries a scope built from its strength and source. Three sub-scores run 0–100. Relevance is the fraction of linked cards whose measured variable falls inside the claim's scope. Sufficiency is coverage: a claim about all of a plant's mass needs a carbon result, a hydrogen result and a total that adds up; a claim about most of the dry mass needs the drying and ashing data. Control subtracts for every rival left lit on the tray with no card excluding it, and for every card whose reported change is below its own instrument resolution. Reviewer B deals an objection each time a rule trips, and it stays open until answered. Crucially the engine can mark a correct conclusion unwarranted, because a right answer reached from one uncontrolled run scores nothing on Control.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Claim strength	Dropdown	All of a plant's mass / Most of a plant's dry mass / Some of a plant's mass	Most of a plant's dry mass	—	Sets how much scope the evidence has to cover
Claim source	Dropdown	Soil / Water / Air / Sunlight / Air and water	Air and water	—	The second half of the sentence in the claim slot
Evidence linked	Multi-select	Willow ledger / Ash bench / Isotope chamber / Bubble rate lab / Dark control / Soil-model board / Starch test	Willow ledger, Isotope chamber	—	Which cards are ribboned to the claim
Rivals to exclude	Multi-select	Minerals from soil / Dust through the lid / Uncollected leaf litter / Balance error / Water alone	Minerals from soil, Water alone	—	Which alternative explanations sit lit on the tray demanding an answer
Reviewer strictness	Radio	Novice / Standard / Strict	Standard	—	How many rules Reviewer B enforces and how far the dials must move to pass
Uncertainty bands	Toggle	On / Off	On	—	Draws each card's instrument resolution onto its chart
Replicates available	Dropdown	n=1 / n=3 / n=12	n=3	—	Structural: how many archived repeats of each run exist as separate cards
Request a measurement	Dropdown	None / Rate lab / Willow rig / Ash bench / Isotope chamber	None	—	Structural: issues a requisition and adds the returned card to the rack
Plot variable	Dropdown	Mass gain / Soil change / Ash percent / Labelled fraction / Bubble rate	Labelled fraction	—	What the plot canvas draws from the currently selected card
Scoring mode	Radio	Practice / Graded	Practice	—	Whether Reviewer B hints at an objection before it fires

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Van Helmont's own conclusion	strength=All of a plant's mass; source=Water; evidence=Willow ledger; rivals=all five; strictness=Standard	This is exactly what he concluded in 1648. What does it score, and which dial fails hardest?

#	Scenario	Preset	The question the student answers
S2	The soil verdict	strength=Most of a plant's dry mass; source=Soil; evidence=Willow ledger, Soil-model board; rivals=all five	Can any card on the rack support this claim at all? What does the missing-evidence detector list?
S3	The full case	strength=Most of a plant's dry mass; source=Air and water; evidence=Willow ledger, Ash bench, Isotope chamber, Dark control; rivals=all five	All three dials should clear. Remove one card at a time: which removal costs the most?
S4	Right for the wrong reasons	strength=All of a plant's mass; source=Air and water; evidence=Starch test; replicates=n=1; strictness=Strict	The claim is essentially correct. Why is it still marked unwarranted?

STUDENT ACTIVITIES

1. Assemble the claim "Most of a plant's dry mass comes from air and water" in the slot and record all three dial readings before linking any evidence.
2. Drag ribbons from the willow ledger and the isotope chamber. Record how each dial moves and which ribbon turned amber, and say why.
3. Answer every objection Reviewer B deals, using either a card or a requisition. Record how many you could not close and what each one demanded.
4. Remove the ash bench card and re-score. Record the drop in Sufficiency and write one sentence on what that card was doing for you.
5. Change the claim to "All of a plant's mass comes from air", re-score with the same evidence, and record which dial collapses.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Warrant score	Live numeric	0–100	Combined score, recomputed the moment any ribbon, card or claim word changes
Three dials	Bar chart	0–100 each	Relevance, Sufficiency and Control shown separately so a failure can be diagnosed rather than guessed
Objections outstanding	Counter	count	Unanswered cards from Reviewer B, each printed with the rule that triggered it
Evidence-claim matrix	Data table	mixed	Every linked card against every element of the claim's scope, marked supports, silent or contradicts
Missing-evidence list	Data table	—	The measurements the claim requires and the rack lacks, each naming the lab that could supply it
Conclusion certificate	Pass-fail badge	—	Warranted, partially warranted or unwarranted, with reasons and dial readings; exportable as a CER document

WHAT THE STUDENT SHOULD REALISE

Students treat a conclusion as sound the moment it is correct, and treat agreement as evidence. Here a right answer from one uncontrolled run scores below a careful, narrower claim, and Van Helmont's conclusion fails the Control dial on his own data. Warrant is a property of the reasoning, not of the answer. The student should be able to say: *"Being right is not the same as being justified — my evidence has to rule out the other explanations too."*

C3 · Cellular respiration · MS-LS1-7

C3.1 · Naming the inputs

Experiment name: Supply Line: What a Mitochondrion Orders

Render mode: 3D Scene

Simulation engine: Molecular + Particle system

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS1-7

THEME & SCENE

Inside the calf muscle of a hiker three hours up a Sierra trail. The camera sits in pale straw cytosol, three-quarters on, with a capillary running diagonally behind a translucent plasma membrane; red blood cells shoulder through it in single file, each flaring from maroon to bright scarlet as it lets go of oxygen. Filling the lower right of the frame is one mitochondrion: a bean the colour of warm ochre, its outer membrane pocked with pale barrel-shaped pores, its interior opened by a cutaway that shows the inner membrane folded into deep parallel shelves. Glucose rings drift in from the left, tumbling. A scale bar reads 1 μm . The control panel docks right; a tally board floats top-left, counting every molecule that crosses in.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Capillary lumen and endothelium	Environment	Tube 7 μm bore running diagonally through the back third of the frame; wall one endothelial cell thick, pale pink, semi-transparent so cargo is visible through it	No
2	Red blood cell	Actor	Biconcave disc 7.5 μm across, 2 μm thick, deforming to squeeze the bore; carries ~270 million haemoglobin tetramers drawn as a dense internal speckle; surface hue drives from scarlet (loaded) to maroon (unloaded) with saturation	Yes: drag
3	Haemoglobin tetramer	Structure	Four-lobed protein, 5.5 nm, each lobe holding one orange-brown haem iron; releases up to 4 O_2 ; lobes rotate 15° on release so unloading is visible in the loupe	No
4	Oxygen molecule (O_2)	Particle	Two red CPK spheres, covalent radius 0.66 Å scaled $\times 40$, joined by a double-bond twin cylinder at 121 pm; countable; Brownian walk at 500 m/s scaled	No
5	Glucose molecule ($\text{C}_6\text{H}_{12}\text{O}_6$)	Particle	Pyranose chair ring: 6 charcoal carbons (0.76 Å), 12 white hydrogens (0.31 Å), 6 red oxygens (0.66 Å); C–C 154 pm, C–H 109 pm, C–O 143 pm; ring puckers slowly; each molecule carries its own atom label on hover	No
6	Muscle-fibre plasma membrane	Structure	Phospholipid bilayer 5 nm thick, heads violet, tails pale yellow, lipids swapping laterally ~1 per 0.4 s; curved across the frame's mid-plane	No
7	GLUT4 glucose transporter	Actor	Twelve-helix clamshell barrel, teal, 4 nm across; binds one glucose, rocks 180°, releases it inside; cycle time 12 ms so it saturates	Yes: place
8	Cytosol and glycolysis bench	Actor	Straw-coloured volume with ten pale enzyme blobs strung in a loose line; a glucose entering the line leaves as two three-carbon pieces plus 2 ATP; sits entirely outside the mitochondrion, and is labelled so	No
9	Pyruvate molecule	Particle	Three charcoal carbons, 3 red oxygens, 4 hydrogens; short kinked chain 0.9 nm; drifts to the mitochondrial surface	No
10	Mitochondrial outer membrane and porins	Structure	Smooth ellipsoid shell 2.4 $\times$ 0.9 μm , 7 nm thick, ochre; studded with 60 pale barrel porins (VDAC), bore 2.5 nm, permanently open to small molecules	No
11	Intermembrane space	Structure	15 nm gap between the two membranes, tinted faintly blue where protons accumulate; the tint deepens as the electron transport chain runs	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
12	Inner membrane folded into cristae	Structure	Continuous sheet folded into 14 lamellar shelves reaching two-thirds across the interior; crista junctions 25 nm; total area 3.5× the outer membrane at default fold setting	Yes: resize
13	Matrix	Environment	Dense amber gel filling between the shelves; contains loose Krebs enzymes as small ochre grains and one circular mtDNA nucleoid	No
14	ATP synthase rotor	Actor	Mushroom 22 nm tall: 10 nm F1 head on a stalk, c-ring of 8 subunits in the membrane; sits in double rows along the crista rims; spins visibly, 3 ATP per revolution	No
15	ATP and ADP tokens	Particle	ATP drawn as a three-phosphate tail with the terminal phosphate glowing amber; ADP identical minus the glow and one phosphate; leave through the porins and stream toward the myosin heads	No
16	Delivery tally and atom inventory	Overlay	Top-left board: glucose in, O ₂ in, CO ₂ out, H ₂ O out, ATP made, plus a live C / H / O element count and a fixed 1 µm scale bar	No

HOW IT WORKS — THE MODEL

Two independent supply chains feed one consumer. Glucose entry rate is set by GLUT4 count × cycle rate and saturates; oxygen entry is pure diffusion, proportional to the blood-to-cytosol partial-pressure difference. Inside, the engine runs strict stoichiometry: one glucose consumed requires six O₂ and yields six CO₂, six H₂O and about 32 ATP. Rate is set by whichever input is currently scarcer, so ATP output tracks the minimum, never the sum. Demand raises the myosin ATP draw; if draw exceeds supply, the ATP pool falls and the myosin heads visibly slow. Below about 60% saturation the cytosolic bench keeps running without the mitochondrion and lactate accumulates, which is why glycolysis is drawn outside the organelle from the first frame. The failure to avoid is any animation showing oxygen being consumed as fuel: oxygen must always be seen ending up in the products, never disappearing.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Blood glucose	Slider	2.0–12.0	5.0	mmol/L	Number of glucose rings in the capillary and the uptake driving force
Blood oxygen saturation	Slider	40–100	98	%	O ₂ carried per red blood cell and the diffusion gradient into the fibre
Altitude	Dropdown	Sea level / 1,500 m / 3,000 m / 4,300 m (Mt Whitney)	Sea level	m	Resets saturation to 98 / 93 / 86 / 70% and tints the sky in the corner vignette
Muscle demand	Slider	1–12	2	× resting	Myosin cycling rate and therefore the ATP draw
GLUT4 transporters	Stepper	0–40	12	count	Maximum glucose uptake rate; transporters appear in the membrane as you add them
Mitochondria in the fibre section	Stepper	1–6	2	count	How many organelles exist to process the supply
Capillary clamp	Toggle	Open / Clamped	Open	—	Halts both deliveries at the same instant
Cutaway depth	Slider	0–100	60	%	Clip plane through the mitochondrion, from intact shell to fully opened matrix
Atom inventory	Toggle	On / Off	On	—	Shows the live C, H and O counts on the tally board
Playback speed	Dial	0.1×–10×	1×	—	Time scaling

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Trail pace	Blood glucose=5.0; Blood oxygen saturation=98; Muscle demand=2; GLUT4 transporters=12	For every one glucose that goes in, how many oxygen molecules go in with it? Read it off the tally, do not guess.
S2	Full tank, no fuel	Blood glucose=2.0; Blood oxygen saturation=98; Muscle demand=6	Oxygen is arriving faster than ever and the ATP rate still collapses. Is oxygen the fuel?
S3	Whitney summit	Altitude=4,300 m (Mt Whitney); Blood glucose=5.0; Muscle demand=6	Saturation falls to 70%. Which input is short now, and what appears in the cytosol that was not there at sea level?
S4	Tourniquet	Capillary clamp=Clamped; Muscle demand=4; Mitochondria in the fibre section=2	Both deliveries stop at once. Which of the two inputs runs out first inside the fibre, and why that one?

STUDENT ACTIVITIES

1. Run S1 for 60 simulated seconds and record the tally totals for glucose in, oxygen in, carbon dioxide out and ATP made. Write the glucose-to-oxygen ratio as a whole number pair.
2. Drag eight more GLUT4 transporters onto the plasma membrane, set demand to 6, re-run, and record the ATP rate before and after.
3. Set glucose to 2.0 with saturation at 98, then glucose to 5.0 with saturation at 55. Record the ATP rate for both runs and write the one sentence that explains both results.
4. Step the altitude dropdown through all four settings at demand 6, recording ATP rate each time, and plot ATP rate against altitude.
5. Open the cutaway to 100% and follow one glucose from the capillary to the point its carbons reach the matrix. List every membrane it crosses, in order, and name the compartment it is in at each stage.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Molecules delivered	Counter	count	Glucose and O ₂ arriving at the fibre, totalled per 10 simulated seconds
Input ratio	Live numeric	O ₂ per glucose	Running ratio of the two consumption counts; settles on 6.0
Atom inventory	Data table	atoms	Live C, H and O totals held in reactants, in products and in transit; the grand total never changes
ATP production rate	Live numeric	ATP/s	Rotor revolutions × 3, summed over all mitochondria, refreshed every 0.5 s
ATP pool	Line graph	ATP tokens vs s	Supply minus demand; falls visibly whenever draw beats delivery
Limiting input flag	Pass-fail badge	—	Names whichever input is currently capping the rate, or reads "neither"
Run log	Data table	mixed	Exportable CSV, one row per simulated 5 s, every counter and every control setting

WHAT THE STUDENT SHOULD REALISE

Students believe oxygen is the fuel, that breathing supplies energy directly, and that food and air are two separate stories. Here the ATP rate collapses in exactly the same way whichever input is starved, and the tally settles stubbornly on six oxygen molecules for every one glucose. The student should be able to say: *"A cell needs both, in a fixed ratio. Oxygen is not the fuel, it is what the fuel gets taken apart with."*

C3.2 · Naming the outputs

Experiment name: The Exhaust Test: Proving What Comes Out

Render mode: Hybrid 2D+3D

Simulation engine: Fluid/thermal + State machine (reagent chemistry)

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS1-7

THEME & SCENE

A school lab bench under cool white light, shot straight on so the whole gas train reads as a line of apparatus left to right. At the left, a squat aquarium pump feeds air through a soda-lime tower; then a sealed borosilicate chamber, one litre, its ground-glass lid clipped down, holding a bed of germinating peas swollen and split with pale radicles. Downstream: a cold finger beaded with condensate, a strip of blue paper turning at its base, a bulb of clear limewater, a vial of orange indicator, and two probe leads running to a data logger. Behind the glass, a vacuum flask with a thermistor in the seeds. A small molecular window floats top-right. Panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Sealed respirometer chamber	Structure	Borosilicate cylinder 90 mm × 160 mm, 1 L, ground-glass lid with two hose barbs and a silicone gasket; contents visible; fogs internally when humidity passes 90%	Yes: swap
2	Germinating pea seeds	Actor	40 wrinkled seeds, 8 mm, pale green swelling to split coats with 5 mm white radicles; respire at 1.2 mL O ₂ per gram per hour at 20 °C; radicles elongate visibly over a long run	Yes: place
3	Boiled seed control	Actor	Identical seeds, dulled and darkened, dosed with antiseptic so nothing microbial grows on them; zero metabolic rate by construction	Yes: swap
4	House crickets	Actor	Six 18 mm insects, tan with darker banding, moving in short hops; respire at 2.6 mL O ₂ per gram per hour; a mesh floor keeps them off the probe	Yes: swap
5	Yeast and glucose flask	Actor	Cloudy beige suspension, 100 mL, with a foam head that builds; produces CO ₂ with or without oxygen, which is the awkward case the student meets in S4	Yes: swap
6	Soda-lime inlet scrubber	Instrument	Clear tower 30 mm × 200 mm of white granules turning violet as they load; removes CO ₂ from incoming air to below 5 ppm	Yes: swap
7	Air pump and flow meter	Instrument	Diaphragm pump with a ball-in-tube rotameter, 0–200 mL/min, ball height tracks the set flow	No
8	Cold finger condenser	Instrument	Chilled glass finger at 4 °C hanging in the gas stream; droplets nucleate, grow and run into a graduated 2 mL collector	No
9	Water detectors	Instrument	Cobalt chloride paper strip, blue when dry and pink above 40% relative humidity, plus a side tube of white anhydrous copper(II) sulfate that turns blue when wetted; both are threshold state machines with a stated trigger	Yes: place
10	Limewater bulb	Instrument	25 mL of clear calcium hydroxide solution in a bubbler; turbidity rises with cumulative CO ₂ passed, from clear through milky to a chalky precipitate	No
11	Hydrogencarbonate indicator vial	Instrument	10 mL of orange-red solution; goes yellow as CO ₂ rises, magenta as CO ₂ falls below atmospheric, with the colour keyed to a printed scale on the vial	Yes: place
12	Carbon dioxide probe	Instrument	NDIR sensor, 0–10,000 ppm, ±30 ppm, 20 s response, mounted in the chamber headspace	No
13	Oxygen probe	Instrument	Galvanic cell, 0–25% O ₂ , ±0.1%, in the same headspace	No
14	Vacuum flask and thermistor	Instrument	250 mL insulated flask holding a second seed sample with a bead thermistor in the middle, 0.05 °C resolution, plus an identical flask of boiled seeds beside it	Yes: swap

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Molecular inset window	Overlay	300 px window on the matrix interior: carbons leaving as CO ₂ , hydrogens and oxygen combining into H ₂ O at the end of the inner membrane, ATP synthase rotors spinning; drawn to the same CPK scheme used elsewhere	No
16	Mass and atom ledger	Overlay	Running dry mass of the sample, cumulative CO ₂ carbon, cumulative water, cumulative joules of heat, and a C / H / O balance that must close	No

HOW IT WORKS — THE MODEL

The chamber is a well-mixed gas volume solved as a mass balance each tick: CO₂ in equals production minus washout, O₂ likewise. Production is mass × specific rate × $Q_{10}^{(T-20)/10}$ with $Q_{10} = 2$, and the respiratory quotient fixes CO₂ made per O₂ used at 1.0 for the starch-fed seeds. Water vapour is produced in the same ratio and condenses on the cold finger once the stream is chilled below its dew point. Metabolic heat is 470 kJ per mole of O₂ consumed, fed into the flask's thermal mass with a loss term to the room. Every reagent is a threshold state machine with a stated trigger, not a decoration. The inlet scrubber is the argument of the whole experiment: with incoming CO₂ at under 5 ppm, any CO₂ the probe reads was made inside the chamber and nowhere else.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Chamber contents	Dropdown	Germinating peas / Boiled peas / Crickets / Yeast and glucose / Empty	Germinating peas	—	Which organism is in the chamber, and therefore which rates exist at all
Sample mass	Slider	2–40	15	g	Scales every production rate proportionally
Chamber temperature	Slider	5–35	22	°C	Respiration rate through $Q_{10} = 2$, and the dew point at the cold finger
Inlet scrubber	Toggle	Soda lime in / Bypassed	Soda lime in	—	Whether incoming air carries 425 ppm CO ₂ or under 5 ppm
Detection train	Multi-select	Cold finger / Cobalt chloride / Copper sulfate / Limewater / Hydrogencarbonate indicator / CO ₂ probe / O ₂ probe	Cold finger, Cobalt chloride, Limewater, CO ₂ probe, O ₂ probe	—	Which detectors are plumbed inline, in that order
Air flow	Slider	0–200	60	mL/min	How fast gas is swept to the detectors; zero seals the chamber
Chamber volume	Dropdown	250 / 500 / 1,000 / 2,000	1,000	mL	How quickly concentrations build for a given production rate
Insulated flask	Toggle	In circuit / Removed	In circuit	—	Whether the thermistor pair is running alongside the gas train
Run length	Slider	5–180	60	min	Simulated duration of the run
Time compression	Dial	1×–600×	120×	—	Simulated minutes per real second

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Living seeds	Chamber contents=Germinating peas; Sample mass=15; Chamber temperature=22; Inlet scrubber=Soda lime in	Name all three outputs and give the evidence for each one: which detector, what change, at what time.

#	Scenario	Preset	The question the student answers
S2	Dead seed control	Chamber contents=Boiled peas; Sample mass=15; Chamber temperature=22	Same seeds, same mass, same air. Nothing changes anywhere. What does that rule out?
S3	Where did the CO ₂ come from	Chamber contents=Germinating peas; Inlet scrubber=Bypassed, then Soda lime in; Air flow=60	With the scrubber in, incoming air has almost no CO ₂ and the probe still climbs. Where was that carbon dioxide made?
S4	Cold store	Chamber contents=Germinating peas; Chamber temperature=5; Run length=180	The outputs still appear but take far longer. Has respiration stopped or slowed? Give the number that decides it.

STUDENT ACTIVITIES

1. Set S1 and run 60 simulated minutes. Record the time at which the limewater first goes milky, the cobalt chloride first goes pink, and the flask thermistor first reads 1.0 °C above the boiled control.
2. Swap the chamber contents to boiled peas and repeat exactly, recording the same three times or writing "never" for each.
3. Bypass the soda lime, run 20 minutes, then re-run with it in circuit, and record the final CO₂ in ppm for both. Write one sentence saying what the scrubber proves.
4. Set the temperature to 5 °C and then 35 °C, holding mass at 15 g, and record CO₂ produced per gram per hour at each. Work out the ratio and compare it with the $Q_{10} = 2$ the model uses.
5. Swap to crickets at the same 15 g and run 30 minutes. Compare CO₂ per gram per hour with the seeds and state which detector was the first to respond in each case.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Chamber CO ₂	Line graph	ppm	Headspace concentration against time, with the 425 ppm atmospheric line drawn for reference
Chamber O ₂	Line graph	%	Falls as CO ₂ rises; the two traces are near mirror images
Water collected	Live numeric	mL	Volume in the cold finger's graduated collector, to 0.05 mL
Temperature difference	Live numeric	°C	Living flask minus boiled flask, to 0.05 °C, updated every simulated minute
Detector state board	Data table	—	Each reagent, its current colour, its trigger threshold and the time it crossed
ATP made	Counter	ATP (×10 ¹⁸)	Read off the molecular inset; the output you cannot collect in a tube
Output ledger	Data table	mixed	Exportable CSV: cumulative CO ₂ , H ₂ O, heat and dry mass lost, one row per simulated 5 min

WHAT THE STUDENT SHOULD REALISE

Students believe respiration's only output is "energy", and that exhaled CO₂ is just the CO₂ breathed in. The scrubbed inlet kills the second idea; the detector board replaces the first. What actually leaves the chamber is carbon dioxide, water and warmth, while the energy that matters is locked into ATP inside. The student should be able to say: *"Carbon dioxide and water come out, and the energy leaves as ATP and heat, not as a gas you can collect."*

C3.3 · Where in the cell this happens

Experiment name: All the Way In: Hummingbird to Crista

Render mode: 3D Scene

Simulation engine: Molecular + State machine (pathway tracker)

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS1-7

THEME & SCENE

An Anna's hummingbird hangs at a manzanita flower in a Bay Area garden, wings a blurred disc, held dead centre in a locked frame. The scene never cuts. A dolly slider drives the camera straight into the bird's chest and keeps going: through iridescent feather, into the deep red slab of flight muscle, into one striped fibre, past the sarcolemma into straw cytosol, then through the ochre wall of a single mitochondrion into an amber matrix roofed by close-packed cristae. Light shifts from garden daylight to a soft internal glow as scale drops. A scale bar in the lower left rewrites itself continuously from centimetres to nanometres. Four labelled step cards wait in a rack on the left. Panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Anna's hummingbird	Actor	11 cm bird, 4.5 g, magenta gorget, wings beating at 42 Hz as a motion-blurred disc; hover cycle loops; body becomes a translucent shell as the camera enters	No
2	Pectoralis flight-muscle block	Structure	Deep red slab, 25% of body mass, fibres running fore-and-aft; cut face shows a mottled texture where mitochondria occupy 35% of fibre volume, the highest known in a vertebrate	No
3	Muscle fibre with sarcolemma	Structure	Cylinder 30 µm across, banded with sarcomeres at 2.2 µm repeat; sheathed by a 5 nm bilayer with T-tubule invaginations every sarcomere	No
4	Cytosol volume	Environment	Straw-tinted gel filling between fibrils; explicitly labelled "cytoplasm — outside the mitochondrion" whenever the glycolysis stage is highlighted	No
5	Glycolysis enzyme cluster	Actor	Ten pale grey-green enzyme bodies, 5–8 nm, strung loosely along a fibril in the cytosol; one glucose enters, two pyruvate and 2 net ATP leave; keeps running when oxygen is switched off	No
6	Mitochondrial outer membrane and porins	Structure	Ellipsoid shell 2.0 × 0.8 µm, 7 nm thick, ochre, 60 pale VDAC barrels of 2.5 nm bore, permanently open	No
7	Intermembrane space	Structure	15 nm gap, tinted with a blue proton haze whose density tracks the pH difference the model computes	No
8	Inner membrane	Structure	Continuous sheet 6 nm thick, darker ochre, impermeable except at named carriers; carries every complex and rotor	No
9	Cristae folds	Structure	Lamellar shelves, 14 at default, spanning two-thirds of the interior; junction necks 25 nm; total area 1×–6× the outer membrane under slider control; rims curve because rotor dimers sit along them	Yes: resize
10	Matrix	Environment	Amber gel between the shelves with loose Krebs enzyme grains, ribosomes and one circular mtDNA nucleoid drawn as a coiled thread	No
11	Electron transport complexes I–IV	Actor	Four distinct membrane bodies, blue, green, violet and rust, plus a small mobile ubiquinone dot and a pink cytochrome c bead shuttling between III and IV; each complex pumps protons on a visible beat	No
12	Proton particles	Particle	1,200 small white spheres with a faint halo; pumped out into the intermembrane space, queue at the rotor inlet, return through it	No
13	ATP synthase rotor	Actor	22 nm mushroom: F1 head 10 nm, stalk, and an 8-subunit c-ring in the membrane; arranged in double rows along crista rims; spins at up to 350 rev/s, releasing 3 ATP per revolution	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
14	Pathway tracker	Overlay	Tints the six carbons of one chosen glucose and follows them through every stage, with a floating caption naming the compartment they are currently in	No
15	Step cards and drop-zones	UI-Probe	Four cards (Glycolysis, Link reaction, Krebs cycle, Electron transport) dragged onto four glowing drop-zones (Cytosol, Matrix, Matrix, Inner membrane); wrong placements bounce back with the compartment named	Yes: place
16	Zoom dolly and scale bar	Instrument	Logarithmic camera rail over eight decades, 0.1 m to 5 nm, with five detented stops; scale bar relabels itself and prints the current magnification	Yes: drag

HOW IT WORKS — THE MODEL

The zoom is one continuous logarithmic camera path with level-of-detail swaps at fixed scales, so the student never sees a cut and never loses the thread between bird and rotor. The pathway is a four-state machine, each state bound to a named compartment: glycolysis in the cytosol, 2 net ATP, no oxygen required; the link reaction and the Krebs cycle in the matrix, releasing the carbons as CO₂; electron transport on the inner membrane, where the complexes pump protons out and the rotors let them back in, yielding roughly 26 further ATP. ATP output scales linearly with crista area because rotor count scales with area. Switching oxygen off freezes the electron chain within two seconds while the cytosolic bench carries on, which is the whole structural argument made visible: if all of respiration lived in the mitochondrion, everything would stop together.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Zoom depth	Slider	0.1 m – 5 nm, logarithmic, eight decades	0.1 m	m	Camera position on the dolly rail and which level of detail is drawn
Cutaway plane	Slider	0–100	0	%	Slices first the fibre, then the mitochondrion, revealing the compartments in order
Crista folding	Slider	1.0–6.0	3.5	× outer-membrane area	Number and depth of shelves, and therefore how many rotors fit
Rotor density	Stepper	500–4,000	2,500	rotors per μm ²	ATP synthase complexes per unit of crista membrane
Stage highlight	Radio	Glycolysis / Link reaction / Krebs cycle / Electron transport / All	All	—	Which stage glows, and which compartment caption is shown
Oxygen supply	Toggle	On / Off	On	—	Stalls the electron chain and the rotors; the cytosolic bench keeps running
Pathway playback	Timeline scrubber	0–120	0	s	Steps the tracked carbons stage by stage, forwards or backwards
Compartment labels	Toggle	On / Off	On	—	Names every compartment the camera is currently inside
Step cards	Drag-handle	4 cards onto 4 drop-zones	Unplaced	—	The placement task; scores each card as it lands
Tracer colour	Dropdown	Magenta / Cyan / Amber	Magenta	—	Tint applied to the tracked carbons

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	The full descent	Zoom depth=0.1 m to 5 nm; Stage highlight=All; Compartment labels=On	Write the five scale stops in order with the size of what you see at each. Where does the first stage of respiration actually happen?

#	Scenario	Preset	The question the student answers
S2	Oxygen off	Oxygen supply=Off; Stage highlight=All; Pathway playback=0–120	Some machinery stops within seconds and some keeps going. Which is which, and what does that tell you about where each stage lives?
S3	Smooth versus folded	Crista folding=1.0, then 6.0; Rotor density=2,500	Nothing about the mitochondrion's outside changes. Why does ATP output change by so much?
S4	Put the stages away	Step cards=drag all four; Compartment labels=Off	Place all four stage cards without the labels showing. Which one is not in the mitochondrion at all?

STUDENT ACTIVITIES

1. Drag the zoom dolly through all five detented stops and record the scale bar reading and the name of the structure filling the frame at each stop.
2. Set stage highlight to Glycolysis and record, in writing, which compartment glows. Repeat for the other three stages and build a four-row table of stage against compartment.
3. Switch oxygen off and run the pathway playback to 120 s. Record which stages still advance and which freeze, with the time each froze.
4. Set crista folding to 1.0, record ATP per second, then step it to 2.0, 3.5, 5.0 and 6.0, recording each time. Plot ATP rate against crista area and describe the shape.
5. Place all four step cards with compartment labels off, then turn labels on and correct any that bounced. Write down which card you got wrong and why.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Scale readout	Live numeric	m	Current field-of-view width, in scientific notation, updated continuously as the dolly moves
Stage-to-compartment table	Data table	—	Four rows built as the student highlights each stage, with the student's own entry checked against the model
ATP production rate	Live numeric	ATP/s	Total across all rotors, refreshed every 0.5 s
ATP rate versus crista area	Line graph	ATP/s vs × outer area	Built point by point as the folding slider is stepped; comes out linear
Proton count in the intermembrane space	Counter	protons	Live population; drops to zero within 2 s of oxygen going off
Card placement score	Pass-fail badge	—	Four green ticks when every stage sits in the right compartment

WHAT THE STUDENT SHOULD REALISE

Students believe respiration happens in the lungs, or that "the mitochondrion does respiration" from start to finish. The oxygen-off run makes both untenable: the first stage runs on in the cytosol without mitochondrion or oxygen, while everything inside the organelle stops. Respiration is a chemical process in cells, and it is spread across two compartments. The student should be able to say: *"Breathing gets oxygen to my cells. Respiration happens inside them, and it starts before the mitochondrion."*

C3.4 · Comparing photosynthesis and respiration side by side

Experiment name: Mirror Image: One Set of Atoms, Two Directions

Render mode: Hybrid 2D+3D

Simulation engine: Molecular + Data-driven model

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS1-7

THEME & SCENE

One coast live oak mesophyll cell fills the screen, its wall a pale straw hexagon, and a fine vertical seam splits it into two working panels. Left, in cool jade light, a chloroplast: a lens 6 μm long, envelope doubled, stacked coins of thylakoid piled into grana, stroma a paler green between them, photons arriving as thin gold darts. Right, in warm ochre, a mitochondrion of the same cell: shelved cristae, amber matrix, rotors turning. Beneath both panels a single horizontal band runs the width of the screen, the shared atom ledger, with three columns of counters. Four ports and four sockets sit at the seam waiting for pipes. Panel docks right; a 1 μm scale bar sits under each organelle.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Mesophyll cell wall and vacuole	Environment	Straw hexagon 60 μm across with a 3 μm cellulose wall showing fibril texture; a large pale vacuole behind both organelles; cytoplasm streams slowly clockwise	No
2	Chloroplast envelope	Structure	Two membranes 10 nm apart wrapping a 6 $\times$ 3 μm lens, outer smooth, inner carrying transporters; jade-tinted and slightly translucent	No
3	Thylakoid discs stacked into grana	Structure	Discs 500 nm across, 20 nm thick, stacked 12 high into 8 grana joined by stroma lamellae; membrane deep green, lumen pale	Yes: resize
4	Stroma	Environment	Pale green gel around the grana, holding starch grains that grow visibly while photosynthesis runs	No
5	Chlorophyll antenna cluster	Structure	Rosettes of 250 pigment discs on each thylakoid face; flash briefly where a photon lands	No
6	Photon particles	Particle	Gold darts 40 px long arriving from the upper left, rate set by the light slider; absorbed on contact with an antenna, never reflected back into the cell	No
7	Rubisco enzyme	Actor	Barrel of eight large and eight small subunits, 12 nm, in the stroma; grabs one CO_2 per cycle and its jaw closes visibly	No
8	Thylakoid ATP synthase	Actor	20 nm rotor in the thylakoid membrane, head facing the stroma; deliberately drawn with the same silhouette as the mitochondrial rotor so the student notices	No
9	Mitochondrial outer membrane and intermembrane space	Structure	Ochre ellipsoid 1.8 $\times$ 0.7 μm with porins and a 15 nm proton-tinted gap	No
10	Cristae and matrix	Structure	12 lamellar shelves in an amber matrix with Krebs enzyme grains; the same asset family used in C3.3, at matching scale	Yes: resize
11	Mitochondrial ATP synthase	Actor	22 nm rotor, head facing the matrix; spins in the opposite screen direction to its chloroplast twin	No
12	Shared molecule pool	Particle	One pool of countable CO_2 (linear, charcoal and red), H_2O (bent, 104.5°), O_2 (red dimer) and glucose (charcoal ring); every molecule drawn once and owned by the cell, not by a panel	Yes: drag
13	Connector pipes	Actor	Four flexible glass tubes the student drags from a port on one panel to a socket on the other; glow when carrying; refuse to connect an output to another output	Yes: connect

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
14	Shared atom ledger band	Overlay	Full-width strip under both panels: C, H and O counters for the left panel, the right panel and the cell total, with the total pinned and highlighted red if it ever moves	No
15	Twin equation strip	Overlay	Two equations printed one above the other, the lower one written right to left so the reversal is visually obvious; terms light up as their molecules are consumed	No
16	Net-flux meter at the cell wall	Instrument	Two gauges on the wall reading net CO ₂ and net O ₂ crossing into or out of the cell, signed, with a green zero mark at the compensation point	No

HOW IT WORKS — THE MODEL

Both panels are stoichiometric engines drawing on one shared molecule pool, so nothing is ever created for a panel's convenience. Photosynthesis takes 6 CO₂ + 6 H₂O and light and returns 1 glucose + 6 O₂, at a rate set by a saturating light curve, a CO₂ term and a temperature optimum near 25 °C. Respiration takes 1 glucose + 6 O₂ and returns 6 CO₂ + 6 H₂O + about 32 ATP, at a rate set by demand and Q₁₀ = 2, and it runs day and night. The ledger is checked every tick and the C, H and O totals for the whole cell must not move by a single atom. The honest limit is stated on screen: this is a bookkeeping mirror, not a mechanical reversal, because the two use different enzymes in different compartments and photosynthesis needs light while respiration does not.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Light level	Slider	0–2,000	900	μmol/m ² /s PAR	Photon arrival rate and therefore the left panel's rate
Carbon dioxide	Slider	100–1,500	425	ppm	Supply to the left panel and the level the right panel adds to
Temperature	Slider	5–40	22	°C	Q ₁₀ = 2 on respiration, a peaked enzyme curve on photosynthesis
Cell demand	Slider	1–8	2	× resting	ATP draw, and therefore the right panel's rate
Organelles present	Multi-select	Chloroplast / Mitochondrion	Chloroplast, Mitochondrion	—	Which panels exist at all; removing one leaves a genuinely one-way ledger
Connector pipes	Drag-handle	4 ports to 4 sockets	None connected	—	Routes one panel's outputs into the other's inputs
Panel mode	Radio	Side by side / Overlay / Ledger only	Side by side	—	Viewport layout; Overlay superimposes the two equations on one axis
Time of day	Timeline scrubber	00:00–24:00	12:00	h	Drives the light slider automatically along a daily curve
Ledger element	Dropdown	Carbon / Hydrogen / Oxygen / All	All	—	Which column of the shared ledger is enlarged and highlighted

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Noon in the canopy	Light level=900; Carbon dioxide=425; Temperature=22; Cell demand=2; Organelles present=Chloroplast, Mitochondrion	Both panels are running. Read the net-flux meter: which way is CO ₂ crossing the wall, and is the right panel switched off?
S2	Midnight	Time of day=00:00; Light level=0; Cell demand=2	With no light the left panel stops. Which way does the ledger run now, and what has the cell been doing all along?

#	Scenario	Preset	The question the student answers
S3	Find the balance point	Light level=step from 0 to 300; Temperature=22; Cell demand=2	Find the light level at which both net-flux gauges read zero. Are both panels stopped, or are they cancelling?
S4	An animal cell	Organelles present=Mitochondrion; Light level=900; Cell demand=4	Remove the chloroplast. Which of the two equations is missing, and what happens to the shared ledger when only one direction exists?

STUDENT ACTIVITIES

1. Run S1 for 60 simulated seconds and record the ledger's C, H and O totals at the start and end. State whether any total changed.
2. Drag all four connector pipes so the left panel's outputs feed the right panel's inputs, re-run, and record how many molecules now cross the cell wall compared with the unconnected run.
3. Set the time-of-day scrubber to 00:00, 06:00, 12:00 and 18:00, recording net CO₂ flux at each, and mark the two times when the gauge crosses zero.
4. Step light from 0 to 300 $\mu\text{mol}/\text{m}^2/\text{s}$ in 25-unit steps at demand 2, recording net O₂ flux each time, and read off the light level where the trace crosses zero.
5. Remove the chloroplast, run 30 s, then put it back and run again. Write one sentence comparing what the ledger's carbon column does in each case.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Shared atom ledger	Data table	atoms	C, H and O per panel and for the whole cell; the cell total is pinned and flashes red if it moves
Net CO ₂ flux	Live numeric	molecules/s	Signed value at the cell wall, positive out; zero at the compensation point
Net O ₂ flux	Live numeric	molecules/s	Signed value at the cell wall, opposite in sign to the CO ₂ gauge
Panel rates	Line graph	reactions/s vs s	Two traces on shared axes, photosynthesis and respiration, showing that both are non-zero in daylight
Compensation point	Live numeric	$\mu\text{mol}/\text{m}^2/\text{s}$ PAR	The light level at which the two rates match, recomputed whenever temperature or demand changes
Glucose pool	Live numeric	molecules	Made by the left panel, spent by the right; rises in light, falls in the dark
Run log	Data table	mixed	Exportable CSV, one row per simulated 2 s, both rates, both fluxes and every ledger column

WHAT THE STUDENT SHOULD REALISE

Students believe photosynthesis and respiration belong to different organisms and are unrelated topics, and that a plant "breathes backwards". The shared ledger removes both ideas: exactly the same carbon, hydrogen and oxygen atoms are being assembled in one panel and taken apart in the other, inside one cell, at the same time. The student should be able to say: *"They are the same atoms rearranged in opposite directions, and one plant cell does both at once."*

C3.5 · Respiration in plants and animals alike

Experiment name: The Plant in the Dark

Render mode: Data Dashboard

Simulation engine: Data-driven model + Fluid/thermal

Interaction level: Argue-from-data

Session length: 18–25 min

NGSS anchor: MS-LS1-7

THEME & SCENE

A growth-room bench, shot slightly from above, with four sealed glass chambers in a row under a dimmable LED panel. Left to right: a coast live oak seedling in damp potting soil, a tray of germinating peas, six crickets on a mesh floor, and a boiled seedling that looks almost identical to the first. Each chamber carries a probe cluster on its lid and a coloured indicator vial clipped to its flank. A heavy black hood hangs on a rail above, ready to be dragged over any one chamber. The upper two-thirds of the screen is the dashboard: four gas traces on shared axes, a day-night band across the top, and a stacked decomposer chart. Control panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Sealed chamber vessel	Structure	Four identical borosilicate cubes 200 mm on a side, 8 L, clamped lids with silicone gaskets and four sealed feedthroughs; interior fogs above 90% relative humidity	Yes: swap
2	Coast live oak seedling	Actor	180 mm seedling, eight leathery leaves with a visible waxy cuticle, roots in 400 g of damp potting soil; a hover inset shows stomatal pores on the underside opening in light and closing in darkness	Yes: place
3	Germinating pea seeds	Actor	60 swollen seeds with 5 mm white radicles and no green tissue anywhere; respire but structurally cannot photosynthesise	Yes: place
4	House crickets	Actor	Six 18 mm insects on a mesh floor above a shallow water dish; hop at intervals; the metabolic comparison animal	Yes: place
5	Boiled seedling	Actor	Visually matched to object 2 but olive-drab and limp, dosed with antiseptic so no microbes colonise it; every rate is zero by construction	Yes: place
6	Bare potting soil	Actor	400 g of the same damp soil with no plant, present to expose the real confound that soil microbes respire too	Yes: place
7	Blackout hood	Structure	Opaque 220 mm fabric box on an overhead rail, dragged onto any chamber slot; drops that chamber's light to 0 without changing its temperature	Yes: drag
8	LED grow-light panel	Instrument	600 mm bar of white and deep-red diodes above all four chambers, dimmable 0–2,000 $\mu\text{mol}/\text{m}^2/\text{s}$, with a quantum sensor reading the delivered level	No
9	Carbon dioxide probe	Instrument	NDIR sensor per chamber, 0–10,000 ppm, ± 30 ppm, 20 s response, lid-mounted	No
10	Oxygen probe	Instrument	Galvanic cell per chamber, 0–25%, $\pm 0.1\%$, lid-mounted alongside the CO_2 probe	No
11	Thermistor and chamber conditioner	Instrument	Bead thermistor to 0.05 $^{\circ}\text{C}$ plus a Peltier plate holding each chamber to the set temperature so heat cannot be confused with light	No
12	Hydrogencarbonate indicator vial	Instrument	10 mL orange-red solution clipped to each chamber, going yellow as CO_2 rises and magenta as it falls, with a printed colour scale on the glass	Yes: place
13	Stirrer fan	Structure	40 mm fan in each lid keeping the gas well mixed so a probe reading means the whole chamber	No
14	Dashboard trace panel	Overlay	Four CO_2 traces and four O_2 traces on shared time axes, colour-keyed to the chambers, with the 425 ppm atmospheric line drawn	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Gross-and-net decomposer	Overlay	Stacked chart splitting a green chamber's measured net flux into a constant respiration band and a light-driven photosynthesis band, so respiration is visible as a band that never disappears	No
16	Day-night timeline scrubber	UI-Probe	Band across the top of the dashboard, dark where the lights are off; draggable playhead that scrubs any completed run	Yes: drag

HOW IT WORKS — THE MODEL

Every chamber is a well-mixed gas volume with net CO₂ flux equal to respiration minus photosynthesis. Respiration is $R = R_{20} \times 2^{((T-20)/10)} \times \text{mass}$, running continuously and completely independently of light. Photosynthesis is $P = P_{\text{max}} \times I/(I + 250)$ scaled by a CO₂ term, and is structurally zero for the peas, the crickets, the boiled control and the bare soil, because none of them contains chlorophyll. The decomposer chart is not a second model: it draws the two terms the engine already computed. The failure the sim must avoid is drawing a plant's gas exchange as one arrow that flips direction at dusk. Both arrows exist at every instant; what the probe measures is only their difference. Bare soil is included because its microbes genuinely respire, and honest data has confounds in it.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Chamber loadout	Multi-select	Four slots from: Oak seedling / Pea seeds / Crickets / Boiled seedling / Bare soil / Empty	Oak seedling, Pea seeds, Crickets, Boiled seedling	—	What is inside each chamber, and therefore which rates can exist
Light level	Slider	0–2,000	600	μmol/m ² /s PAR	Photon delivery to every uncovered chamber
Blackout hood	Drag-handle	Hood onto slot 1, 2, 3, 4 or the rail	On the rail	—	Darkens exactly one chamber while the others stay lit
Chamber temperature	Slider	2–35	20	°C	Respiration rate in every chamber through Q ₁₀ = 2
Sample mass per chamber	Slider	1–50	12	g	Living tissue mass, which scales respiration proportionally
Day length	Slider	0–24	12	h	Length of the lit part of the simulated daily cycle
Time compression	Dial	60×–3,600×	600×	—	Simulated seconds per real second
Gross-and-net decomposer	Toggle	On / Off	Off	—	Splits the green chambers' net traces into their two component bands
Probe set	Multi-select	CO ₂ / O ₂ / Thermistor / Indicator vial	CO ₂ , O ₂ , Indicator vial	—	Which instruments report to the dashboard
Trace units	Radio	Raw ppm / Per gram of tissue / Net beside gross	Raw ppm	—	How the traces are normalised for comparison

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Lights out	Chamber loadout=Oak seedling, Pea seeds, Crickets, Boiled seedling; Light level=0; Chamber temperature=20; Day length=0	In total darkness, which chambers show CO ₂ rising? Does the plant chamber behave like the cricket chamber or like the dead one?

#	Scenario	Preset	The question the student answers
S2	One hood, three lit	Blackout hood=on slot 1; Light level=600; Chamber temperature=20	The hooded seedling makes CO ₂ and the lit one absorbs it. Are these two different plants, or the same plant doing the same thing under different lighting?
S3	Salinas cold store	Chamber loadout=Pea seeds, Pea seeds, Empty, Empty; Chamber temperature=2, then 20; Light level=0	Harvested produce is chilled before shipping. Using your CO ₂ numbers, say exactly what refrigeration is slowing down.
S4	Is it really the plant	Chamber loadout=Oak seedling, Bare soil, Boiled seedling, Empty; Light level=0; Sample mass per chamber=12	The soil chamber makes CO ₂ too. How much of chamber 1's CO ₂ came from the seedling itself? Show the subtraction.

STUDENT ACTIVITIES

1. Run S1 for 6 simulated hours and record each chamber's CO₂ at 0, 2 and 6 hours. Rank the four chambers by CO₂ produced per gram.
2. Drag the blackout hood onto the oak seedling with the lights at 600, run 4 hours, and record net CO₂ flux for the hooded and the unhooded seedling.
3. Turn the decomposer on with light at 600 and record the height of the respiration band at four different light levels. State whether that band ever reaches zero.
4. Set the temperature to 2 °C and then 20 °C for identical pea chambers, recording CO₂ per gram per hour for each, and calculate the ratio.
5. Run S4 and subtract the bare-soil chamber's CO₂ from the seedling chamber's. Record the corrected figure and write one sentence on why the control was necessary.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Chamber CO ₂ traces	Line graph	ppm vs h	Four colour-keyed traces with the day-night band overhead and the 425 ppm line marked
Chamber O ₂ traces	Line graph	% vs h	Four traces; falls wherever CO ₂ climbs
Net flux per chamber	Live numeric	μmol CO ₂ per gram per hour	Signed, positive when the chamber is a net CO ₂ source
Gross respiration band	Bar chart	μmol CO ₂ per gram per hour	Respiration only, drawn for every chamber including the lit plant, where it is otherwise hidden
Indicator vial states	Data table	—	Colour of each vial with the time it changed and the CO ₂ level at that moment
Control comparison	Pass-fail badge	—	Green once the student has logged both the dead control and the soil control for the run
Run log	Data table	mixed	Exportable CSV, one row per simulated 10 min, every chamber and every probe

WHAT THE STUDENT SHOULD REALISE

Students believe plants photosynthesise instead of respiring, and that gas exchange in a plant only ever runs one way. A seedling sealed in the dark raises the CO₂ in its chamber exactly as the crickets do, while the boiled seedling beside it does nothing, and the decomposer shows the respiration band still present at full light. The student should be able to say: *"Plants respire all the time. In daylight photosynthesis is just faster, so it hides it."*

C4 · Food molecules rearranged · MS-LS1-7

C4.1 · Modeling a food molecule broken down

Experiment name: Taking Glucose Apart

Render mode: 3D Scene

Simulation engine: Molecular

Interaction level: Manipulate

Session length: 18–25 min

NGSS anchor: MS-LS1-7

THEME & SCENE

A dark slate workbench under a single bright ring light, everything else in shadow. Centre stage, held in a soft magnetic cradle and slowly turning, is one glucose molecule built as ball and stick: a charcoal six-carbon ring with white hydrogens bristling off it and red oxygens at every hydroxyl, catching the ring light in glossy highlights. A rack on the left holds three tools: bond scissors with fine chrome jaws, a capping dispenser, and blunt tweezers. Along the bottom, four shallow parts-tray wells wait, each numbered. On the right, a tally board shows three enormous counters reading C 6, H 12, O 6. A ghost outline of the original molecule hangs faintly behind the cradle. Panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Carbon atom	Particle	Charcoal sphere, covalent radius 0.76 Å rendered at ×80, matte with a faint specular; four tetrahedral sockets at 109.5° drawn as recessed dimples when unfilled	Yes: drag
2	Hydrogen atom	Particle	White sphere, 0.31 Å at the same scale, one socket; the smallest object on the bench and deliberately hard to lose	Yes: drag
3	Oxygen atom	Particle	Red sphere, 0.66 Å, two sockets at 104.5°, with two faint lone-pair lobes on the reverse face	Yes: drag
4	Bond cylinders	Structure	Single bond: one 0.18 Å-wide chrome cylinder, drawn at true length, C–C 154 pm, C–H 109 pm, C–O 143 pm, O–H 96 pm. Double bond: twin parallel cylinders at 123 pm. Cylinders stretch and redden as the scissors close on them	Yes: swap
5	Glucose molecule assembly	Actor	Pyranose chair ring by default: 5 C and 1 O in the ring, one exocyclic CH ₂ OH, four ring hydroxyls; a toggle unrolls it into the open-chain aldehyde form with 5 C–C, 7 C–H, 5 C–O, 5 O–H and 1 C=O	Yes: swap
6	Magnetic cradle	Structure	Ring of six pale field emitters holding the molecule at bench centre; releases any fragment the tweezers pull free	No
7	Bond scissors	Instrument	Chrome tool with a 2 Å jaw; hovering over a bond highlights it and prints its type and length; a cut leaves two glowing open sockets and adds to the scissors charge	Yes: drag
8	Capping dispenser	Instrument	Twin-barrelled tool loaded with H caps and OH caps, each drawn from a water molecule in the reservoir, so a cap always costs the tally board something	Yes: drag
9	Enzyme stamp set	Actor	Three translucent enzyme bodies, 6 nm, each with a shaped recess: hexokinase, phosphofructokinase and aldolase; stamping the molecule performs that enzyme's real cut and nothing else	Yes: place
10	Fragment tray wells	Structure	Numbered shallow wells along the bench front; a fragment parked in a well is counted and labelled with its own formula	Yes: place
11	Valence halo	Overlay	Red pulsing ring around any atom with an unfilled socket, and a small printed count of sockets still open on that atom	No
12	Per-element tally board	Overlay	Three large counters, C, H and O, split into "on the bench", "in the wells" and "total"; the total is boxed and cannot change	No
13	Ghost of the original molecule	Overlay	Faint wireframe of the intact glucose left in place behind the cradle so every fragment can be traced to where it came from	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
14	Scissors charge counter	Instrument	A meter that only ever increases, labelled "energy spent cutting, kJ per mole", accumulating the real bond energy of each bond broken	No
15	Water reservoir	Structure	Small glass boat of 12 bent water molecules feeding the capping dispenser; empties visibly as caps are spent	Yes: place
16	Rewind scrubber and scale readout	UI-Probe	Timeline of every action taken, draggable backwards to undo cuts one at a time, with a live ångström scale printed beside the cursor	Yes: drag

HOW IT WORKS — THE MODEL

The molecule is a graph: atoms are nodes with a valence budget of C 4, O 2, H 1, and bonds are edges with a type and a real length. The scissors delete an edge and leave two unfilled sockets, which glow red, and the bench refuses to be declared finished while any socket is open. Caps can close sockets, but every cap is drawn from the water reservoir and every atom in it is counted, so nothing is ever free. Cutting the C3–C4 bond gives the split that aldolase actually makes and lights the three-carbon ghost template. The scissors charge accumulates real mean bond energies, 347 kJ/mol for C–C, 413 for C–H, 358 for C–O, 464 for O–H, and it only ever rises, which quietly sets up the next two experiments. The failure to avoid is any animation of energy pouring out of a bond as it is cut.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Molecule on the cradle	Dropdown	Glucose ring / Glucose open chain / Fructose / Ribose	Glucose ring	—	Which molecule is built in the cradle and what the tally board starts at
Tool in hand	Radio	Scissors / Hydrogen cap / Hydroxyl cap / Tweezers / Enzyme stamp	Scissors	—	What clicking on the molecule does
Enzyme stamp	Dropdown	Hexokinase / Phosphofructokinase / Aldolase	Aldolase	—	Which real cut the stamp performs
Water available	Stepper	0–12	6	molecules	How many caps can be paid for before the reservoir is empty
Display style	Radio	Ball and stick / Space filling / Skeletal	Ball and stick	—	How the molecule is drawn; space filling shows why the ring is bulky
Valence lamps	Toggle	On / Off	On	—	Whether open sockets glow red
Fragment tray wells	Stepper	2–8	4	count	How many places exist to park fragments, which caps how far you can shred
Tally board	Toggle	On / Off	On	—	Shows or hides the live element counters
Rewind	Timeline scrubber	0–40	0	actions	Steps backwards through every cut and cap made
Scale readout	Dropdown	Ångström / Picometre / Off	Ångström	—	Units printed beside the cursor and on every bond label

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Cut it in half	Molecule on the cradle=Glucose open chain; Tool in hand=Scissors; Fragment tray wells=4	Cut one C–C bond so the ring becomes two three-carbon pieces. What does the tally board read afterwards, and what did it read before?

#	Scenario	Preset	The question the student answers
S2	Shred it	Molecule on the cradle=Glucose open chain; Fragment tray wells=8; Water available=12	Cut every carbon-to-carbon bond. You now have six separate pieces. How many carbons, hydrogens and oxygens are on the bench?
S3	The vanishing act	Tool in hand=Scissors; Water available=0; Valence lamps=On	Try to make one hydrogen atom disappear from the bench. Report what the tool does and what the tally board does.
S4	The real route	Tool in hand=Enzyme stamp; Enzyme stamp=Aldolase; Molecule on the cradle=Glucose open chain	Use the enzyme rather than the scissors. Where does aldolase cut, and how does its cut differ from the one you chose in S1?

STUDENT ACTIVITIES

1. Record the tally board's three numbers before touching anything, then cut the molecule in half and record them again. Write down which numbers changed.
2. Cut every C–C bond in turn, parking each fragment in a numbered well, and write the formula of each fragment from the labels.
3. Set water available to 0, cut a C–H bond, and record what the valence halo shows and what the bench refuses to let you do next.
4. Read the scissors charge after one cut, after three cuts and after all five C–C cuts. Say whether it ever goes down, and what that suggests about breaking bonds.
5. Stamp the molecule with aldolase, then rewind fully and stamp with hexokinase. Record what each enzyme did, in one sentence each.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Per-element tally	Live numeric	atoms	C, H and O on the bench, in the wells and in total; the total is boxed and never moves
Fragment inventory	Data table	—	One row per fragment: formula, carbon count, which bond it came from, which well it sits in
Open sockets	Counter	count	Unsatisfied valences currently on the bench; must reach zero before a build can be declared finished
Scissors charge	Live numeric	kJ/mol	Cumulative bond energy spent cutting, using real mean bond energies; monotonic increasing
Bond log	Data table	mixed	Every bond broken or made, with its type, its length and its energy; exportable CSV
Conservation badge	Pass-fail badge	—	Green while the total atom counts equal the starting counts; there is no way to turn it red, which is the point

WHAT THE STUDENT SHOULD REALISE

Students believe food is "used up", "burned away" or "turned into energy", so that at the end there is less stuff than there was. Twenty minutes of cutting leaves exactly six carbons, twelve hydrogens and six oxygens on the bench, every single time, and the only meter that moved was the one counting energy spent. The student should be able to say: *"Every atom that was in the glucose is still here. I only changed how they are joined together."*

C4.2 · Modeling the atoms rearranged into products

Experiment name: Building the Exhaust

Render mode: 3D Scene

Simulation engine: Molecular

Interaction level: Design

Session length: 18–25 min

NGSS anchor: MS-LS1-7

THEME & SCENE

The same slate bench, swung ninety degrees so it now runs left to right as an assembly line. At the left, the loose parts from the previous experiment lie in an open tray: charcoal carbons, white hydrogens, red oxygens, still tinted the blue of things that came from food. Overhead hangs an oxygen dispenser, a chrome hopper of red dimers tinted green, with a numbered dial on its face. At the right, a rack of empty product templates glows faintly: linear ghosts waiting for carbon dioxide, bent ghosts waiting for water. Beneath the line runs a see-saw ledger, reactants on the left arm, products on the right, currently tilted. A dark commit button sits under the rack. A small breath inset floats top-right. Panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Loose atom pool	Particle	The 6 C, 12 H and 6 O carried over from C4.1, lying in an open tray with their sockets visible; each atom keeps a persistent identity number	Yes: drag
2	Oxygen dispenser	Instrument	Chrome hopper with a numbered dial 0–12; each pull releases one O ₂ , drawn as two red spheres with a twin cylinder at 121 pm, tinted green to mark it as air-derived	Yes: drag
3	Carbon dioxide template	Structure	Linear ghost, O=C=O at exactly 180°, two double-bond slots at 116 pm; six of them in the rack; snaps shut with a click when correctly filled	Yes: place
4	Water template	Structure	Bent ghost at 104.5° with two O–H slots at 96 pm; six in the rack; refuses any atom that is not one oxygen and two hydrogens	Yes: place
5	Snap magnet field	Field	Invisible attractor around each empty slot with a 1.5 Å capture radius; pulls a correct atom home and repels a wrong one with a small shudder	No
6	Valence halo	Overlay	Red pulsing ring on any atom with an open socket anywhere on the bench, including inside a half-built product	No
7	Source tint shader	Overlay	Tints every atom by origin: blue for atoms that came from glucose, green for oxygen that came from the dispenser; survives assembly, so a finished molecule shows its parentage	No
8	Carbon tracer tag	UI-Probe	A heavy-isotope tag the student paints onto any one carbon; that carbon keeps a bright halo through every subsequent step and into the breath inset	Yes: place
9	Per-element see-saw ledger	Overlay	Physical balance beam under the line: reactant counts on the left pan, product counts on the right, tilting visibly until C, H and O all match, then levelling with a chime	No
10	Coefficient counters	Instrument	Three printed counters above the rack reading how many O ₂ used, how many CO ₂ built and how many H ₂ O built, updating as products complete	No
11	Commit gate	Actor	Heavy switch under the rack, dark and physically immovable until every socket is filled and the see-saw is level; throwing it sends the products to the bin and logs the run	Yes: place
12	Product bin and breath inset	Overlay	Committed products drop into a bin, then appear in a small inset of a nose and mouth exhaling, where the tagged carbon is visible leaving the body	No
13	Leftover shelf	Structure	A shelf beside the rack where any unused atoms sit; if it is not empty the commit gate stays locked and the shelf is labelled with what is stranded	Yes: place
14	Failure log	Overlay	Running list of blocked attempts with the reason printed: "3 oxygens short", "carbon has 3 bonds, needs 4", "2 hydrogens left on the shelf"	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Electron transport cutaway	Actor	Optional panel showing where inhaled O ₂ really ends up: at complex IV on the inner membrane, taken up into water rather than into carbon dioxide; opens as a side window without disturbing the bench	Yes: swap
16	Before and after inventory	Overlay	Two columns, reactants and products, listing C, H and O with a printed difference of zero on every row	No

HOW IT WORKS — THE MODEL

Snap assembly under strict valence enforcement. Each template exposes typed slots; a magnet field pulls a matching atom in and rejects anything else. The commit gate is a hard gate with three conditions: no open sockets, no atoms on the leftover shelf, and a level see-saw. The arithmetic drives everything: glucose supplies only 6 oxygen atoms, while 6 CO₂ needs 12 and 6 H₂O needs 6, so 18 are required and exactly 6 O₂ must be dialled in. Five leaves the ledger short, seven leaves two oxygens stranded. Source tint runs alongside as a tracer argument: every carbon in every committed CO₂ is blue, meaning it came from food. The optional cutaway keeps the model honest by showing that, in the real pathway, the inhaled oxygen is reduced to water at complex IV rather than being bolted directly onto carbon.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Oxygen molecules dialled in	Stepper	0–12	0	molecules	How many O ₂ the dispenser releases onto the bench
Starting fragments	Dropdown	Intact glucose / Two three-carbon pieces / Six one-carbon pieces	Two three-carbon pieces	—	What the tray begins with, and therefore how much building is left to do
Product to build	Radio	Carbon dioxide / Water / Auto	Carbon dioxide	—	Which template the rack presents to the cursor
Snap assist	Toggle	On / Off	On	—	Whether the magnet field pulls correct atoms home or you must place them exactly
Source tint	Toggle	On / Off	On	—	Colours atoms blue for food-derived and green for air-derived
Carbon tracer	Drag-handle	Paint any 1 of the 6 carbons	None painted	—	Which carbon carries a halo through assembly and into the breath inset
Ledger element	Dropdown	Carbon / Hydrogen / Oxygen / All	All	—	Which row of the see-saw is enlarged
Where the inhaled oxygen goes	Toggle	Overall equation / Real pathway	Overall equation	—	Opens the electron transport cutaway alongside the bench
Breath inset	Toggle	On / Off	On	—	Shows committed products leaving the body

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Honest build	Starting fragments=Two three-carbon pieces; Oxygen molecules dialled in=0, then raise it; Snap assist=On	Build six carbon dioxides and six waters. How many O ₂ did you have to dial in, and why not fewer?
S2	No air	Oxygen molecules dialled in=0; Starting fragments=Intact glucose; Product to build=Auto	Using only the atoms glucose brought, build as many complete products as you can. What is left stranded, and why can you not finish?
S3	Too much oxygen	Oxygen molecules dialled in=8; Starting fragments=Two three-carbon pieces	The gate will not throw. Read the failure log and the leftover shelf, then say exactly how many oxygen atoms are surplus.

#	Scenario	Preset	The question the student answers
S4	Follow one carbon	Carbon tracer=paint carbon 1; Oxygen molecules dialled in=6; Breath inset=On	Tag a carbon, commit, and watch the inset. Where did the carbon in your exhaled CO ₂ come from, the food or the air?

STUDENT ACTIVITIES

1. Count the oxygen atoms glucose brings and the oxygen atoms the six CO₂ and six H₂O will need. Write both numbers, then predict how many O₂ you must dial in before you touch the dial.
2. Build all six carbon dioxides, recording the coefficient counters after each one, then build all six waters and record again.
3. Set oxygen to 5 and try to commit. Copy the exact wording from the failure log and say which element is short.
4. Paint the tracer on one carbon, commit the run, and record where that carbon appears in the breath inset and what colour it is.
5. Switch "Where the inhaled oxygen goes" to Real pathway and write one sentence on how the real route differs from the overall equation you just built.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
See-saw balance	Pass-fail badge	—	Level and green only when C, H and O all match across the beam
Coefficient counters	Live numeric	molecules	O ₂ used, CO ₂ built and H ₂ O built, updated as each product completes
Before and after inventory	Data table	atoms	Reactant and product columns for C, H and O with a difference column pinned at zero
Open sockets	Counter	count	Unsatisfied valences anywhere on the bench, including inside part-built products
Leftover shelf contents	Data table	atoms	What is stranded and why, refreshed on every failed commit
Failure log	Data table	—	Every blocked commit with its timestamp and printed reason; exportable CSV
Tracer destination	Live numeric	—	Names the product molecule the tagged carbon ended up in

WHAT THE STUDENT SHOULD REALISE

Students believe the carbon dioxide they breathe out is the air they breathed in coming back, and that oxygen is somehow converted into carbon dioxide. The source tint makes that untenable: every committed carbon is food-blue, and the six O₂ were needed because glucose does not carry enough oxygen of its own. The student should be able to say: *"The carbon in my breath came from my food. The oxygen from the air joined it, it did not become it."*

C4.3 · Energy released, not created

Experiment name: The Bond Bank

Render mode: Hybrid 2D+3D

Simulation engine: Molecular + Data-driven model

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS1-7

THEME & SCENE

The screen is split horizontally. The upper half is the molecular bench, dimly lit: glucose and six oxygen molecules on the left, six carbon dioxides and six waters on the right, and every single bond in both groups drawn as a small coiled spring with its energy printed on a paper tag hanging off it. The lower half is a side-on track shaped like a hill with a long drop beyond it, drawn in cool blueprint lines, with a small brass cart parked at the left shoulder. To the right of the track stand two receivers: a tall vault marked ATP with thirty-two slots, and a broad radiator that glows dull orange as it takes waste heat. A three-column ledger sits under the track. Panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Reactant bond springs	Structure	Glucose's 5 C–C at 347, 7 C–H at 413, 5 C–O at 358, 5 O–H at 464 and 1 C=O at 745, plus 6 O=O at 498, each a coiled spring with a printed kJ/mol tag; springs stretch and whiten as they are pulled apart	Yes: swap
2	Product bond springs	Structure	12 C=O in carbon dioxide at 805 and 12 O–H in water at 464, drawn shorter and visibly stiffer than the reactant springs, which is the whole reason there is a surplus	No
3	Bond-energy label tags	Overlay	Small paper tags on every spring; hovering enlarges one and names the bond type, its length and its mean energy	No
4	Reaction profile track	Structure	Blueprint-line rail: a starting shoulder, an activation hump, and a lower finishing shelf; the vertical axis is graduated in kJ/mol with real numbers, not a cartoon	No
5	Reaction cart	Actor	30 px brass cart carrying a tiny glucose icon; rolls only if given enough energy to clear the hump, then coasts to the lower shelf	Yes: drag
6	Activation hump	Structure	Adjustable ridge on the track, 1,100 kJ/mol high with no enzymes and 55 kJ/mol with the full enzyme set; its height is the only thing enzymes change	Yes: resize
7	Enzyme actors	Actor	Up to thirty translucent enzyme bodies placed along the track; each one placed lowers the hump and adds one step to the staircase	Yes: place
8	Staircase step blocks	Structure	The long drop redrawn as a flight of small steps, one per enzyme, each labelled with its own modest energy drop; a single cliff when no enzymes are present	Yes: resize
9	ATP vault	Instrument	Glass tower with thirty-two numbered slots; ATP tokens drop in as they are made, each slot labelled 30.5 kJ/mol	No
10	ATP and ADP tokens	Particle	ATP with three phosphates and a glowing terminal one, ADP with two; ADP and phosphate feed in at the base of the vault as ATP leaves the top	No
11	Heat radiator	Instrument	Finned block beside the vault with a running joule counter; glows from grey to dull orange in proportion to the energy it has taken	No
12	Three-column ledger	Overlay	"Spent breaking", "paid back forming" and "net", each a large running total in kJ/mol; the net column is computed, never typed	No
13	Free-energy machine	UI-Probe	A brass gadget the student may bolt to the vault to try to draw more out than went in; it always stalls, and prints which ledger column it ran out of	Yes: place
14	Bond-energy lookup table	Overlay	Printed table of the mean bond energies used, so every number on screen can be checked against a source	No
15	Conservation badge	Instrument	Continuous check that spent minus paid-back equals net and that vault plus radiator equals net; goes red if either fails	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Cell thermistor	Instrument	Bead probe in a small cell inset, showing the temperature rise that the radiator's joules actually produce in a real cell	No

HOW IT WORKS — THE MODEL

Straight bond-energy accounting, with every number drawn from the printed table. Breaking all of glucose's bonds costs 9,481 kJ/mol and breaking six O=O costs 2,988, so 12,469 kJ/mol must be spent before anything is gained. Forming twelve C=O in carbon dioxide pays back 9,660 and twelve O–H in water pays 5,568, so 15,228 comes back. The net is 2,759 kJ/mol released, close to the accepted 2,803 for glucose because mean bond energies are averages, and the sim says so rather than hiding the gap. Of that net, the coupling slider sends a fraction to the ATP vault, 34% by default, giving about 32 ATP, and the remainder goes to the radiator. The absolute rule is that no animation may ever show energy emerging from a bond as it breaks: breaking always costs, forming always pays.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
View	Radio	Bond springs / Energy staircase / Both	Both	—	Which half of the screen is active and enlarged
Stage	Radio	Break only / Form only / Whole reaction	Whole reaction	—	Runs half the accounting so the two halves can be read separately
Bond types shown	Multi-select	C–C / C–H / C–O / O–H / C=O / O=O	All six	—	Which springs are drawn and which rows of the ledger are itemised
Enzymes present	Stepper	0–30	30	steps catalysed	Activation hump height and how many steps the staircase has
ATP coupling	Slider	0–60	34	% of net	How much of the released energy reaches the vault rather than the radiator
Cell temperature	Slider	20–45	37	°C	Enzyme activity, and the temperature the radiator's joules produce in the cell inset
Free-energy machine	Toggle	Armed / Off	Off	—	Bolts the trap gadget to the vault and lets the student try to beat the ledger
Ledger units	Dropdown	kJ per mole / kJ per gram / kcal per gram	kJ per mole	—	Units on all three columns and on every spring tag
Playback	Timeline scrubber	0–60	0	s	Steps through breaking, then forming, then coupling

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	The whole ledger	Stage=Whole reaction; Enzymes present=30; ATP coupling=34; View=Both	Read all three ledger columns. Which is bigger, spent or paid back, and by how much?
S2	Breaking only	Stage=Break only; Enzymes present=30; Playback=0–60	Stop after every bond is broken. Is the ledger positive or negative at that moment? What does that say about "energy is released when bonds break"?
S3	No enzymes	Enzymes present=0; Cell temperature=37; Stage=Whole reaction	The hump is 1,100 kJ/mol and the cart will not move. The sugar just sits there. Did the net energy change when you removed the enzymes?
S4	Beat the bank	Free-energy machine=Armed; ATP coupling=60; Stage=Whole reaction	Try to fill more than thirty-two ATP slots. Which ledger column stops you, and what is the most the vault could ever hold?

STUDENT ACTIVITIES

1. Record the "spent breaking" total, the "paid back forming" total and the net, in kJ per mole, and check that the first two really do subtract to the third.
2. Run Stage=Break only and record the ledger. Write one sentence saying whether breaking bonds gave energy out or took energy in.
3. Set enzymes to 0, 5, 15 and 30, recording the hump height and the net energy each time. State which of the two changed.
4. Move ATP coupling from 0% to 60% in steps of 15, recording ATP slots filled and radiator joules at each setting, and check that they always add to the same total.
5. Arm the free-energy machine, try to overfill the vault, and copy down the message it prints when it stalls.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Three-column ledger	Live numeric	kJ/mol	Spent breaking, paid back forming and net, all updating as the playback advances
Bond itemisation	Data table	kJ/mol	Every bond type, how many of them, and the subtotal for breaking and for forming
Energy profile	Line graph	kJ/mol vs reaction progress	The hump and the drop, redrawn as a staircase when enzymes are present
ATP slots filled	Counter	ATP	Out of 32; tracks the coupling slider
Radiator total	Live numeric	kJ/mol	Waste heat, always equal to net minus what reached the vault
Conservation badge	Pass-fail badge	—	Green while vault plus radiator equals net; it cannot be made to go red by any legal setting
Cell temperature rise	Live numeric	°C	What the radiator's joules do to the cell inset, so waste heat is a real quantity and not a shrug

WHAT THE STUDENT SHOULD REALISE

Students believe energy is created in respiration, and repeat that "energy is released when bonds break". Breaking every bond in this reaction costs 12,469 kJ/mol and the ledger stays negative until products form. The surplus exists only because the new bonds pay back more than the old cost. The student should be able to say: *"Nothing made new energy. Building the new bonds gave back more than taking the old ones apart cost, and that difference is what I get."*

C4.4 · Comparing respiration to a familiar reaction

Experiment name: Two Ways to Burn the Same Sugar

Render mode: Hybrid 2D+3D

Simulation engine: Fluid/thermal + Molecular + Data-driven model

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS1-7

THEME & SCENE

One long lab bench behind a sheet of safety glass, lit hard from above, with two rigs facing each other across a shared central ledger. On the left, a bomb calorimeter: a squat steel pressure vessel with ignition leads, sunk in a stirred water jacket, a Beckmann thermometer standing in the water, and a small armoured viewport through which a 4.00 g pressed glucose tablet sits in a nickel crucible. On the right, a conical respirometer flask of cloudy beige yeast suspension holding a second, identical tablet, a thermistor and two gas probes through its bung, and no flame anywhere. Two wall clocks above the rigs run at wildly different speeds, each with its compression factor printed large in red. Panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Safety glass and twin bench	Environment	2.4 m stainless bench split by a central ledger plinth; 8 mm polycarbonate screen with a faint reflection so the scene reads as behind glass	No
2	Bomb calorimeter vessel	Structure	Steel cylinder 65 mm × 100 mm with a screw head, oxygen fill valve, two insulated ignition posts and a 20 mm armoured viewport; cutaway view available	Yes: swap
3	Water jacket, stirrer and thermometer	Instrument	2.00 kg of water in an insulated pail with a paddle stirrer and a Beckmann thermometer reading to 0.01 °C; the jacket is the whole measurement	No
4	Ignition coil and fuse wire	Actor	Nichrome loop across the crucible; glows orange then white on firing; the only energy input to the left rig, logged at 50 J and subtracted from the result	Yes: place
5	Glucose tablet	Actor	Pressed dextrose cube, 4.00 g, 18 mm, chalky white with a pressed seam; two identical tablets exist, one per rig; shrinks, browns and chars in the left rig, dissolves in the right	Yes: resize
6	Flame	Particle	Luminous yellow-white plume with a blue base, 1,100 °C in air and up to 1,600 °C in pure oxygen; emits light and soot; the temperature figure is printed on the plume	No
7	Soot and ash residue	Structure	Grey-black crust in the crucible whose mass is weighed and logged; a reminder that combustion in poor oxygen is incomplete	No
8	Respirometer flask	Structure	1 L conical flask with a three-hole bung, magnetic stirrer bar, and a water bath around it holding 37 °C	Yes: swap
9	Yeast suspension	Actor	600 mL cloudy beige culture, 8 g of cells, aerated by a fine sparger; swappable for a minced-muscle homogenate that gives the same chemistry with a different rate	Yes: swap
10	Thermistor	Instrument	Bead probe in the flask reading to 0.01 °C against a matched reference flask, so a rise of a fraction of a degree is still measurable	No
11	Carbon dioxide and oxygen probes	Instrument	NDIR CO ₂ sensor 0–10,000 ppm and a galvanic O ₂ cell 0–25%, both in the flask headspace; a matching pair samples the bomb's gases after firing	No
12	ATP counter	Instrument	Digital totaliser on the right rig only, reading ATP made in moles and in kJ; the left rig's equivalent readout is present and permanently pinned at zero	No
13	Reaction profile overlay	Overlay	One pair of axes carrying both routes: a single tall cliff for combustion and a flight of thirty small steps for respiration, drawn to the same vertical scale in kJ/mol	Yes: swap

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
14	Shared product ledger	Overlay	Central plinth display: CO ₂ made, H ₂ O made and total energy released, one column per rig, converging on identical figures	No
15	Energy pie	Overlay	Two pies below the ledger splitting each rig's output into heat, light and ATP; the left pie has no ATP wedge at all	No
16	Twin timeline with compression badges	UI-Probe	Two scrubbable timelines with their compression factors printed in red; a lock button forces both to real time so the rate difference is felt rather than read	Yes: drag

HOW IT WORKS — THE MODEL

Both rigs run identical stoichiometry, $\text{C}_6\text{H}_{12}\text{O}_6 + 6 \text{O}_2 \rightarrow 6 \text{CO}_2 + 6 \text{H}_2\text{O}$, and identical total energy, 15.56 kJ per gram, so 4.00 g releases 62.2 kJ on either side. Everything else differs. Combustion is one uncatalysed step gated by an ignition temperature near 400 °C; once lit, it is self-sustaining and oxygen-limited, finishing in about 35 s with a flame at 1,100 °C, and all 62.2 kJ leaves as heat and light, raising the 2.00 kg jacket by 7.4 °C. Respiration is thirty enzyme-catalysed steps at 37 °C, rate-limited by enzyme number, finishing in about six hours; 34% of the energy is captured as roughly 0.71 mol of ATP and the remaining 41 kJ trickles out at under 2 W, raising the flask by about 1 °C. The rule the engine must never break is that the two product ledgers end identical.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Glucose mass	Slider	0.50–10.00	4.00	g	Fuel in both rigs at once; they always get the same
Oxygen supply to the bomb	Slider	20–100	100	% O ₂	Flame temperature and how much soot is left in the crucible
Ignition	Toggle	Armed / Off	Off	—	Fires the coil; nothing happens on the left rig until this is thrown
Enzyme steps active	Stepper	0–30	30	steps	How many catalysed steps the right rig has, and therefore its rate
Cell rig temperature	Slider	5–45	37	°C	Enzyme rate, with denaturation collapsing the rate above 42 °C
Water jacket insulation	Toggle	On / Off	On	—	Whether the left rig's heat is retained for measurement or lost to the room
Left rig time compression	Dial	0.1×–10×	1×	—	Playback speed of the combustion rig
Right rig time compression	Dial	1×–5,000×	1,000×	—	Playback speed of the cell rig, so six hours fits in the lesson
View	Radio	Bomb only / Cell only / Synchronised split	Synchronised split	—	Which rig fills the frame; Synchronised locks both clocks to real time
Ledger units	Dropdown	kJ / kJ per gram / kcal	kJ	—	Units on the shared ledger and both pies

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Same fuel, both rigs	Glucose mass=4.00; Ignition=Armed; Enzyme steps active=30; Cell rig temperature=37; View=Synchronised split	Record peak temperature, duration and total energy for each rig. Which of those three numbers is the same on both sides?
S2	A cell at flame temperature	Cell rig temperature=45; Enzyme steps active=30; Glucose mass=4.00	Push the cell rig hot. The rate collapses instead of rising. What broke, and why can a cell not simply run hotter to go faster?

#	Scenario	Preset	The question the student answers
S3	No enzymes, no ignition	Enzyme steps active=0; Ignition=Off; Glucose mass=4.00; Cell rig temperature=37	Both tablets just sit there unchanged. If glucose really has 62.2 kJ in it, why is nothing happening on either side?
S4	Where the carbon ends up	Ignition=Armed; Oxygen supply to the bomb=20; Enzyme steps active=30	Starve the flame of oxygen and compare the ledgers. Which rig accounted for every carbon, and what is sitting in the crucible?

STUDENT ACTIVITIES

1. Run S1 and record, for each rig, the peak temperature, the time to finish, the CO₂ made and the total energy released. Circle every figure that matches across the two rigs.
2. Lock both timelines to real time using the View control, watch for thirty seconds, and write one sentence describing what each rig has managed in that period.
3. Set enzyme steps to 0, 10, 20 and 30, recording the time the right rig takes to finish each run, and state whether the total energy released changed.
4. Read both energy pies and record the percentage that ended as ATP in each rig. Explain the left rig's figure.
5. Drop the bomb's oxygen to 20%, fire it, and record the ash mass and the CO₂ shortfall. Say what happened to the missing carbon.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Peak temperature per rig	Live numeric	°C	Flame temperature on the left, flask temperature on the right, side by side
Temperature traces	Line graph	°C vs s	Jacket and flask on shared axes with a logarithmic time axis, so a 35 s event and a 6 h event both fit
Shared product ledger	Data table	mol and g	CO ₂ and H ₂ O made by each rig, plus total energy released; the columns converge
Energy split	Bar chart	%	Heat, light and ATP for each rig; ATP is zero on the left by construction
Duration	Timer	s	Elapsed simulated time to completion for each rig, printed with its compression factor
Reaction profile	Line graph	kJ/mol vs progress	The cliff and the staircase drawn to one vertical scale
Run log	Data table	mixed	Exportable CSV: every control setting and every readout, one row per run

WHAT THE STUDENT SHOULD REALISE

Students think respiration is nothing like burning, or that cells burn food and get hot. Both rigs take the same glucose and oxygen and release the same 62.2 kJ, yet one runs at 1,100 °C for 35 seconds and the other at 37 °C for six hours, banking a third as ATP. The student should be able to say: *"Same reaction, same products, same energy. The cell just does it in small steps, slowly and cool enough to keep."*

C4.5 · Applying the model to an unfamiliar food molecule

Experiment name: Mystery Fuel: Balancing a Fat

Render mode: Hybrid 2D+3D

Simulation engine: Molecular + Data-driven model

Interaction level: Argue-from-data

Session length: 18–25 min

NGSS anchor: MS-LS1-7

THEME & SCENE

The molecular bench returns, but the cradle now holds something the student has never seen: a triglyceride, three long charcoal hydrocarbon tails hanging from a short three-carbon backbone like a comb with only three teeth, fifty-one carbons in all, so large it has to be framed wide. Behind the cradle a brass carousel carries five other sample vials on a turntable. To the right, the familiar product rack and oxygen dial, with one new template greyed out and padlocked. Along the bottom, a prediction card is pinned to the bench with three empty boxes waiting for numbers, and beside it a dashboard strip carries an RQ dial, an energy-per-gram bar chart and a small water gauge. A field inset window sits top-right, currently dark. Panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Substrate carousel	Instrument	Brass turntable of six labelled vials; rotating it lifts the chosen molecule into the cradle and rebuilds the tally board's starting counts	Yes: swap
2	Glucose reference molecule	Actor	The pyranose ring from C4.1, kept as the known case; $C_6H_{12}O_6$, 24 atoms, drawn at the same scale as everything else so size comparisons are honest	Yes: drag
3	Butanoic acid molecule	Actor	$C_4H_8O_2$, a four-carbon chain ending in a carboxyl group with one C=O and one O–H; small enough to count by hand, and the rung the student starts on	Yes: drag
4	Lauric acid molecule	Actor	$C_{12}H_{24}O_2$, a twelve-carbon saturated chain with a single carboxyl head; a real coconut and palm-kernel fatty acid; 38 atoms	Yes: drag
5	Tripalmitin molecule	Actor	$C_{51}H_{98}O_6$: glycerol backbone of 3 carbons carrying three ester linkages, each to a sixteen-carbon palmitate tail; ester oxygens highlighted so the three joins are findable; 155 atoms	Yes: drag
6	Alanine molecule	Actor	$C_3H_7NO_2$: a carboxyl group, a central carbon with a methyl branch, and an amino group whose blue nitrogen is the only blue atom on the bench	Yes: drag
7	Nitrogen atom	Particle	CPK blue sphere, covalent radius 0.71 Å at the same ×80 scale, three sockets; the atom the CO ₂ -and-water model has no home for	Yes: drag
8	Oxygen dial and dispenser	Instrument	The hopper from C4.2 with its range widened to 0–160, because a triglyceride needs seventy-two and a half oxygen molecules and the student has to meet that half	Yes: drag
9	Product templates	Structure	Linear CO ₂ ghosts and bent H ₂ O ghosts as before, plus a padlocked urea template, O=C(NH ₂) ₂ , that unlocks only when nitrogen appears on the bench	Yes: place
10	Per-element tally with nitrogen column	Overlay	C, H and O counters as before, with an N column that fades in the moment a nitrogen-bearing substrate is loaded, and a flag reading "this model has no product for N" until the urea route is chosen	No
11	Prediction card	UI-Probe	Pinned card with three numeric boxes for O ₂ , CO ₂ and H ₂ O, plus a signature line; the run button stays locked until the card is filled in	Yes: place
12	Scoring badge	Instrument	Marks the submitted prediction against the balanced equation, element by element, and prints which element was wrong rather than just "incorrect"	No
13	Respiratory quotient dial	Instrument	Analogue dial reading CO ₂ made divided by O ₂ used, 0.60 to 1.10, with printed bands: fat near 0.70, protein near 0.80, carbohydrate at 1.00	No
14	Energy-per-gram bar chart	Overlay	Bars for each substrate in kJ per gram: glucose 15.6, butanoic acid 24.8, lauric acid 36.8, tripalmitin 38.9, alanine 17.7, with the food-label values printed alongside	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Metabolic water gauge	Instrument	Column gauge reading grams of water made per gram of fuel: glucose 0.60, lauric acid 1.08, tripalmitin 1.09; fills as the run proceeds	No
16	Field inset	Overlay	Small window showing a Merriam's kangaroo rat in a Mojave burrow that never drinks, an overwintering monarch cluster at Pismo Beach, or an Anna's hummingbird in overnight torpor; each captioned with the fuel it is living on	Yes: swap

HOW IT WORKS — THE MODEL

The balancer is general, not scripted. For any formula $C_xH_yO_z$ it solves the same three conservation equations the student is asked to solve: carbon fixes CO_2 at x , hydrogen fixes H_2O at $y/2$, and oxygen fixes O_2 at $(2x + y/2 - z)/2$. Tripalmitin returns 72.5, which forces the student to double the whole equation, and the sim says so rather than quietly rounding. Nitrogen breaks the pattern deliberately: with alanine loaded there is no product in the rack that can take an N, and the tally board flags it. Choosing the calorimeter route balances to N_2 ; choosing the body route unlocks urea and drops the usable energy from the bomb value of 23 kJ per gram for protein to the 17 kJ per gram a food label prints, because the nitrogen leaves the body still carrying energy. Energy and RQ come from a lookup of real measured values, not from the bond ledger, and the panel says which.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Substrate	Dropdown	Glucose / Butanoic acid / Lauric acid / Tripalmitin / Alanine / Mystery fuel	Glucose	—	Which molecule the carousel lifts into the cradle and what the tally starts at
Predicted O_2	Numeric field	0–160	0	molecules	The student's own coefficient, submitted before the run is allowed to start
Predicted CO_2	Numeric field	0–120	0	molecules	As above, for carbon dioxide
Predicted H_2O	Numeric field	0–120	0	molecules	As above, for water
Balance help	Radio	Off / Hint / Solve	Off	—	Off scores blind; Hint names the element that is short; Solve fills the card and is logged as such
Nitrogen route	Radio	Nitrogen gas (calorimeter) / Urea (body)	Urea (body)	—	Which product template unlocks, and which energy figure the chart reports
Sample mass	Slider	0.5–20.0	5.0	g	Drives the per-gram outputs, the water gauge and the total energy figure
View	Radio	Molecule bench / Dashboard / Both	Both	—	Whether the cradle, the readouts, or both fill the frame
Tally columns	Multi-select	C / H / O / N	C, H, O	—	Which element counters are shown; N appears automatically when needed
Field inset	Dropdown	None / Mojave kangaroo rat / Pismo monarch cluster / Hummingbird in torpor	None	—	Which real animal is shown alongside, with the fuel it is running on

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Check the tool	Substrate=Glucose; Balance help=Off; Predicted O_2 =6; Predicted CO_2 =6; Predicted H_2O =6	Start with the molecule you already know. Does the bench give the same answer you got in C4.2?
S2	Climbing the fat ladder	Substrate=Butanoic acid, then Lauric acid, then Tripalmitin; Balance help=Off; Sample mass=5.0	Balance each in turn. Why does oxygen needed per gram keep climbing as the tails get longer?
S3	Name that fuel	Substrate=Mystery fuel; Balance help=Off; Sample mass=5.0; Field inset=Mojave kangaroo rat	The RQ dial reads 0.70 and the water gauge reads 1.08 g per gram. Which of the five known substrates is this animal running on, and how do you know?

#	Scenario	Preset	The question the student answers
S4	The nitrogen problem	Substrate=Alanine; Tally columns=C, H, O, N; Nitrogen route=Nitrogen gas (calorimeter), then Urea (body)	The rack has no product that can hold a nitrogen. Where does the model break, and what has to be added to fix it?

STUDENT ACTIVITIES

1. Load lauric acid and fill in the prediction card before running anything. Record your three numbers, then run and copy the scoring badge's verdict, including which element it named.
2. Balance all three fats in turn and build a table of substrate, O₂ per molecule, CO₂ per molecule and H₂O per molecule.
3. Set sample mass to 5.0 g for glucose and then for tripalmitin, recording energy released and water made for each. Work out how many times more water the fat produced per gram.
4. Run the mystery fuel and record the RQ dial and the water gauge. Name the fuel and give both readings as your evidence.
5. Load alanine and try to balance it with only CO₂ and H₂O templates. Record what the tally board's nitrogen column does, then switch the nitrogen route to Urea and record how the energy figure changes.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Balanced equation	Live numeric	molecules	Solved coefficients for O ₂ , CO ₂ , H ₂ O and, where relevant, urea or N ₂ , printed as a full equation
Prediction score	Pass-fail badge	—	Element-by-element check of the student's card, naming any element that failed
Per-element tally	Data table	atoms	C, H, O and N before and after, with a difference column pinned at zero
Respiratory quotient	Live numeric	CO ₂ per O ₂	Computed from the balanced coefficients, to 2 dp, with its band label printed beside it
Energy per gram	Bar chart	kJ/g	All six substrates on one axis, with bomb and food-label values shown separately for the protein
Metabolic water yield	Live numeric	g water per g fuel	From the balanced equation and the molar masses, to 2 dp
Oxygen demand	Live numeric	mol O ₂ per gram	Shows fat needing roughly 2.7 times as much oxygen per gram as glucose
Substrate comparison log	Data table	mixed	Exportable CSV, one row per substrate, every coefficient and every readout

WHAT THE STUDENT SHOULD REALISE

Students believe the respiration equation is a fact about sugar, and that a model that fails is worthless. Balancing a fifty-one-carbon fat with the same rules shows the model generalises; the fat's oxygen-poor tails explain its extra oxygen demand and energy per gram; and alanine's nitrogen shows where it runs out. The student should be able to say: "Same rules, different atom counts, different answer. When the atoms include nitrogen, I need a new product to put it in."

C5 · Tracing carbon through an organism

C5.1 · A carbon atom enters a plant

Experiment name: Through the Stoma: One Carbon Comes In

Render mode: 3D Scene

Simulation engine: Molecular + State machine

Interaction level: Explore

Session length: 18–25 min

NGSS anchor: Supporting — MS-LS1-6

THEME & SCENE

The camera hangs beneath a valley oak leaf in the Central Valley, looking up at its pale, downy underside, backlit so the blade glows lime-green and the veins read as dark ribs. Ahead sits one stoma: two kidney-shaped guard cells, inner walls thickened and darker, bowed apart around a slot 12 μm long. Air molecules drift past as faint grey pairs and trios. One CO_2 molecule carries a carbon drawn as a bright gold sphere with a pulsing white ring, and the whole scene is lit to serve it. Depth-of-field fog hides the mesophyll beyond the pore until the student dives through. A scale bar at bottom left reads 10 μm and rescales at every stage. The control panel docks right; the Tag Tracker strip runs along the top.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Valley oak leaf underside	Environment	Lower epidermis as a sheet of jigsaw-shaped pavement cells under a waxy cuticle with a faint sheen, tufts of star-shaped trichomes; backlit lime-green with dark vein ribs; about 120 stomata per 0.25 mm ² patch, only the central one built at full detail	No
2	Guard cell pair	Actor	Two kidney-shaped cells 35 μm long; thick inner wall drawn darker, thin outer wall, radial cellulose micro-fibrils drawn as fine hoops; each holds five small chloroplasts and a vacuole that swells with turgor; pore width animates 0–8 μm ; the student can pull them apart or push them together	Yes: drag
3	Stomatal pore and substomatal chamber	Structure	Slot 12 μm by 0–8 μm opening into an air-filled cavity 60 μm across, lined by wet spongy-mesophyll surfaces drawn with a glossy water film	Yes: resize
4	Spongy mesophyll cell	Structure	Irregular lobed cell 40 μm across; primary cell wall 0.5 μm as a woven cellulose mat; plasma membrane beneath it as a thin bilayer; streaming cytoplasm, one large pale vacuole; 20 chloroplasts pressed against the wall	No
5	Chloroplast	Structure	Lens 6 $\times$ 3 μm ; double envelope 10 nm apart; 40 grana of 10–20 thylakoid discs each, joined by stroma lamellae; stroma a pale-green gel; two or three white starch grains that grow during the run	Yes: drag
6	Thylakoid membrane detail	Structure	At the nanometre stage the disc surface shows chlorophyll antenna rosettes as green discs, two photosystem blocks, and a teal ATP synthase rotor 20 nm tall that spins while photons land; water molecules split at photosystem II and O_2 dimers leave from there	No
7	Rubisco	Actor	Barrel about 11 nm across, eight large teal subunits and eight small violet ones; 2,000 faint instances in the stroma, one at full detail near the camera; its jaw closes on a CO_2 and opens on two 3-PGA beads	No
8	Calvin cycle carousel	Overlay	Ring of bead molecules in the stroma (RuBP 5 beads, 3-PGA 3 beads, G3P 3 beads) turning slowly; every three CO_2 that enter release one G3P; two G3P join into a glucose ring at the exit	No
9	Tagged carbon atom	Particle	Charcoal CPK carbon overdrawn as a gold sphere $r=0.7$ u with a white pulsing halo and a 2 s comet trail; keeps its identity through every reaction; the student drops the tag onto any CO_2 in the boundary layer	Yes: place

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
10	CO ₂ , H ₂ O and O ₂ pool	Particle	Linear O=C=O (red, charcoal, red), bent H ₂ O (red with two white, 104.5°), O ₂ red dimers; about 400 CO ₂ , 1,500 H ₂ O and 250 O ₂ in view, Brownian walk; all countable in the ledger	No
11	Photon darts	Particle	Gold darts 30 px long arriving from above at a rate set by light; absorbed on contact with an antenna, never reflected back; drive guard-cell opening and the carousel speed	No
12	Product molecules	Structure	Glucose as a six-carbon chair ring with red oxygens and white hydrogens; starch as a helix of glucose rings coiling into the grain; sucrose as a glucose-fructose pair that exits the cell; cellulose as straight glucose chains bundled into a 3 nm microfibril in the wall	Yes: swap
13	Cellulose synthase rosette	Actor	Six-lobed rosette 25 nm across in the plasma membrane; spins and extrudes a microfibril into the wall that lengthens visibly, one glucose unit per turn	No
14	Aperture gauge and chamber CO ₂ probe	Instrument	Micrometer readout across the pore giving width in μm ; a small infrared probe the student drags into the substomatal chamber reading ppm	Yes: drag
15	Tag Tracker strip	Overlay	Top strip naming the tagged atom's current molecule, compartment and elapsed time, with a breadcrumb chain of every molecule it has been part of	No
16	Stage camera and scale bar	UI-Probe	Six-stage camera rig (leaf surface, stoma, mesophyll cell, chloroplast, thylakoid and stroma, cell wall) with a lock-to-atom button; scale bar auto-rescales from 1 mm to 10 nm	Yes: drag

HOW IT WORKS — THE MODEL

Pore width w follows guard-cell turgor: light drives K⁺ into the guard cells, water follows, and w rises toward 8 μm ; darkness or dry air reverses it, with a ten-minute time constant compressed to seconds. CO₂ entry per tick is proportional to $w \times (C_{\text{outside}} - C_{\text{chamber}})$. Inside, the tagged CO₂ dissolves into the wet wall, diffuses straight through the membrane (small and nonpolar, so no channel is needed), crosses the double envelope and reaches the stroma. The Calvin ring is a state machine: Rubisco joins one CO₂ to a five-carbon RuBP, giving two three-carbon 3-PGA; every three CO₂ fixed release one G3P; two G3P make one glucose. The tagged carbon becomes one of that glucose's six carbons and keeps its gold halo through starch, sucrose or cellulose. Failure to avoid: the tagged carbon must never vanish and must never appear in O₂; the O₂ that leaves comes only from water molecules split on the thylakoid membrane, which is why the oxygen tags exist.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Light level	Slider	0–2,000	800	$\mu\text{mol}/\text{m}^2/\text{s}$ PAR	Photon arrival rate, guard-cell opening and carousel speed
Outside CO ₂	Slider	150–1,000	425	ppm	Number of CO ₂ molecules in the boundary-layer air and the entry rate
Air humidity	Slider	10–95	55	% RH	Vapour-pressure deficit; dry air closes the pore even in full light
Guard-cell turgor override	Dial	Auto / 0–100	Auto	%	Forces pore width by hand regardless of light and humidity
Tag element	Dropdown	Carbon of CO ₂ / Oxygen of CO ₂ / Oxygen of H ₂ O	Carbon of CO ₂	—	Which atom carries the gold halo, and therefore where the tag ends up
Product fate	Radio	Starch grain / Cellulose in wall / Sucrose to phloem	Starch grain	—	Where the tagged glucose is built once it leaves the carousel

Control	Widget	Range / options	Default	Unit	What it changes
Camera stage	Dropdown	Leaf surface / Stoma / Mesophyll cell / Chloroplast / Thylakoid and stroma / Cell wall	Stoma	—	Scene scale, fog depth and the scale bar
Lock camera to atom	Toggle	On / Off	On	—	Camera follows the tagged atom through every stage automatically
Molecule counters	Toggle	On / Off	On	—	Shows the countable CO ₂ , H ₂ O, O ₂ and glucose ledger
Playback speed	Dial	0.25×–8×	1×	—	Time scaling

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Morning in the Valley	Light level=800; Outside CO ₂ =425; Air humidity=55; Tag element=Carbon of CO ₂ ; Product fate=Starch grain	Follow the gold atom from the air to the starch grain. How many different molecules is it part of on the way, and does it ever stop being a carbon atom?
S2	Door shut at noon	Light level=800; Air humidity=15; Guard-cell turgor override=Auto	It is a hot, dry afternoon and the pore closes though the sun is bright. What happens to the chamber CO ₂ reading and to the carousel speed?
S3	Berkeley, 1941	Tag element=Oxygen of CO ₂ , then Oxygen of H ₂ O; Light level=800	Ruben and Kamen at Berkeley tagged oxygen atoms. Which tag turns up in the O ₂ leaving the leaf, and which stays in the sugar?
S4	Building a wall	Product fate=Cellulose in wall; Camera stage=Cell wall; Light level=1,200	Your atom began as a gas. How long does it take to become part of a solid microfibril, and how many glucose units are in the chain when it arrives?

STUDENT ACTIVITIES

1. Drag the gold tag onto any CO₂ molecule at the leaf surface, run at 1×, and record every molecule listed on the Tag Tracker breadcrumb until the atom reaches a glucose.
2. Set Air humidity to 15 % with Light level at 800 and record pore width and chamber CO₂ every 10 s for one minute; note the moment the carousel slows.
3. Pull the guard cells apart by hand with the override on, then release them, and record how pore width and the CO₂ entry rate change.
4. Predict, then test, which tag (carbon of CO₂, oxygen of CO₂, oxygen of H₂O) appears in the O₂ leaving the pore; record the ledger counts for each run.
5. Swap Product fate to Cellulose in wall, run until the atom is fixed in the wall, and record the glucose count in the microfibril and the air-to-solid timer.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Tag Tracker breadcrumb	Data table	—	Ordered list of every molecule and compartment the tagged atom has occupied, with the time of each hand-over
Pore width	Live numeric	μm	Stomatal aperture, updated every frame
Chamber CO ₂	Live numeric	ppm	Substomatal probe reading, updated every 0.5 s
CO ₂ entry rate	Line graph	molecules/s vs s	Entry through the modelled stoma; falls to zero as the pore shuts
Molecule ledger	Counter	molecules	CO ₂ in, H ₂ O in, O ₂ out and glucose made, running totals that settle at 6:6:6:1
Air-to-solid timer	Timer	s	Simulated time from tagging to the atom's fixing in starch or cellulose

Readout	Type	Unit	Description
Run log	Data table	mixed	Exportable CSV of every run, with tag element, fate and timings

WHAT THE STUDENT SHOULD REALISE

Students believe a plant turns carbon dioxide into oxygen, as if one gas were swapped for the other. The tagged atom makes that untenable: the carbon never leaves as O₂, the oxygen that leaves comes from water, and the carbon ends up fixed in a solid starch grain or a cellulose wall. The student should be able to say: *"The plant keeps the carbon. That gas atom is now part of the leaf, and the oxygen it gave off came from water."*

C5.2 · A carbon atom moves into an animal

Experiment name: Swallowed: One Carbon Changes Owner

Render mode: 2.5D Layered

Simulation engine: Agent-based + State machine

Interaction level: Manipulate

Session length: 18–25 min

NGSS anchor: Supporting — MS-LS1-7

THEME & SCENE

A valley oak branch at the woodland edge, and on it the eater the student has chosen: a California oakworm caterpillar, 25 mm long, olive-green with yellow and black stripes, shearing a leaf edge with dark mandibles, or a black-tailed mule deer with its head down in the same branch. The animal is drawn whole and small at the left; its gut unrolls to the right as a glowing cutaway strip, pink-red and moist, built in sliding layers: lumen, gut wall, blood, muscle. The gold carbon glows inside the leaf, in a starch grain or a cellulose wall, waiting to be bitten. Leaf tissue keeps a cool green tint, the animal's body is warm crimson, and the gut lumen is deliberately left green because it is still outside. Scale bar re-labels per layer, cm to nm. Panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Eater body	Actor	Two full-body models: a 25 mm California oakworm caterpillar (olive-green, yellow and black stripes, brown head capsule) chewing a leaf edge, and a 1.4 m black-tailed mule deer browsing an oak twig with a sideways grinding jaw; the cutaway strip re-plans itself when the eater is swapped	Yes: swap
2	Valley oak leaf fragment	Structure	Leaf piece 20 × 20 mm showing mesophyll cells with white starch grains, a woven cellulose wall and a phloem vein carrying sucrose; the gold carbon sits in whichever the student chooses; glucose units drawn as countable rings	Yes: place
3	Mouthparts	Structure	Caterpillar mandibles as two dark chitin blades that shear the leaf into 0.5 mm strips; deer molars as ridged enamel that grind to 0.1–2 mm particles; particle size is read by the model	No
4	Foregut and crop, or rumen	Structure	Caterpillar: a short muscular crop; deer: a rumen about 6 L, olive-brown, holding a floating fibre mat; both animate a slow churn	No
5	Rumen microbes	Actor	Deer only: cellulose-digesting bacteria as 2 µm cream rods, 3,000 instanced, and ciliate protozoa 50 µm long, translucent, that attach to fibres and release volatile fatty acid beads (acetate, propionate, butyrate)	Yes: swap
6	Midgut or small intestine lumen	Environment	Tube 2 mm across (caterpillar, pH 10, violet tint) or 25 mm across (deer, pH 7); a peristalsis wave every 4 s carries the food bolus along	No
7	Digestive enzymes	Actor	Amylase as a blue clamshell that cuts starch chains to maltose; maltase as a teal block fixed on the brush border; cellulase as an orange wedge that exists only inside rumen microbes; 200 of each, drifting and snapping onto matching substrates	Yes: swap
8	Gut wall epithelium	Structure	Deer: villi 0.5 mm tall; caterpillar: flat columnar cells; every cell carries a brush border of 1 µm microvilli, a nucleus, and tight junctions drawn as zips to its neighbours; zooms to a bilayer at the nm stage	No
9	Sugar transporter and acid route	Actor	SGLT1 as a violet barrel in the microvillus membrane that carries one glucose with Na ⁺ per cycle and therefore saturates; in the deer's rumen wall a pale-yellow passive route for the fatty-acid beads	No
10	Capillary and blood	Environment	8 µm capillary under the epithelium with plasma pale yellow and biconcave red blood cells 7 µm across, 100 instanced, moving at 1 mm/s toward the hepatic portal vein; in the caterpillar an open haemolymph space with a dorsal tube heart	No
11	Liver way-station	Structure	Lobule cutaway where glucose may be parked as branched glycogen and later released; in the deer it also rebuilds glucose from propionate; adds a visible delay when the detour is on	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
12	Heart and arterial route	Structure	Deer: four-chamber heart at 60 bpm sending blood down an artery to a leg muscle; caterpillar: dorsal vessel pulsing haemolymph forward through the body cavity	No
13	Muscle cell	Structure	Striated fibre 50 μm across; sarcolemma studded with green sugar-transporter proteins; cytoplasm with glycogen granules and dark, inactive mitochondria that light up only in C5.3	No
14	Frass and dung turnstile	Instrument	Counter at the hindgut or rectum ticking for every tagged atom that leaves without entering the body; black 2 mm frass pellets or deer pellets pile up beneath	No
15	Tagged carbon atom	Particle	Gold sphere with white halo and comet trail, identical to C5.1; keeps its identity through every molecule; the camera lock follows it across layers; the student may re-tag any glucose in the leaf	Yes: drag
16	Sugar meter, pH probe and Tag Tracker	Instrument	A meter reading blood glucose or haemolymph trehalose in mmol/L; a draggable pH probe along the gut strip; the top Tag Tracker strip listing molecule, compartment and owner (leaf, gut lumen, animal body) with a transit stopwatch	Yes: drag

HOW IT WORKS — THE MODEL

The gut is a chain of compartments run as a state machine, and the tagged atom's current molecule decides which transitions it may take. Chewing sets particle size, and digestion rate scales with surface area, so rate $\propto 1/\text{diameter}$. Starch is cut by amylase, then maltase, to glucose, at a rate set by pH and enzyme count; glucose crosses the epithelium only through SGLT1 carriers, so absorption saturates when a big meal arrives. Cellulose has no enzyme in either animal's own set: in the caterpillar it goes straight to frass, while in the deer rumen microbes carrying cellulase ferment it to volatile fatty acids that cross the rumen wall and reach the same muscle. In the deer the microbes ferment most starch too, and the liver rebuilds glucose from propionate; in the caterpillar absorbed sugar travels the haemolymph as trehalose. Transit times use real data (caterpillar 2–4 h, deer 24–72 h), compressed. Every tagged atom ends in muscle, liver store or dung; none is ever created, destroyed or "turned into energy".

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Eater	Radio	California oakworm caterpillar / Mule deer	California oakworm caterpillar	—	Swaps the body, the gut plan, the circulation and the enzyme set
Tag location in leaf	Dropdown	Starch grain / Cellulose wall / Sucrose in phloem sap	Starch grain	—	Which molecule the gold atom starts in
Chewing thoroughness	Slider	0.1–3.0	1.0	mm particle size	Surface area available to enzymes and therefore digestion rate
Gut microbes	Toggle	Present / Absent	Present	—	Whether cellulase-carrying microbes live in the rumen (deer) or the midgut (caterpillar, very few)
Meal size	Slider	1–30	10	g leaf	Number of glucose units in play and the height of the sugar peak
Muscle activity	Slider	1–8	2	$\times$ resting	Rate at which muscle transporters pull sugar out of the blood
Liver detour	Toggle	On / Off	Off	—	Whether the tagged glucose is parked as glycogen before release
Camera layer	Dropdown	Whole animal / Gut lumen / Gut wall / Bloodstream / Muscle cell	Whole animal	—	Which layer slides to the front, and the scale bar
Playback speed	Dial	1 $\times$ –600 $\times$	60 $\times$	—	Time compression of the gut transit

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Caterpillar eats starch	Eater=California oakworm caterpillar; Tag location in leaf=Starch grain; Gut microbes=Present	Track the gold atom from leaf to muscle. At which single step does it stop being outside the animal and start being inside it?
S2	Caterpillar eats wall	Eater=California oakworm caterpillar; Tag location in leaf=Cellulose wall	The atom is eaten but never absorbed. Where does it come out, and why could the caterpillar not use it?
S3	The deer's helpers	Eater=Mule deer; Tag location in leaf=Cellulose wall; Gut microbes=Present, then Absent	Same wall carbon, different animal. Which molecule carries the atom into the deer's blood, and what happens when the microbes are switched off?
S4	Big meal, fast run	Eater=Mule deer; Tag location in leaf=Starch grain; Meal size=30; Muscle activity=8	Blood sugar spikes and then falls. Where did the atoms go, and does a bigger meal make each transporter work any faster?

STUDENT ACTIVITIES

1. Set Eater to the caterpillar and the tag in a starch grain, run at 60×, and record every molecule and compartment on the Tag Tracker, marking the row where the owner column changes from gut lumen to animal body.
2. Move the tag to the cellulose wall, run again, and record whether the atom ever reaches the blood; read the frass turnstile at the end.
3. Swap Eater to the mule deer with the tag still in cellulose and record the molecule that finally crosses the rumen wall; switch Gut microbes to Absent and re-run.
4. Set Chewing thoroughness to 3.0 mm and then 0.1 mm at Meal size 10, and record the transit stopwatch and the sugar-meter peak for each.
5. Drag the pH probe along the caterpillar's gut and record the pH where amylase works fastest; compare with the deer's small intestine.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Tag Tracker breadcrumb	Data table	—	Molecule, compartment, owner and time for every hand-over of the tagged atom
Blood sugar	Line graph	mmol/L vs min	Meter trace of glucose (deer) or trehalose (caterpillar); rises after absorption, falls with muscle uptake
Absorption rate	Live numeric	molecules/s	Crossings through the transporters per second, showing saturation at large meals
Frass and dung tagged count	Counter	atoms	Tagged atoms that left without ever entering the body
Gut transit time	Timer	h	From first bite to exit or to the muscle cell, per run
Gut pH	Live numeric	pH	Reading at the draggable probe position
Run log	Data table	mixed	Exportable CSV with eater, tag location, particle size and final fate

WHAT THE STUDENT SHOULD REALISE

Students believe an eaten leaf simply becomes part of the animal, or is "used up as energy". The tagged atom shows neither: the leaf's carbon changes owner only at the instant a small molecule crosses the gut wall into the blood, and carbon locked in cellulose passes straight through a caterpillar unless a microbe frees it. The student should be able to say: *"Eating doesn't move carbon into an animal; absorbing does, and only small molecules get absorbed."*

C5.3 · A carbon atom returns to the air

Experiment name: Exhaled: One Carbon Leaves the Deer

Render mode: Hybrid 2D+3D

Simulation engine: Molecular + Fluid/thermal

Interaction level: Investigate

Session length: 12–18 min

NGSS anchor: Supporting — MS-LS1-7

THEME & SCENE

A black-tailed mule deer stands in morning fog at Point Reyes, breath steaming, drawn as a small live silhouette in the top-left corner whose ribs rise and fall at the breathing rate. The main view is a ride inside one of its leg-muscle fibres: amber cytoplasm banded with striations, and in front a mitochondrion sliced open, outer membrane smooth, inner membrane folded into cristae like a rack of shelves, the matrix a warm honey gel. The gold carbon arrives inside a glucose ring and the camera follows it in. Along the bottom a 2D route map, drawn like a transit line, lists the stations ahead: cytosol, mitochondrion, capillary, vein, heart, lung capillary, alveolus, nostril. A capnograph trace glows at the deer's nose. Scale bar starts at 1 μm ; panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Deer silhouette with capnograph	Actor	2D silhouette in fog, ribcage animating at the breathing rate; a probe at the nostril traces exhaled CO_2 percentage breath by breath	No
2	Muscle fibre cytosol	Environment	50 μm fibre with sarcomere banding, glycogen rosettes, and ATP drawn as yellow triple-phosphate tokens consumed by sliding filaments at a rate set by activity	No
3	Glycolysis lane	Overlay	Ten-step conveyor in the cytosol reduced to three visible beads: one glucose (6 carbons) splitting into two pyruvate (3 carbons each); the gold carbon rides one of them	No
4	Mitochondrial outer membrane	Structure	Smooth ellipsoid $2 \times 0.8 \mu\text{m}$ with grey porin barrels that pyruvate and CO_2 pass through; intermembrane space 15 nm tinted proton-blue	No
5	Inner membrane and cristae	Structure	Fourteen lamellar shelves in deep amber; electron-transport complexes I to IV as embedded blocks; teal ATP synthase rotors 22 nm tall that spin faster with demand; O_2 dimers are consumed at complex IV and water molecules appear there	Yes: resize
6	Matrix and Krebs carousel	Overlay	Honey gel holding a ring of eight bead molecules from citrate (6 beads) round to oxaloacetate (4 beads); every turn ejects two CO_2 from the ring into the matrix	No
7	Pyruvate gate complex	Actor	Large violet complex at the matrix entry that snips one carbon off each pyruvate as CO_2 and hands the two-carbon remainder to the carousel	No
8	Tagged carbon atom	Particle	Gold sphere with white halo and trail, identical to C5.1 and C5.2; the student chooses which of glucose's six carbons is gold, which sets the station where it leaves as CO_2	Yes: place
9	O_2 , CO_2 and H_2O pool	Particle	O_2 red dimers arriving from the capillary; CO_2 made in the matrix as red-charcoal-red rods; H_2O formed at complex IV; all countable, settling at 6 O_2 in to 6 CO_2 out per glucose	No
10	Sarcolemma and interstitial fluid	Structure	Muscle-cell bilayer that CO_2 crosses directly with no channel; a thin water layer between the fibre and the capillary	No
11	Capillary with red blood cells	Environment	8 μm vessel; biconcave red cells 7 μm across with haemoglobin; inside each, carbonic anhydrase drawn as a green disc converting $\text{CO}_2 + \text{H}_2\text{O}$ into a bicarbonate bead and H^+ ; three carriage forms colour-coded (dissolved, bicarbonate, bound to haemoglobin)	Yes: swap

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
12	Venous return and heart	Structure	Transit-map stations with a four-chamber heart beating at 60–180 bpm; the right ventricle sends blood to the lung capillary; blood speed scales with heart rate	No
13	Alveolus and barrier	Structure	Air sac 200 μm across lined by flat type I cells with a surfactant sheen; total barrier 0.5 μm to the capillary; CO_2 diffuses from blood to air across it while O_2 crosses the other way	No
14	Airway and nostril turnstile	Instrument	Bronchiole to trachea to nostril; a turnstile counts every tagged atom leaving the body, per breath, and stamps a gold tick on the capnograph	No
15	Gas ledger	Overlay	Live O_2 consumed, CO_2 produced, H_2O made and ATP made, with $\text{RQ} = \text{CO}_2 \div \text{O}_2$ over the last 10 s	No
16	Station map and scale bar	UI-Probe	Bottom transit line with eight stations and a sliding gold marker; dragging the marker jumps the camera to that station; scale bar auto-rescales from 1 μm to 1 mm	Yes: drag

HOW IT WORKS — THE MODEL

The glucose carrying the tag is split in the cytosol into two pyruvate, releasing no CO_2 yet. Each pyruvate enters the matrix, where the gate complex snips one carbon off as CO_2 and the Krebs carousel releases two more per turn, so all six glucose carbons leave within a few turns; the tag position sets the exit station from a lookup table, a fair simplification of a ring that scrambles positions. O_2 is consumed only at complex IV on the cristae, where it becomes water: the sim must never show O_2 turning into CO_2 , because that is the misconception. ATP synthase rate rises with demand, and demand sets O_2 delivery, heart rate and breathing rate. CO_2 leaves the cell by diffusion down its gradient and travels 70 % as bicarbonate, 23 % bound to haemoglobin and 7 % dissolved, with carbonic anhydrase setting the conversion speed. At the alveolus CO_2 crosses into air because its partial pressure in blood is higher. RQ is 1.0 for glucose and 0.7 for fat.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Activity level	Slider	1–10	2	× resting	Muscle ATP demand, O_2 delivery, heart rate (60–180 bpm) and breathing rate
Fuel	Radio	Glucose / Fatty acid / Acetate from the rumen	Glucose	—	Which molecule carries the tag in; changes the entry route and the RQ
Tag carbon position	Stepper	C1–C6	C3	—	Which carbon of glucose is gold; sets the station where it leaves as CO_2
Carbonic anhydrase	Toggle	Active / Blocked	Active	—	Speed of CO_2 to bicarbonate conversion inside red blood cells
Breathing	Radio	Auto / Hold breath	Auto	—	Hold breath stops exit at the alveolus; blood CO_2 climbs until release
Cristae per mitochondrion	Slider	4–24	14	count	ATP synthase capacity and the maximum respiration rate
Camera station	Dropdown	Cytosol / Mitochondrion / Capillary / Heart / Alveolus / Nostril	Mitochondrion	—	Camera position and scale bar
Playback speed	Dial	0.25×–60×	4×	—	Time compression
Gas ledger	Toggle	On / Off	On	—	Shows the O_2 , CO_2 , H_2O , ATP counters and RQ

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Resting deer	Activity level=2; Fuel=Glucose; Tag carbon position=C3	Follow the atom from glucose to the nostril. At which station does it become CO ₂ , and does any O ₂ molecule ever touch it?
S2	Uphill sprint	Activity level=9; Fuel=Glucose	The deer bolts. How much faster does the gold atom leave, and which numbers on the gas ledger change together?
S3	Held breath	Breathing=Hold breath; Activity level=4	The atom reaches the lung and stops. Where does CO ₂ pile up, and what does that tell you about why an animal must breathe out?
S4	Slow blood	Carbonic anhydrase=Blocked; Activity level=4	With the red-cell enzyme blocked, most CO ₂ must travel dissolved. Compare capillary CO ₂ and the transit time to the lung with the S1 run.

STUDENT ACTIVITIES

1. Run S1 at 4× and record the station where the gold atom leaves as CO₂ and the elapsed time to the nostril turnstile.
2. Step Tag carbon position from C1 to C6 and record the exit station for each; note which carbons leave first.
3. Set Activity level to 2, then 9, and record O₂ consumed, CO₂ produced and heart rate over 30 simulated seconds each; calculate RQ for both.
4. Swap Fuel to Fatty acid, run 30 s, and record how RQ changes; explain it from the ledger counts.
5. Hold breath for 20 simulated seconds, record capillary CO₂ before and after, then release and record the height of the first exhaled CO₂ peak on the capnograph.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Tag station log	Data table	—	Station, molecule and time for every hand-over from glucose to exhaled CO ₂
Gas ledger	Live numeric	molecules	O ₂ in, CO ₂ out, H ₂ O made and ATP made, updated every tick
Respiratory quotient	Live numeric	—	CO ₂ produced divided by O ₂ consumed over the last 10 s
Capnograph	Line graph	% CO ₂ vs s	Exhaled CO ₂ per breath at the nostril, with tagged exits marked as gold ticks
Blood CO ₂ carriage	Bar chart	%	Share travelling dissolved, as bicarbonate, and bound to haemoglobin
Heart rate and breathing rate	Live numeric	bpm and breaths/min	Both driven by activity level
Run log	Data table	mixed	Exportable CSV of every run

WHAT THE STUDENT SHOULD REALISE

Students believe the CO₂ an animal breathes out is the air it breathed in with the oxygen taken away, and that food mass "becomes energy". The tagged atom leaves the mitochondrion as CO₂ built from food carbon, while every O₂ molecule ends up in water, never in CO₂. The student should be able to say: *"The carbon I breathe out was in my food, and before that in a leaf. Food mass leaves through the lungs as gas; energy is what the bonds gave up on the way."*

C5.4 · Photosynthesis and respiration as linked opposite processes

Experiment name: Round Trip: Atoms Cycle, Energy Doesn't

Render mode: 2D Canvas

Simulation engine: Molecular + Data-driven model

Interaction level: Argue-from-data

Session length: 12–18 min

NGSS anchor: Supporting — MS-LS1-6, MS-LS1-7

THEME & SCENE

A flat, high-contrast bench on an ink-dark background, split by one bright vertical seam. Left, in cool jade: a glass reaction box labelled PHOTOSYNTHESIS, six CO₂ and six H₂O tiles on its input shelf, one glucose ring and six O₂ on its output shelf, a thick arrow between, and an anglepoise lamp above pouring gold photons in. Right, in warm ochre: an identical box labelled RESPIRATION with its arrow pointing the other way, heat shimmer rising and yellow ATP tokens dropping into a tray. Between them the shared ledger: one column of atom counters, C 6, H 12, O 18, that both boxes draw on, and beneath it an energy ledger with a light-in counter and a heat-out counter. The gold tagged carbon glows in whichever tile holds it. Panel docks right; the claim-builder tray runs along the bottom.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Photosynthesis reaction box	Structure	Glass box 480 × 300 px with a jade rim, input shelf on the left, output shelf on the right, and a labelled arrow; runs the forward reaction whenever its input shelf holds a full set	No
2	Respiration reaction box	Structure	Identical geometry with an ochre rim and the arrow reversed; runs the breakdown reaction whenever a glucose and six O ₂ sit on its input shelf	No
3	Reversible reaction arrows	Actor	Two thick arrows with draggable heads; flipping one redraws that box's shelves and re-runs it backwards; a red padlock appears on the respiration arrow when reversed, because the box cannot emit light	Yes: drag
4	Molecule tiles	Particle	CPK ball-and-stick tiles: C charcoal r=0.7 u, O red r=0.6 u, H white r=0.3 u, bonds as grey cylinders; per set exactly 6 CO ₂ , 6 H ₂ O, 1 glucose and 6 O ₂ exist at any moment, in one form or the other; tiles drag between shelves and across the seam	Yes: drag
5	Tagged carbon	Particle	One of the six carbons drawn gold with a white halo; it appears exactly once in the whole scene at every instant and the ledger checks for it	Yes: place
6	Shared atom ledger	Overlay	Central column of C, H and O counters for the whole scene plus per-box sub-counts; totals pinned; flashes red if a tile is ever duplicated or lost	No
7	Energy ledger	Overlay	Below the atom ledger: a photon counter (light in, gold) and a heat-plus-ATP counter (energy out, red and yellow); the first increments only on the left, the second only on the right, and neither ever counts down	No
8	Lamp and photon stream	Environment	Anglepoise lamp over the left box; photons as gold darts at a rate set by light that vanish into the box and never come back out	No
9	Heat shimmer and ATP tokens	Particle	Rising heat waves from the right box and yellow ATP tokens dropping into a tray, about 32 per glucose; both feed the energy ledger	No
10	Bond-energy bars	Overlay	Side bars showing the stored bond energy of each shelf's molecules; higher for glucose plus O ₂ than for CO ₂ plus H ₂ O by about 2,800 kJ per mole of glucose	No
11	Loop conveyor	Actor	Optional belt the student connects from the left output shelf to the right input shelf and back again, so the same tiles circulate; a lap counter shows completed round trips	Yes: connect
12	Location and enzyme cards	Structure	Two small cards per box, hidden until revealed: where (chloroplast stroma and thylakoid, or mitochondrial matrix and cristae) and which enzymes (Rubisco and ATP synthase, or dehydrogenases and ATP synthase); deliberately not mirror images	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
13	Claim-builder tray	UI-Probe	Bottom tray holding the claim "respiration is photosynthesis run backwards" and draggable evidence cards (atom ledger, energy ledger, location, light, enzymes); scores each card as supporting or contradicting	Yes: drag
14	Rate dials	Instrument	One dial per box reading reactions per minute; the dials drive how fast tiles move between shelves	No
15	Net-rate meter	Instrument	Reads left rate minus right rate as net CO ₂ consumed or made per minute, with a green zero mark	No
16	Journey replay scrubber	UI-Probe	Timeline that replays the tagged atom's full C5.1 to C5.3 journey as one lap on the conveyor, with the leaf, the deer's gut and the deer's lung drawn as small icons on the belt	Yes: drag

HOW IT WORKS — THE MODEL

Both boxes are stoichiometric engines over one shared tile set: the left consumes 6 CO₂ + 6 H₂O and light to yield 1 glucose + 6 O₂, the right consumes 1 glucose + 6 O₂ to yield 6 CO₂ + 6 H₂O plus heat and about 32 ATP. The atom ledger is recomputed every tick from the tiles on screen and must never change; the tagged carbon is a persistent object that swaps its parent molecule but never its identity. Rates come from the panel: the left box follows a saturating light curve, the right follows demand. The energy ledger is one-way by construction: light enters only on the left, heat and ATP leave only on the right, and neither counter ever decrements, which is why reversing the respiration arrow to "make light" locks with a padlock. This is honest at this grade: the two processes are exact opposites in atom bookkeeping but not in energy, machinery or location. The failure to avoid is a single double-headed arrow implying one reaction that simply runs backwards.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Light on the left box	Slider	0–2,000	800	μmol/m ² /s PAR	Photon rate and the photosynthesis dial
Energy demand on the right box	Slider	0–8	2	× resting	Respiration dial, ATP and heat output
Arrow to reverse	Radio	None / Left (photosynthesis) / Right (respiration)	None	—	Attempts to run one box backwards; the atom ledger allows it, the energy ledger locks the right box
Loop conveyor	Toggle	Connected / Open	Open	—	Whether each box's outputs feed the other's inputs, enabling laps
Tile sets	Stepper	1–4	1	glucose equivalents	How many sets of 6 CO ₂ + 6 H ₂ O are on the bench
Ledger view	Dropdown	Atoms / Energy / Both	Both	—	Which ledger is enlarged in the centre
Location and enzyme cards	Toggle	Hidden / Revealed	Hidden	—	Shows the non-mirror facts about where and how each process runs
Journey replay	Timeline scrubber	Leaf (C5.1) to Lung (C5.3)	Leaf (C5.1)	—	Scrubs the tagged atom around one full lap of its earlier journey
Playback speed	Dial	0.25×–8×	1×	—	Time scaling

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Mirror check	Light on the left box=800; Energy demand on the right box=2; Ledger view=Atoms	Read the atom ledger before and after ten reactions in each box. Which numbers never change, and whose outputs exactly match whose inputs?

#	Scenario	Preset	The question the student answers
S2	Flip the arrow	Arrow to reverse=Right (respiration); Ledger view=Energy	You reversed respiration, so it ought to make light. The atoms allow it. What locks, and what does that say about the two processes?
S3	Ten laps	Loop conveyor=Connected; Light on the left box=800; Energy demand on the right box=2; Tile sets=1	Run the gold atom round ten laps. How many atoms did you use up, and how much light did you have to add?
S4	Not a mirror after all	Location and enzyme cards=Revealed; Ledger view=Both	Build the claim "respiration is photosynthesis in reverse". Which evidence cards support it and which contradict it? Write the corrected claim.

STUDENT ACTIVITIES

1. Run S1 and record the C, H and O totals at the start and after ten reactions in each box, noting which tiles crossed the seam.
2. Drag the head of the right box's arrow to reverse it and record what the energy ledger shows and why the padlock appears.
3. Connect the loop conveyor, run ten laps, and record the lap counter, the atom totals, and the light-in and heat-out counters at the end.
4. Predict, before revealing the cards, whether the same enzymes run both boxes; then reveal them and record both location cards.
5. Drag evidence cards into the claim-builder until it accepts a claim, and write the corrected one-sentence claim it accepted.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Shared atom ledger	Data table	atoms	C, H and O totals for the scene and per box; pinned, flashing red if any total moves
Energy ledger	Live numeric	photons and kJ	Light in (left only) and heat plus ATP out (right only), cumulative and one-way
Bond energy per shelf	Bar chart	kJ per mol glucose	Stored bond energy of inputs versus outputs in each box
Lap counter	Counter	laps	Complete round trips of the tagged atom on the conveyor
Net rate	Live numeric	CO ₂ /min	Left rate minus right rate
Claim score	Pass-fail badge	—	Green when the evidence dragged onto the claim agrees with the ledgers and cards
Run log	Data table	mixed	Exportable CSV of every run

WHAT THE STUDENT SHOULD REALISE

Students believe respiration is photosynthesis played backwards, so the two cancel out and nothing is needed to keep them going. The two ledgers split that idea in half: the atoms do reverse exactly and can go round for ever, but the energy does not, since light must be added on every lap and heat leaves on every lap, and the machinery and location differ. The student should be able to say: *"The atoms go round and round; the energy goes through once, in as light and out as heat."*

C5.5 · Why plants run both processes

Experiment name: Night Shift: A Sealed Seedling, 24 Hours

Render mode: Hybrid 2D+3D

Simulation engine: Data-driven model + Particle system

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: Supporting — MS-LS1-7

THEME & SCENE

A single glass chamber on a dark bench under a dimmable LED bar that brightens and fades with a 24-hour clock in the corner. Inside stands a valley oak seedling 25 cm tall with five lobed leaves, its acorn still attached at the base, the pot cut away so the tap root shows against dark soil. A second, lower glass box can seal the root zone on its own. Two CO₂ sensors glow on the lids, a third sensor labelled TAGGED FRACTION reads a gold number, and a gas syringe on the top lid holds a charge of gold-tagged CO₂. Gas molecules drift as countable specks, gold where tagged. The right half of the screen is a 24-hour strip chart with the night shaded. A 5 cm scale bar sits by the pot; a leaf inset rescales to 20 µm. Panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Shoot chamber	Structure	Sealed borosilicate box 300 × 300 × 400 mm (36 L) with a silicone-gasketed lid, four feedthroughs and a lid fan; the glass fogs slightly at night when the leaves stop transpiring	No
2	Root chamber	Structure	Lower box 300 × 300 × 200 mm (18 L) sealed around the stem with a split rubber collar; holds the pot and soil; merges with the shoot chamber when the split is off	Yes: swap
3	Valley oak seedling	Actor	250 mm seedling with five lobed matte-green leaves on petioles, a woody stem, shrivelling acorn cotyledons at the base, a 200 mm tap root with fine laterals; leaves tilt toward the lamp by day; grows measurably along the run; three age variants	Yes: swap
4	Leaf inset	Overlay	20 µm-scale inset of a mesophyll cell with stomata opening by day, chloroplasts whose starch grains grow by day and shrink by night, and mitochondria drawn active at every hour	No
5	Root cell inset	Overlay	Inset of a root cortex cell with no chloroplasts, many mitochondria, sucrose arriving from the phloem and ATP tokens made continuously	No
6	Phloem sugar stream	Particle	Sucrose beads travelling down the stem from leaves to roots at a scaled 0.5 mm/s, drawn gold when they carry a tagged carbon	No
7	Chamber gas molecules	Particle	Countable CO ₂ rods (red, charcoal, red) and O ₂ red dimers on a Brownian walk; tagged CO ₂ drawn gold; the total count scales with the sensor reading	No
8	Tagged CO ₂ syringe	Instrument	60 mL gas syringe on the lid pre-filled with gold-tagged CO ₂ ; pushing the plunger injects the pulse at the set clock time	Yes: drag
9	Infrared CO ₂ sensors	Instrument	One per chamber, 0–5,000 ppm, ±20 ppm, 20 s response, lid-mounted with a small LED display	No
10	Tagged-fraction sensor	Instrument	Laser cell that reads the percentage of chamber CO ₂ carrying the tag, ±0.5 %; can be moved between the two chambers	Yes: drag
11	LED grow bar and clock	Environment	500 mm bar dimmable 0–2,000 µmol/m ² /s following the 24-hour clock's programme; a quantum sensor reads delivered light	No
12	Thermistor and Peltier plate	Instrument	Holds each chamber at the set day and night temperature so heat cannot be confused with light	No
13	Balance under the pot	Instrument	0.01 g platform balance reporting the plant's dry-mass gain computed from net carbon	No
14	Plant energy gauge	Overlay	Three ATP dials (leaf, stem, root) driven only by respiration, and a growth-and-transport bar that stalls when any dial reaches zero	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Respiration lever	UI-Probe	A red lever labelled "Pause respiration (impossible in a real plant)"; when pulled, every mitochondrion dims and the ATP dials fall	Yes: drag
16	24-hour strip chart	Overlay	Right-hand chart of CO ₂ per chamber, tagged fraction, light level and the ATP dials against clock time, with the night band shaded	No

HOW IT WORKS — THE MODEL

Each chamber is a well-mixed gas volume where net CO₂ change equals respiration minus photosynthesis. Photosynthesis runs in leaves only, $P = P_{\text{max}} \times I/(I + 250)$ times a CO₂ term; respiration runs in every living cell at every hour, roots included, $R = R_{20} \times 2^{((T-20)/10)} \times \text{tissue mass}$. Sugar made in the leaves is split by a fixed rule: 40 % respired in the leaf, 30 % sent down the phloem to stem and roots, 30 % stored as starch and drawn down at night. The tagged pulse is tracked as a labelled fraction through every pool, so the sim can report what share of tonight's exhaled CO₂ was fixed today and when tagged carbon first reaches the roots, after a phloem delay of a few hours. Growth and transport cost ATP, which only respiration supplies. The virtual respiration lever is honest because it is labelled impossible: it exists to show what would fail. The failure to avoid is one arrow that flips at dusk; both arrows exist at every hour.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Day length	Slider	0–24	14	h	Hours the LED bar is on in each 24-hour cycle
Peak light	Slider	0–2,000	900	μmol/m ² /s PAR	Midday photon level at the leaves
Day temperature	Slider	5–35	25	°C	Respiration rate while the lamp is on, through Q ₁₀ = 2
Night temperature	Slider	5–35	15	°C	Respiration rate while the lamp is off
Tagged CO ₂ pulse time	Timeline scrubber	00:00–24:00	06:00	h	Clock time at which the syringe fires
Pulse size	Slider	0–400	200	ppm added	Amount of tagged CO ₂ injected into the shoot chamber
Chamber layout	Radio	Single sealed / Shoot and root split	Single sealed	—	Whether the roots have their own gas volume and sensor
Seedling stage	Dropdown	Acorn just sprouted (no leaves) / 4-week seedling / 12-week seedling	4-week seedling	—	Leaf area and root mass, and whether photosynthesis is possible at all
Respiration lever	Toggle	Running / Paused (virtual)	Running	—	Thought experiment that pauses every mitochondrion
Time compression	Dial	60×–3,600×	600×	—	Simulated seconds per real second

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	One full day	Day length=14; Peak light=900; Chamber layout=Single sealed; Respiration lever=Running	CO ₂ falls from 08:00 and rises after 20:00. Does the night rise mean photosynthesis has reversed, or that something else was running all along?
S2	Tag at dawn, sniff at midnight	Tagged CO ₂ pulse time=06:00; Pulse size=200; Chamber layout=Single sealed	The gold CO ₂ disappears by noon. At midnight the sensor finds gold CO ₂ again. Where did it spend the afternoon, and what does its return prove?
S3	Roots never see the sun	Chamber layout=Shoot and root split; Tagged CO ₂ pulse time=06:00; Pulse size=200	The root chamber's CO ₂ climbs all day and all night. When does tagged carbon first appear down there, and how did it get there?

#	Scenario	Preset	The question the student answers
S4	The impossible plant	Respiration lever=Paused (virtual); Day length=14; Peak light=900	Photosynthesis still runs and sugar piles up, yet the ATP dials fall to zero. What stops first, and why can a plant not live on photosynthesis alone?

STUDENT ACTIVITIES

1. Run S1 for one 24-hour cycle at 600× and record shoot-chamber CO₂ at 06:00, 12:00, 18:00 and 00:00; mark the two clock times when the trace changes direction.
2. Set the pulse to 06:00, push the syringe plunger, and record the tagged fraction at 08:00, 12:00, 18:00 and 00:00; note the earliest night hour at which it rises again.
3. Switch to the split layout, re-run with the same pulse, and record the hour at which the root chamber's tagged fraction first exceeds zero, plus root-chamber CO₂ at noon and at midnight.
4. Pull the respiration lever at 08:00 and record the leaf, stem and root ATP dials every two simulated hours until the growth bar stops; write down what failed first.
5. Swap Seedling stage to Acorn just sprouted, run 24 hours, and record the CO₂ trace and the balance reading; state what the seedling is respiring when it has no leaves at all.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Shoot chamber CO ₂	Line graph	ppm vs h	24-hour trace with the night band; falls by day and rises by night
Root chamber CO ₂	Line graph	ppm vs h	Rises at every hour; drawn only in the split layout
Tagged CO ₂ fraction	Line graph	% vs h	Gold trace that drops after the pulse and returns at night; read per chamber
Tagged carbon location	Bar chart	%	Share of the pulse now in air, leaf sugar, starch, phloem, roots, or respired back out
Net daily carbon balance	Live numeric	mg C per day	Photosynthesis gain minus 24-hour respiration; matches the balance reading
ATP dials	Live numeric	% of demand	Leaf, stem and root, driven only by respiration
Run log	Data table	mixed	Exportable CSV, one row per simulated 10 min

WHAT THE STUDENT SHOULD REALISE

Students believe photosynthesis is what plants do and respiration is what animals do, or that a plant respire only at night. The sealed seedling fixes tagged carbon by day and breathes part of it back out that same night, its roots release CO₂ at every hour, and the virtual lever shows that a plant without respiration cannot grow or feed its roots. The student should be able to say: *"Photosynthesis makes the plant's food; respiration is how every cell in the plant, day and night, uses it."*